

Cardiovascular System

August 2025!

Lecture contents of Cardiovascular system

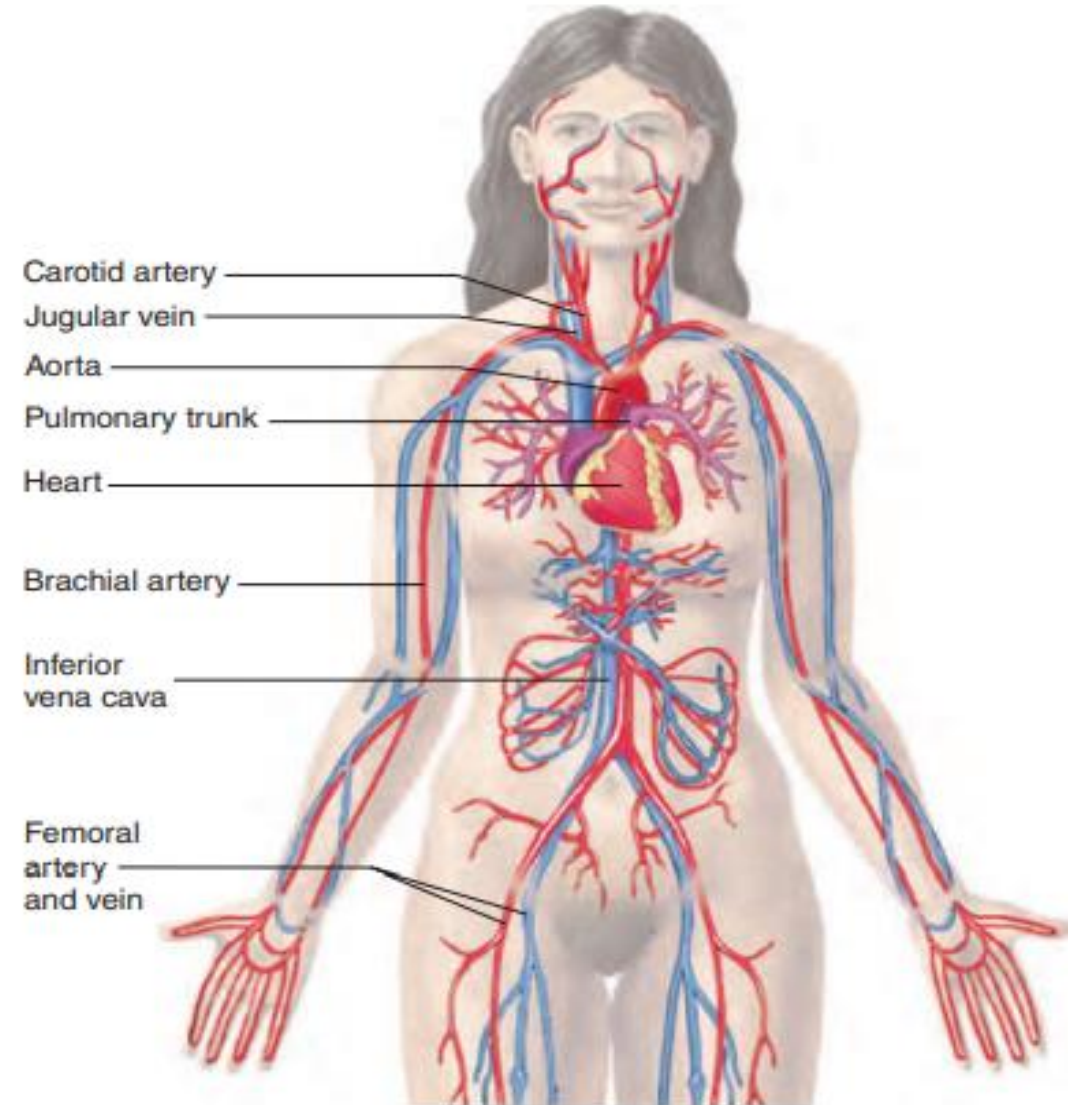
- ✓ Introduction to Cardiovascular System
- ✓ *Thoracic cavity*
- ✓ Position, Size, and Shape of the Heart
- ✓ Gross Anatomy of the Heart
- ✓ The wall of heart chambers
- ✓ Heart Valves
- ✓ Vasculature of heart(*Coronary vasculature*)
- ✓ Conducting system of the heart
- ✓ Innervation of the heart

Lecture contents of Cardiovascular system,,

- ✓ Surface anatomy of the heart & great vessels**
- ✓ Arteries of the Head and Neck**
- ✓ Arteries of the Upper Limbs**
- ✓ Arteries of the Abdomen**
- ✓ Arteries of the pelvis and lower limbs**
- ✓ Venous drainage of the body**
- ✓ Circulatory Routes**
- ✓ Anatomy of the Lymphatic System**

Introduction to Cardiovascular System

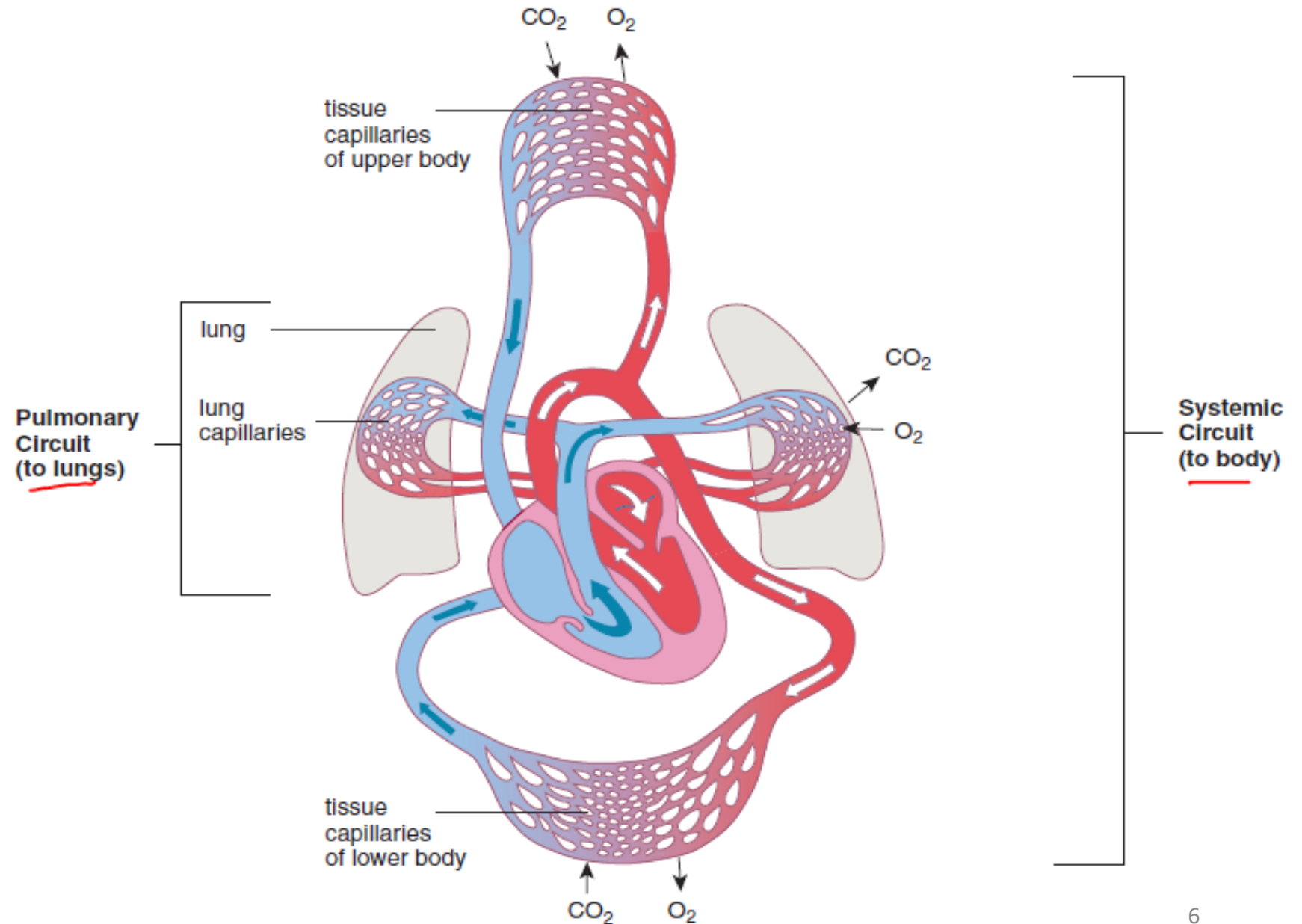
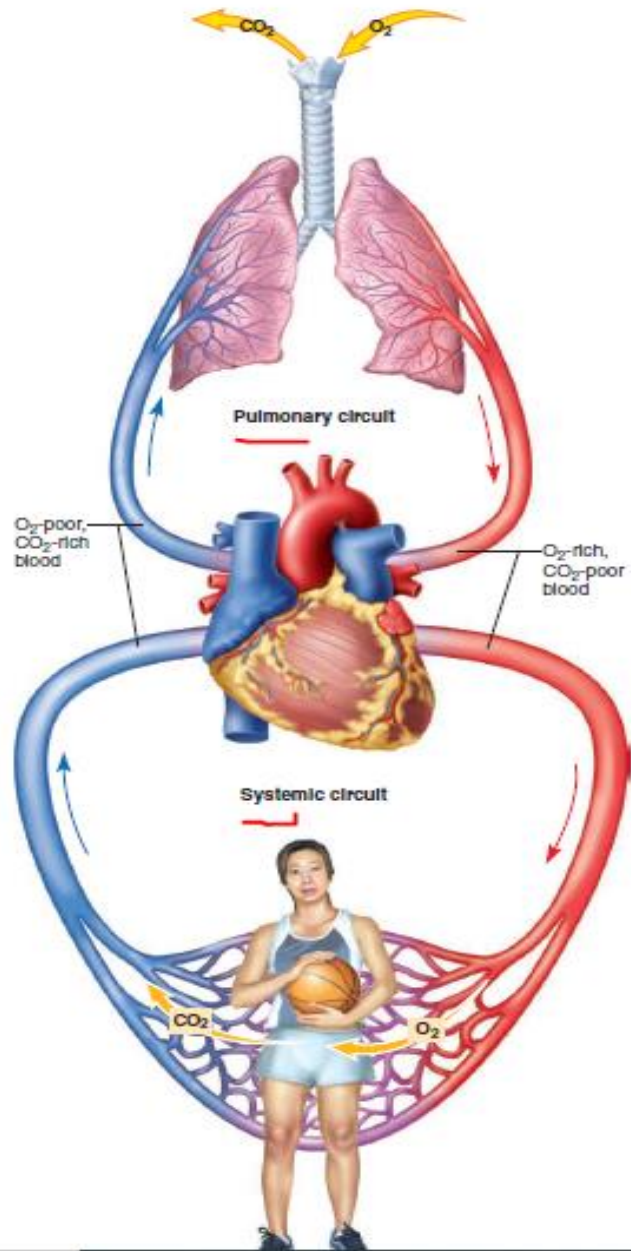
- The **cardiovascular system** (*cardio*- heart; *vascular* blood or blood vessels) consists of the **heart and the blood vessels**.
- The **heart is a muscular pump** that keeps blood flowing through the vessels.
- The **vessels deliver** the blood to all the body's organs and then return it to the heart.



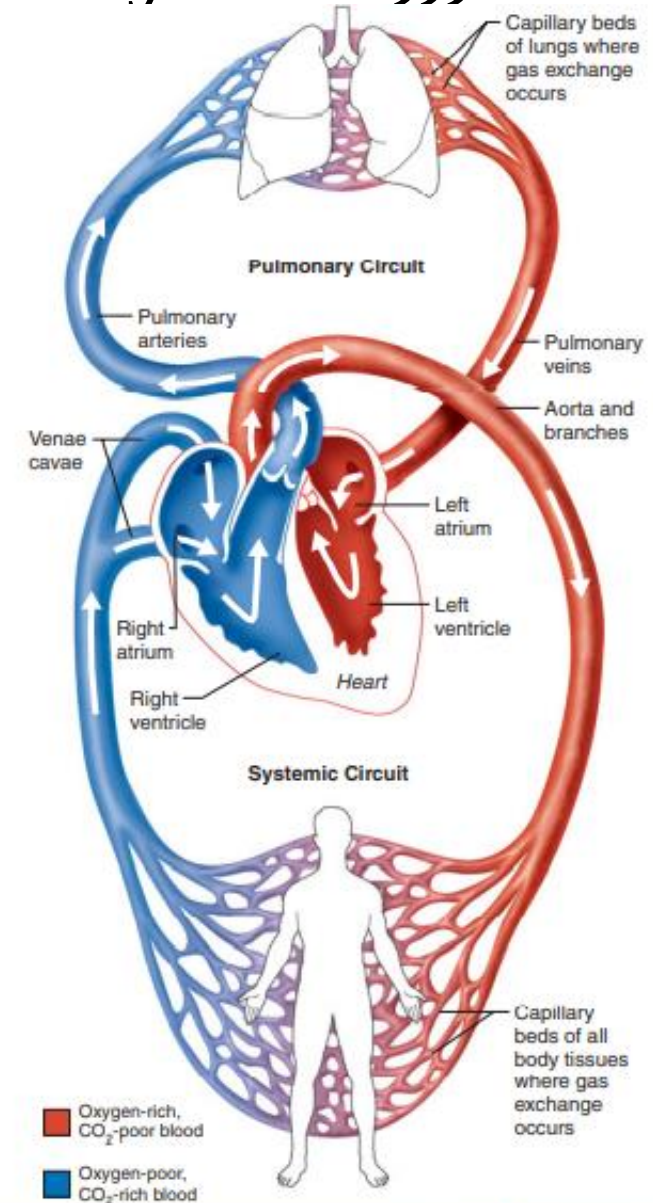
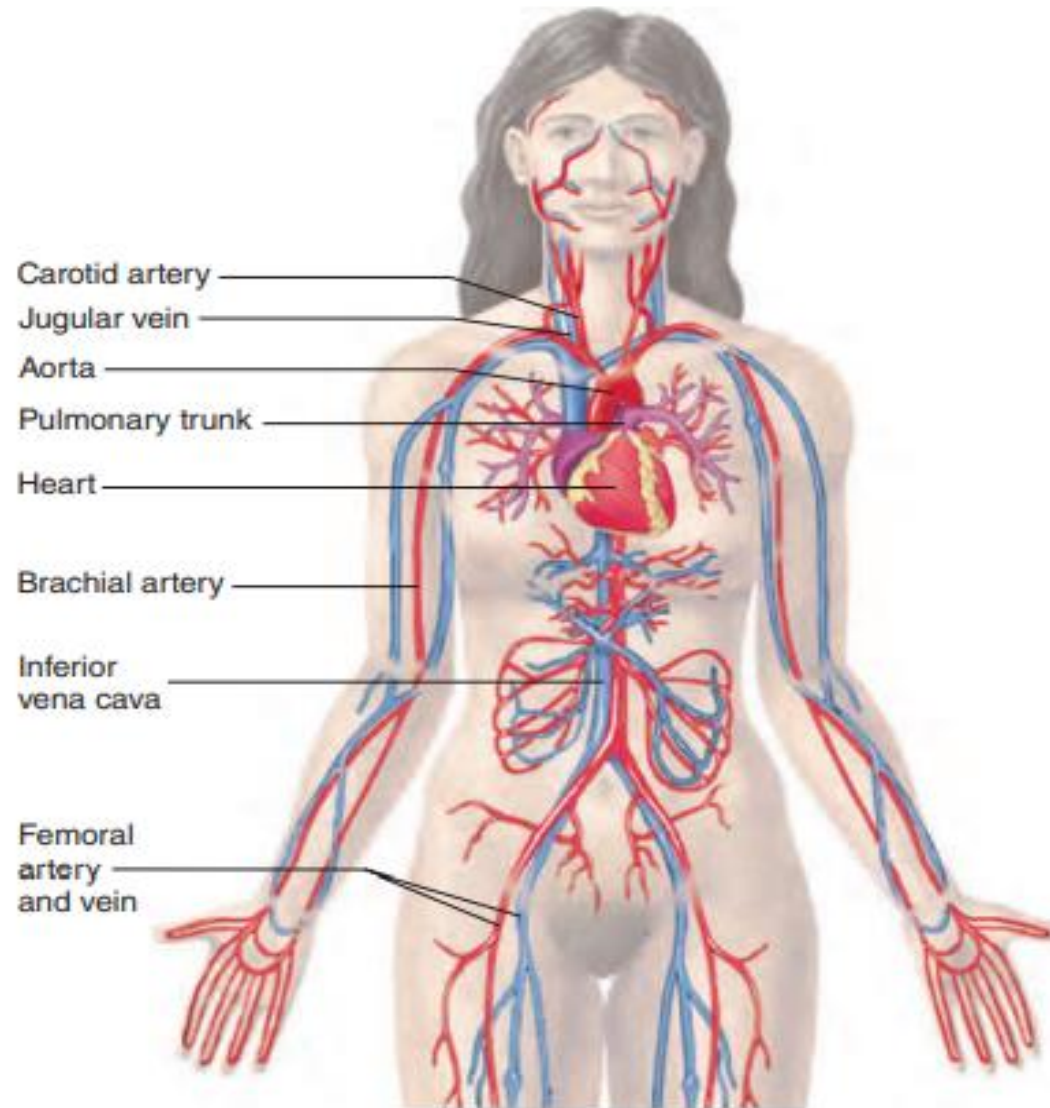
Introduction to Cardiovascular System,,

- **Cardiology** is a medical specialty that deals with the study of **structure, function, prevention, and treatment** of disorders of the heart.
- **The heart is a vital organ responsible for pumping** blood throughout the body, delivering oxygen and nutrients to tissues and organs while removing waste products.
- This module focuses on the anatomy of **the 2 components of cardiovascular system:**
 - ❖ **The heart**, which pumps blood so that it flows to tissue capillaries and lung capillaries.
 - ❖ **The blood** vessels through which the blood flows.

Introduction to Cardiovascular System,,



Introduction to Cardiovascular System,,,



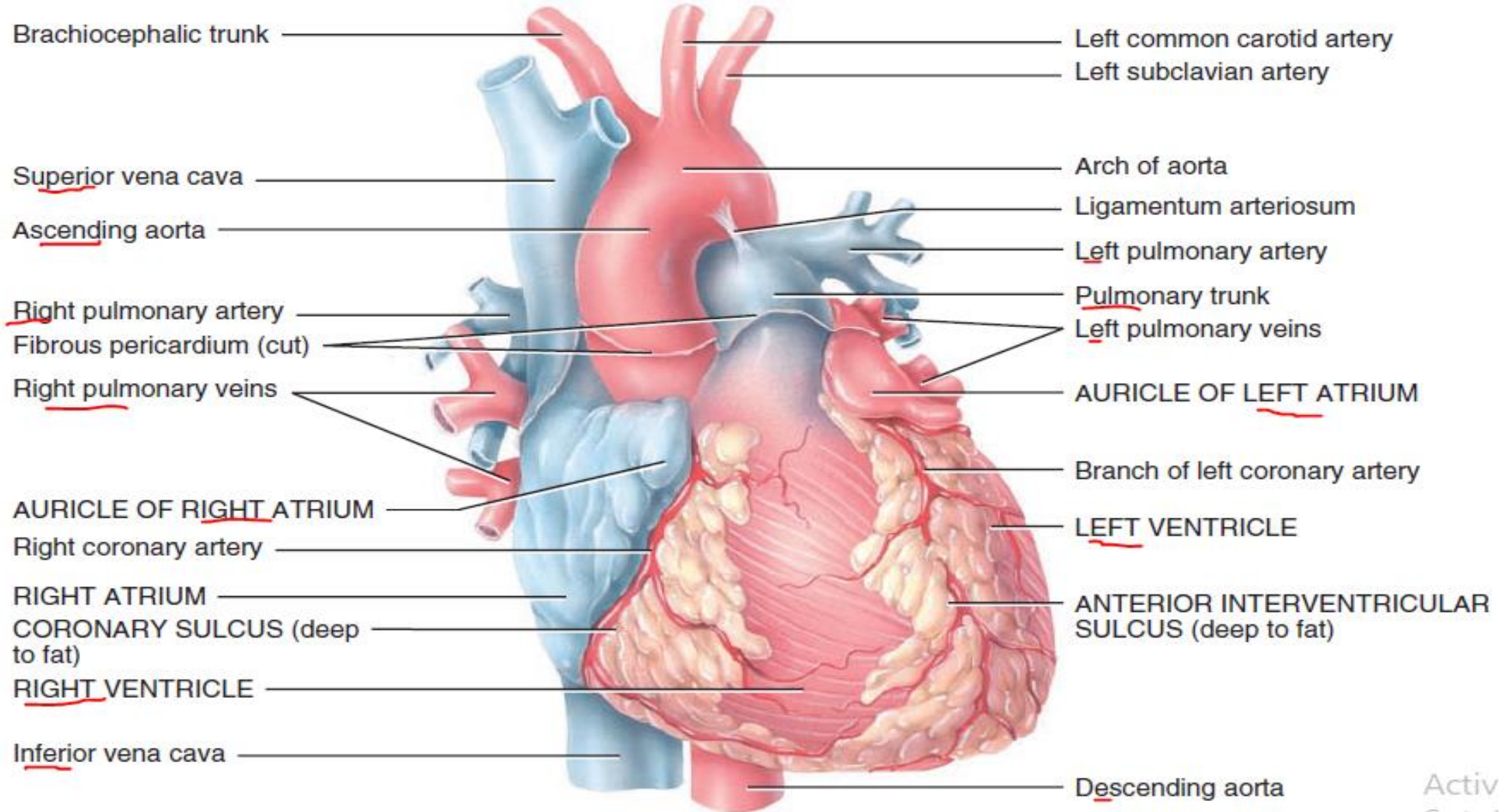
Introduction to Cardiovascular System,,

- The heart is a muscular pump that keeps blood flowing through the vessels.
- The vessels deliver the blood to all the body's organs and then return it to the heart.
- **The cardiovascular system has two major divisions: a pulmonary circuit,** which carries blood to the lungs for gas exchange and returns it to the heart, and **a systemic circuit,** which supplies blood to every organ of the body, including other parts of the lungs and the wall of the heart itself.

Introduction to Cardiovascular System,,

- ❖ **The right half of the heart supplies the pulmonary circuit.** It receives blood that has circulated through the body, unloaded its oxygen and nutrients, and picked up a load of carbon dioxide and other wastes.
- It pumps **this oxygen-poor blood** into a large artery, the *pulmonary trunk*, which immediately divides into right and left *pulmonary arteries*.
- These transport blood to the air sacs (*alveoli*) of the lungs, where carbon dioxide is **unloaded and oxygen is picked up**.
- The oxygen-rich blood then flows by way of **the *pulmonary veins*** to the left side of the heart.

Introduction to Cardiovascular System,,



(a) Anterior external view showing surface features

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Introduction to Cardiovascular System,,

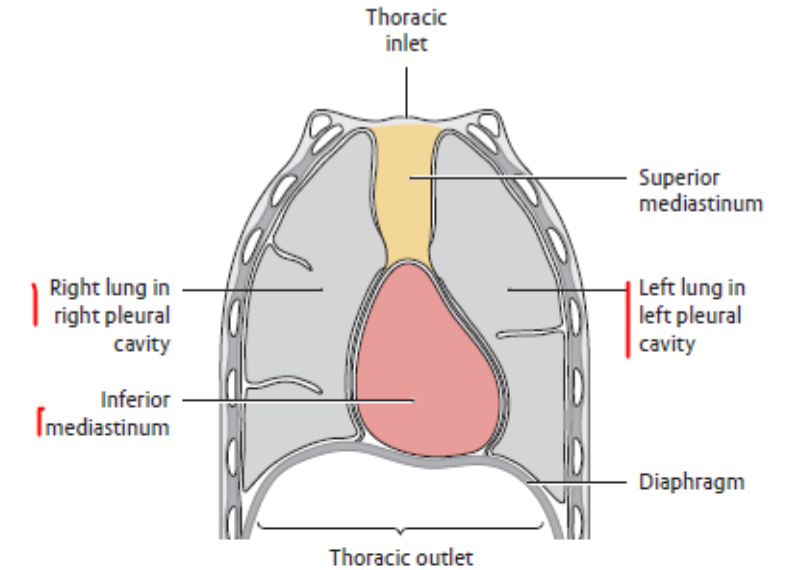
- ❖ **The left half of the heart supplies the systemic circuit.** Blood leaves it by way of another large artery, **the *aorta*.**
- ❑ **The aortic arch gives off** arteries that supply the head, neck, and upper limbs.
- The aorta then travels through the thoracic and abdominal cavities and issues smaller arteries to the other organs before branching into the lower limbs.
- After circulating through the body, the now **deoxygenated systemic** blood returns to the right side of the heart mainly by way of 2 large veins: **the *superior vena cava* and *inferior vena cava*.**
- **The major arteries & veins entering & leaving the heart are called the *great vessels*.**

Thoracic cavity

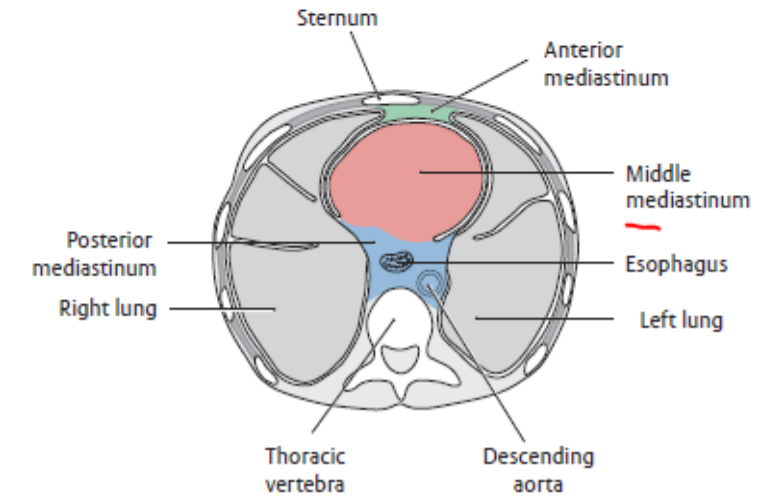
❑ *The thoracic cavity is divided into 3 compartments:*

❖ Right and left **pulmonary cavities**, bilateral compartments that contain the lungs and *pleurae* (lining membranes) and occupy the majority of the thoracic cavity.

❖ **A central mediastinum:** contains all other thoracic structures ,**the heart, thoracic parts of the great vessels**, thoracic part of the trachea, esophagus, thymus, and lymph nodes.

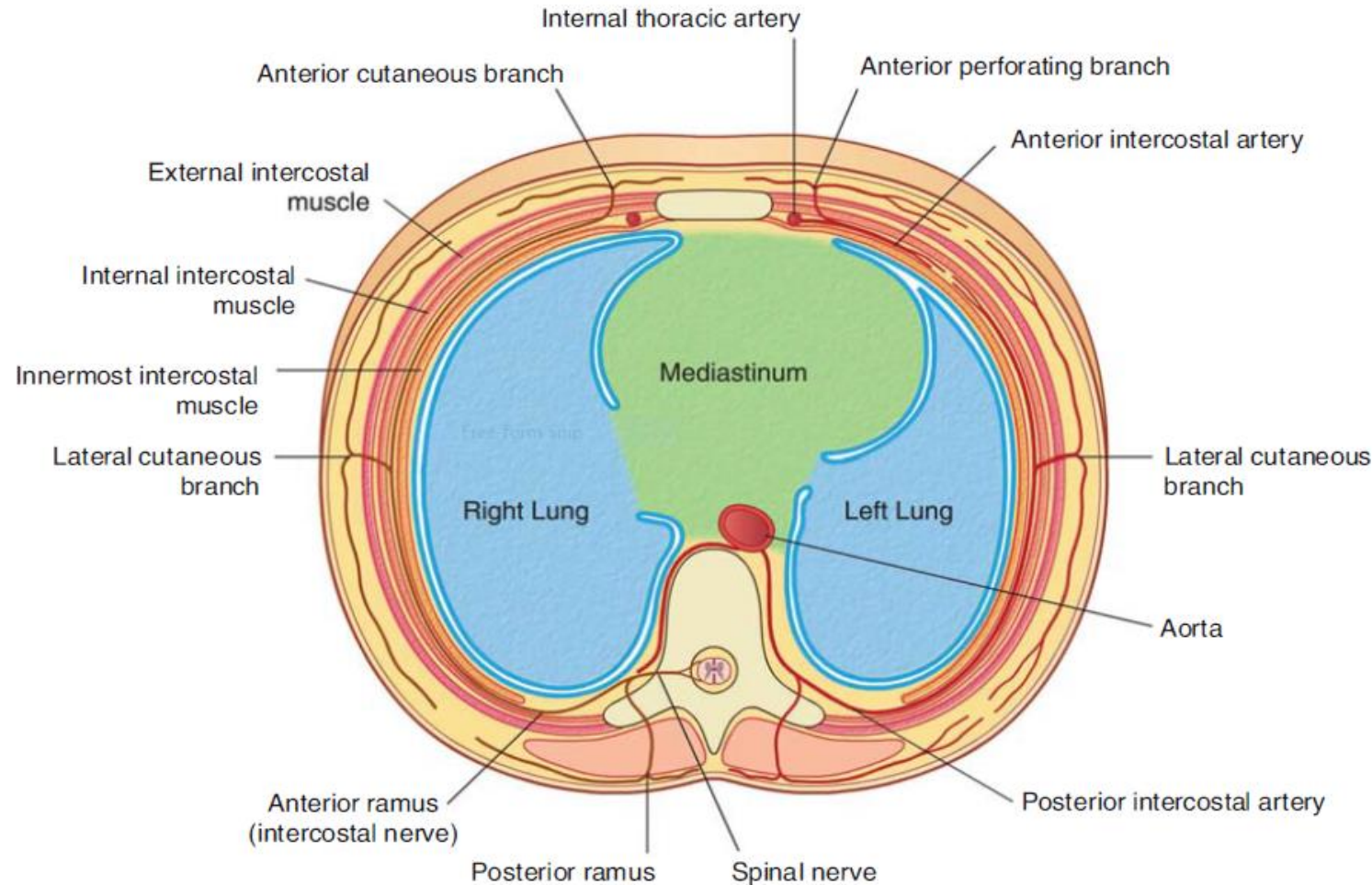


A Anterior view (coronal section).

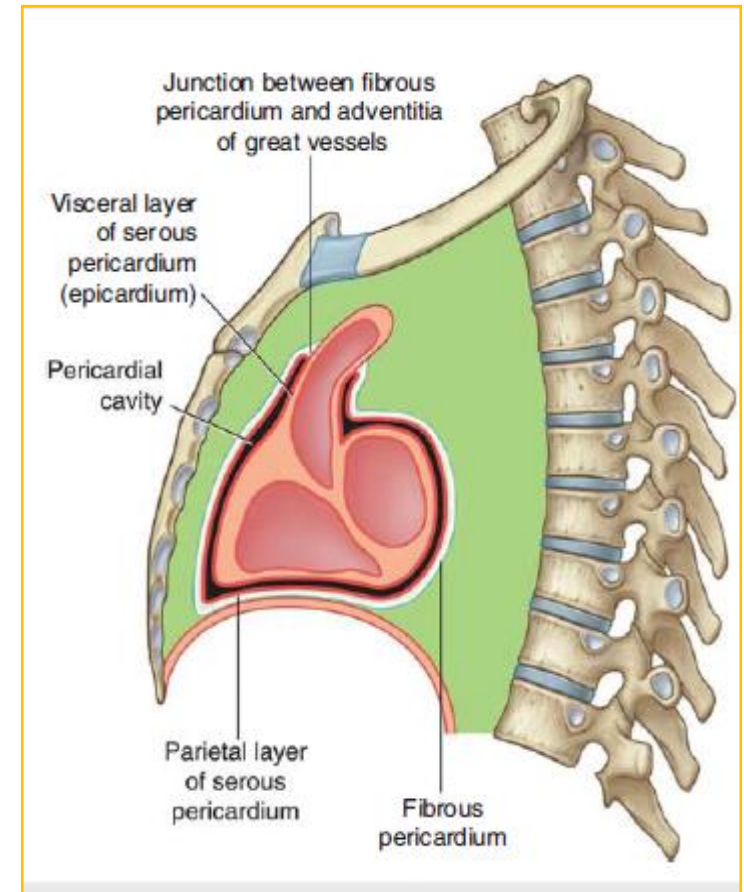


Thoracic cavity,,

Transverse section of the thoracic cavity



Sagittal section of the thoracic cavity

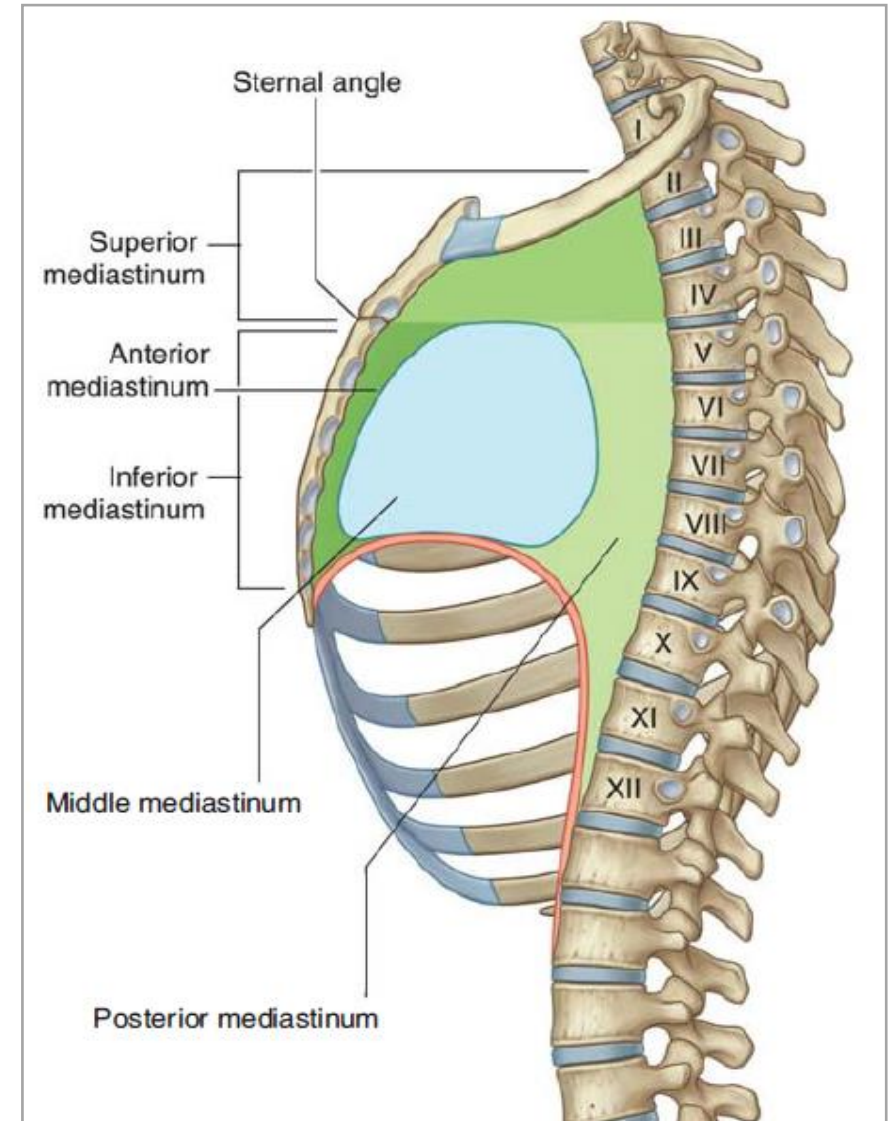


The mediastinum

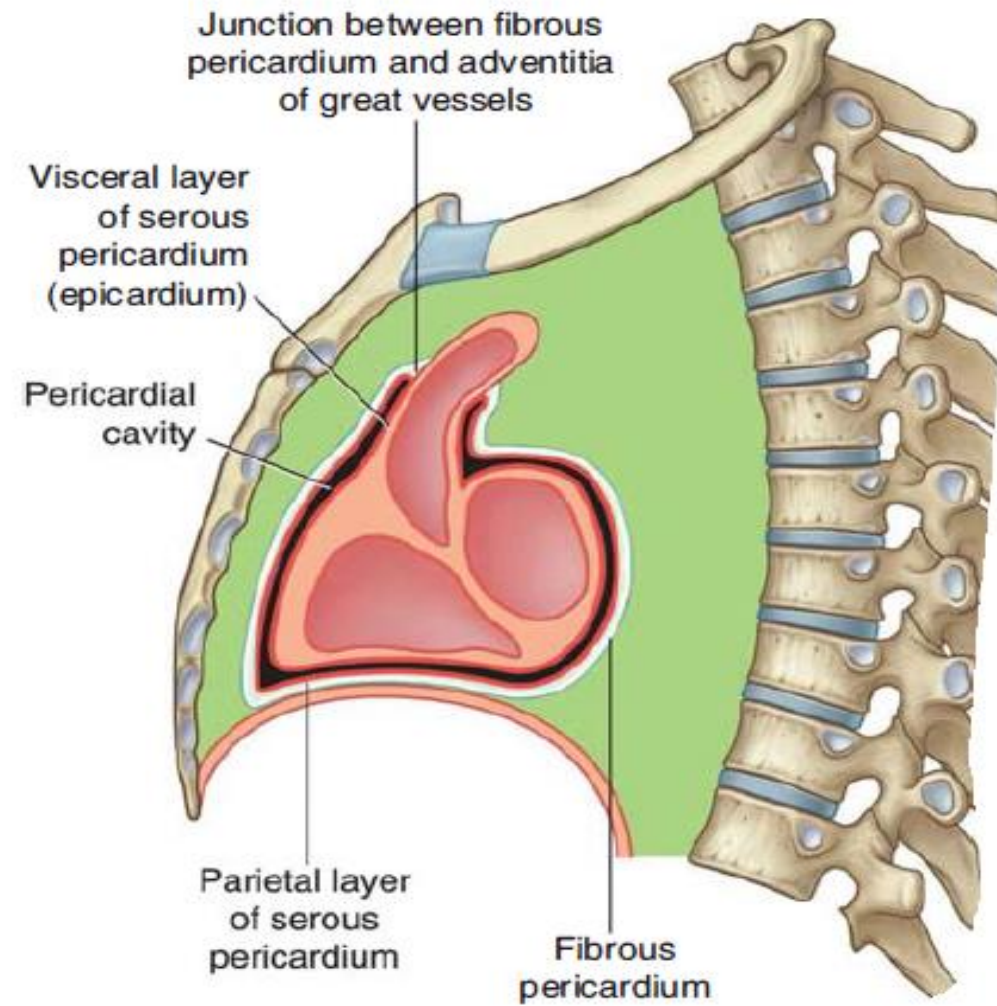
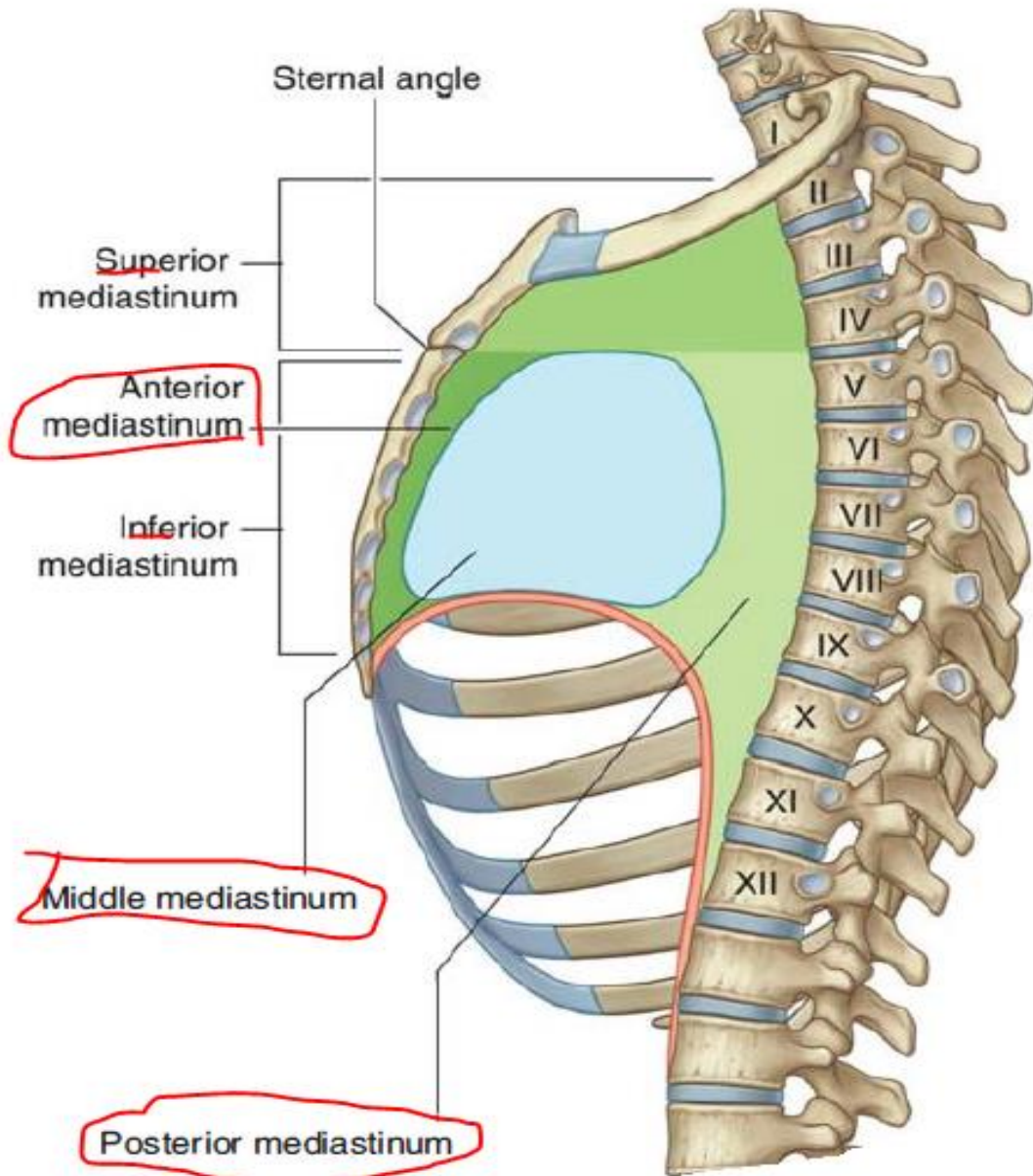
- The **mediastinum** (L. middle septum), is the central compartment of the thoracic cavity.
- Occupied by the mass of tissue.
- It extends from the superior thoracic aperture to the diaphragm & from the sternum to the bodies of the thoracic vertebrae.
- It serves as a passageway for structures from the neck & head to the abdomen.
- Is covered on each side by mediastinal pleura.
- It contains all the thoracic viscera except the lungs.

The mediastinum,,,

- A horizontal plane passing through the sternal angle & to the intervertebral disc b/n vertebrae TIV and TV divides the mediastinum into **superior** and **inferior** parts.
- **inferior mediastinum** further partitioned into the **anterior**, **middle**, and **posterior** mediastinum by the pericardial sac.

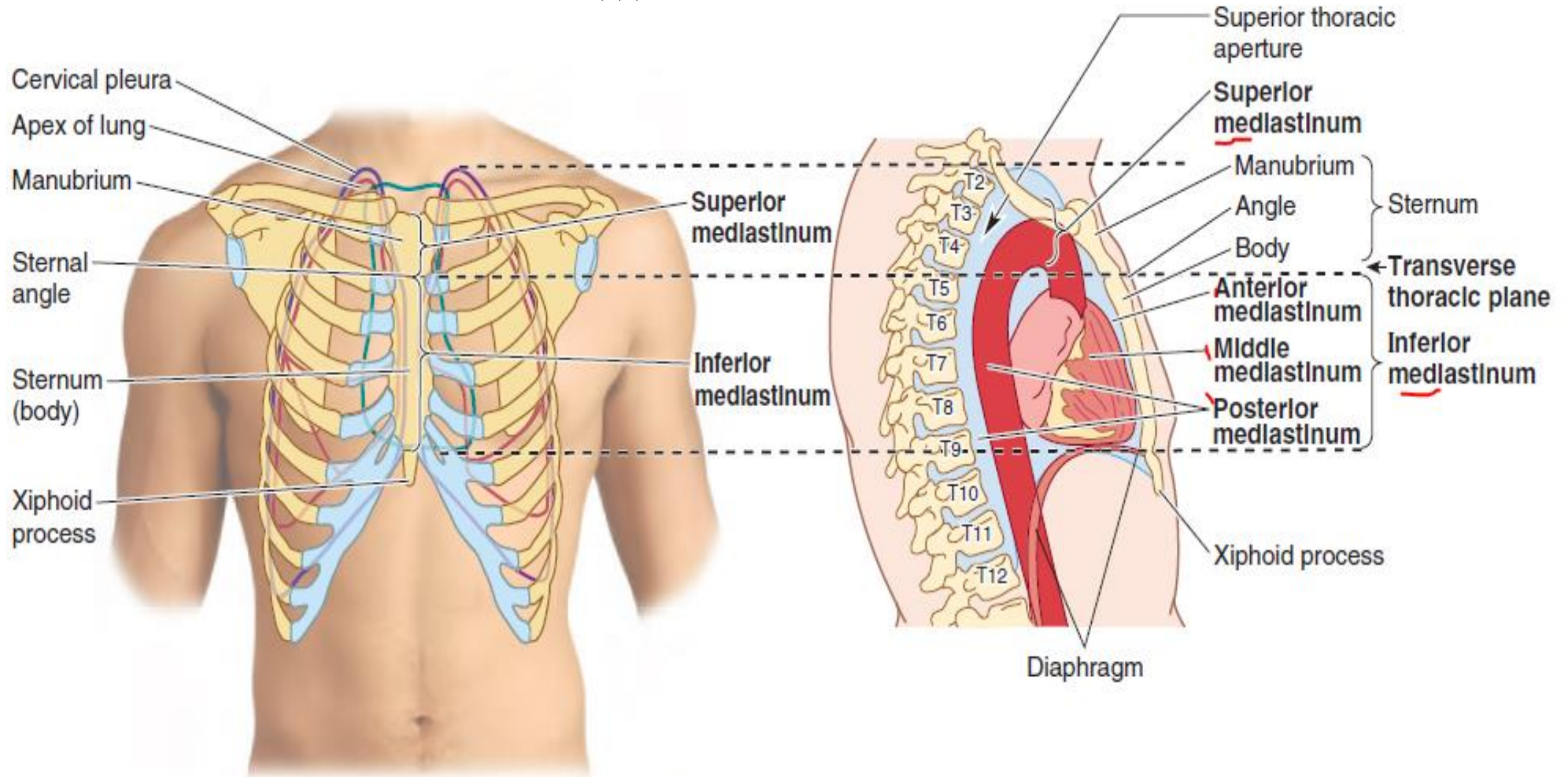


The mediastinum,,



Sagittal section of the pericardium.

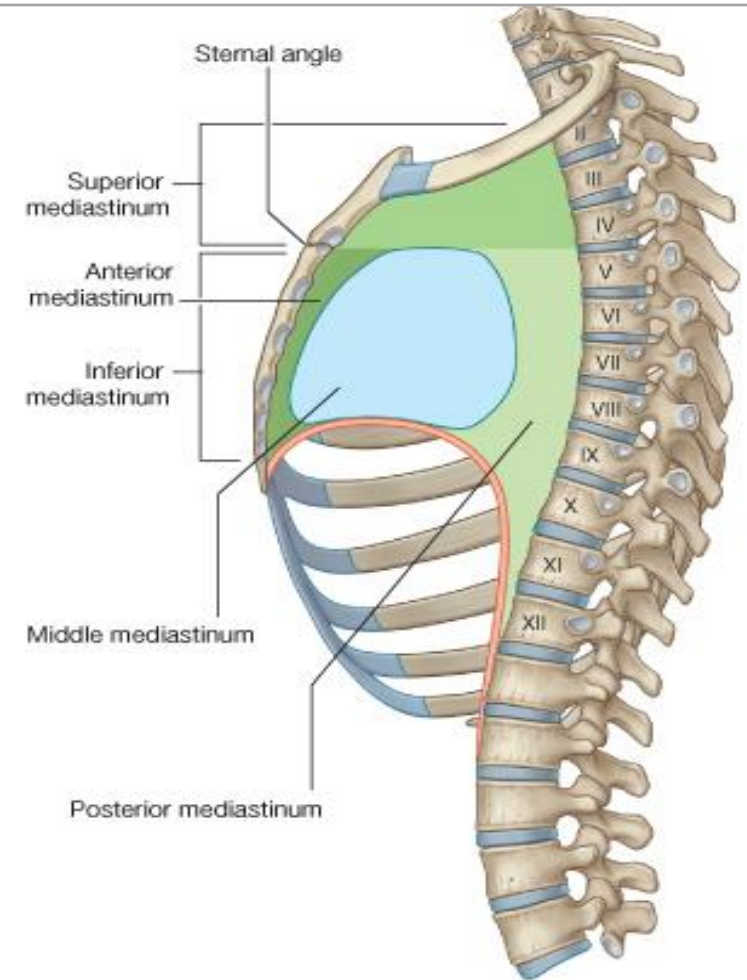
The mediastinum,,



The mediastinum,,

Boundaries of superior mediastinum:

- **upper boundary** - jugular notch, 1st rib, T1 vertebra.
- **lower boundary** --plane of sternal angle(sternal angle to lower border of 4th thoracic vertebra)
- **lateral boundaries** -- mediastinal pleurae
- **anterior boundary** -- manubrium of sternum
- **posterior boundary** -- bodies of thoracic vertebrae: T1-T4.



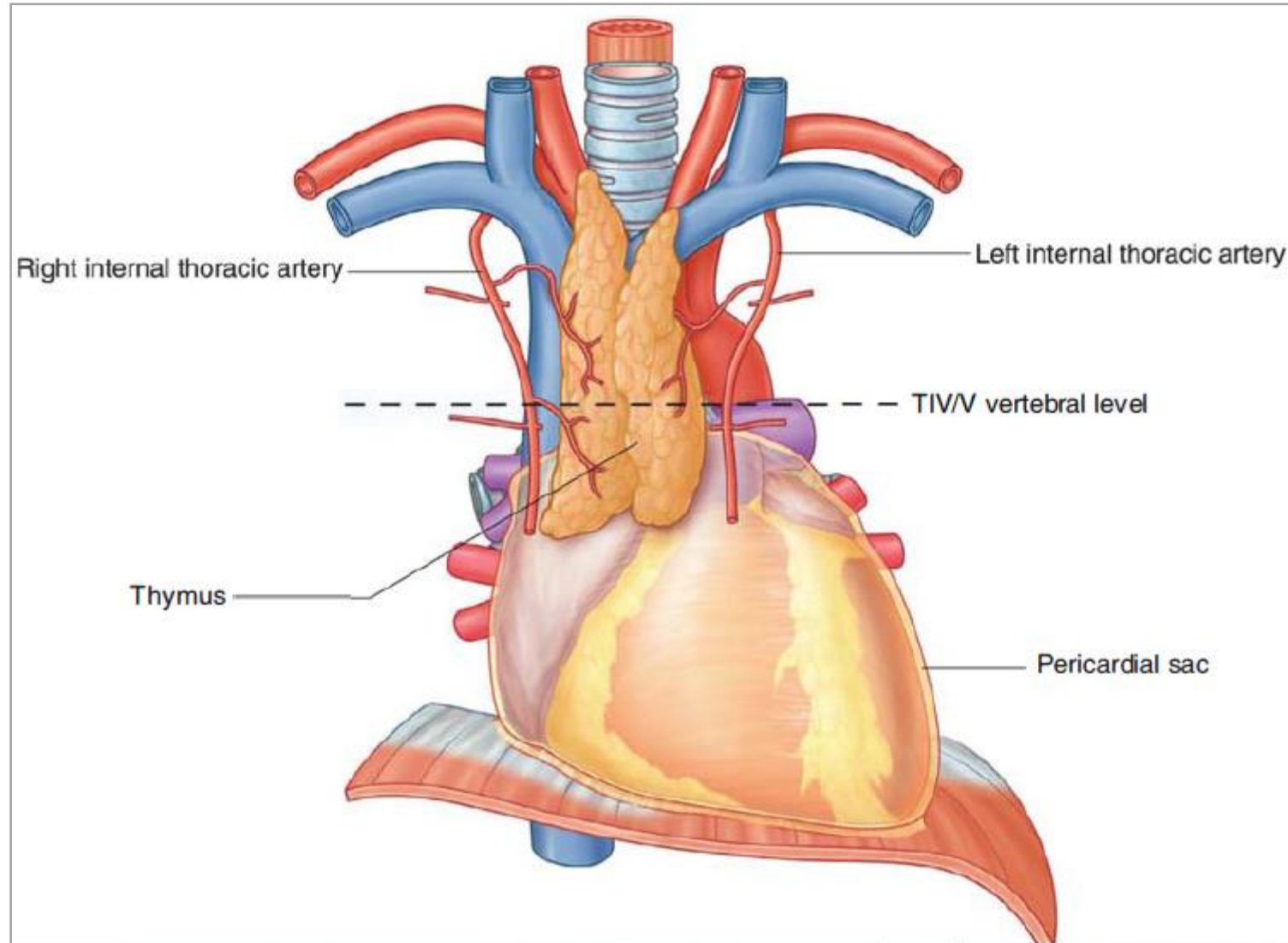
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Contents of the superior mediastinum

❑ From anterior to posterior, the contents of the superior mediastinum are.

- ✓ **glandular plane** (Thymus).
- ✓ **venous plane contains** (brachiocephalic veins and SVC).
- ✓ **arterial-nervous plane contains** arteries (arch of aorta & roots of its major branches—the brachiocephalic trunk, left common carotid artery & left subclavian artery) & related nerves (vagus & phrenic nerves & the cardiac plexus of nerves).
- ✓ **visceral plane contains** (trachea anteriorly and esophagus posteriorly) and related nerves (left recurrent laryngeal nerve).
- ✓ **lymphatic plane.**

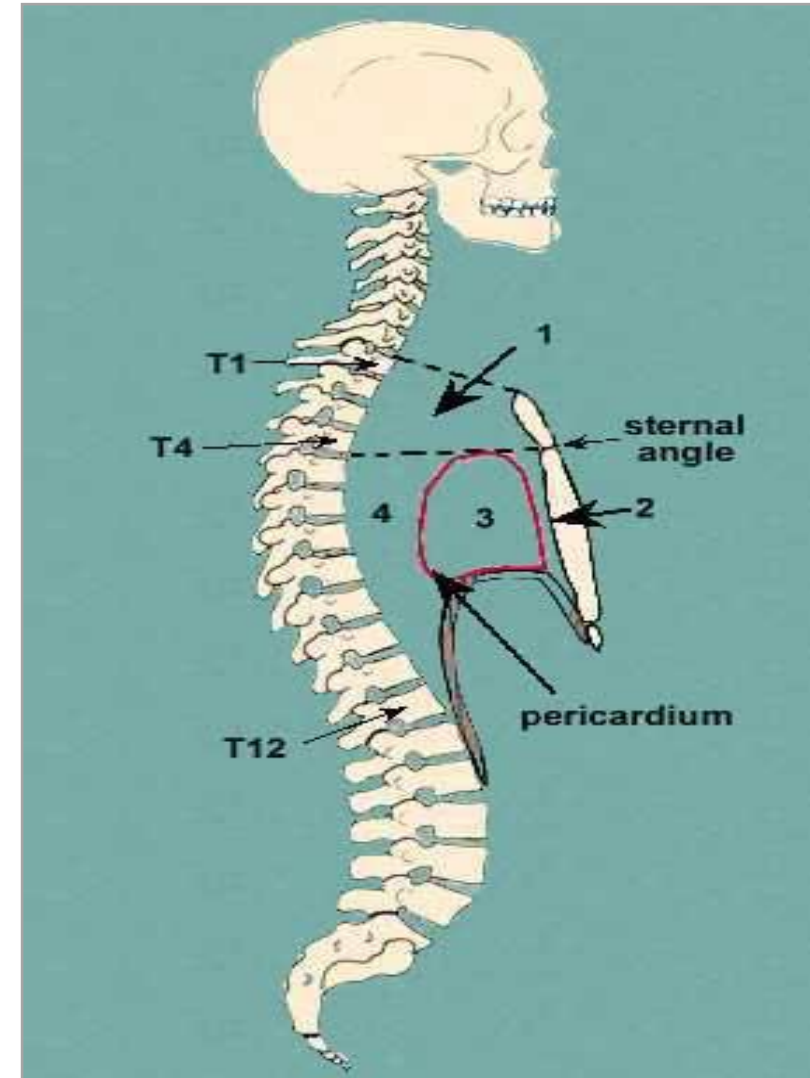
Contents of the superior mediastinum...



The inferior mediastinum

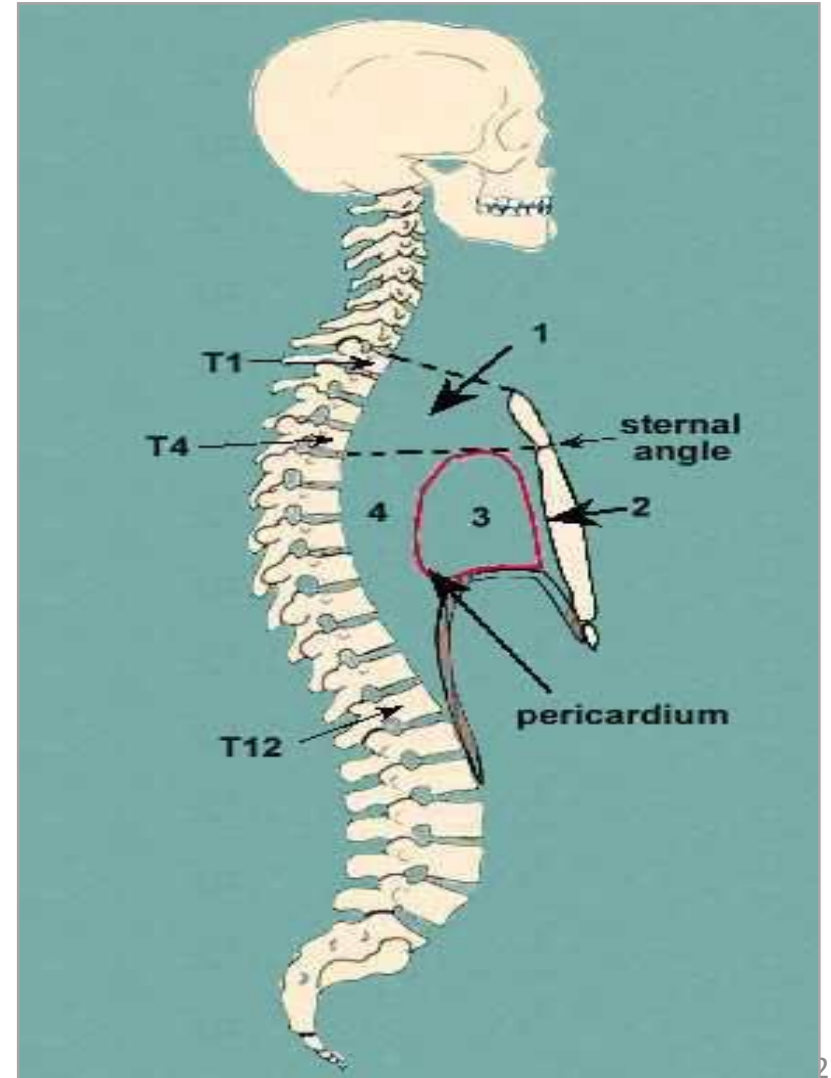
❑ The inferior mediastinum,
is divided by the pericardial
sac into the:

- anterior(2)
- middle (3)
- posterior (4)



Anterior mediastinum

- **Boundaries of *anterior* mediastinum(2):**
- anteriorly -- body and xiphoid of sternum
- posteriorly -- pericardium
- laterally -- mediastinal pleura
- superiorly -- plane of sternal angle
- inferiorly ---diaphragm



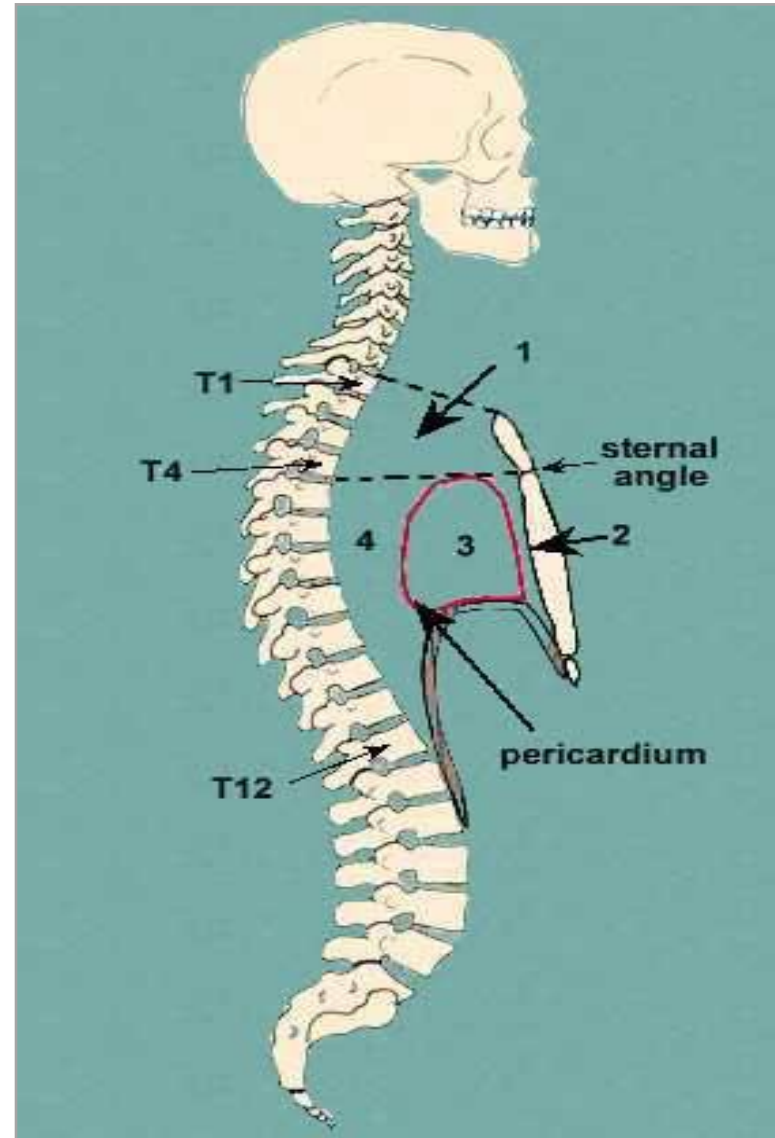
Anterior mediastinum,,

- The anterior mediastinum, is the smallest subdivision of the mediastinum.
- It is continuous with the superior mediastinum at the sternal angle and is limited inferiorly by the diaphragm.
- **It consists of** loose connective tissue (**sternopericardial ligaments**), **fat, lymphatic vessels, a few lymph nodes** & branches of the **internal thoracic vessels**.
- In infants and children, the anterior mediastinum contains the inferior part of the thymus.

Middle mediastinum

❑ ***Middle mediastinum:***

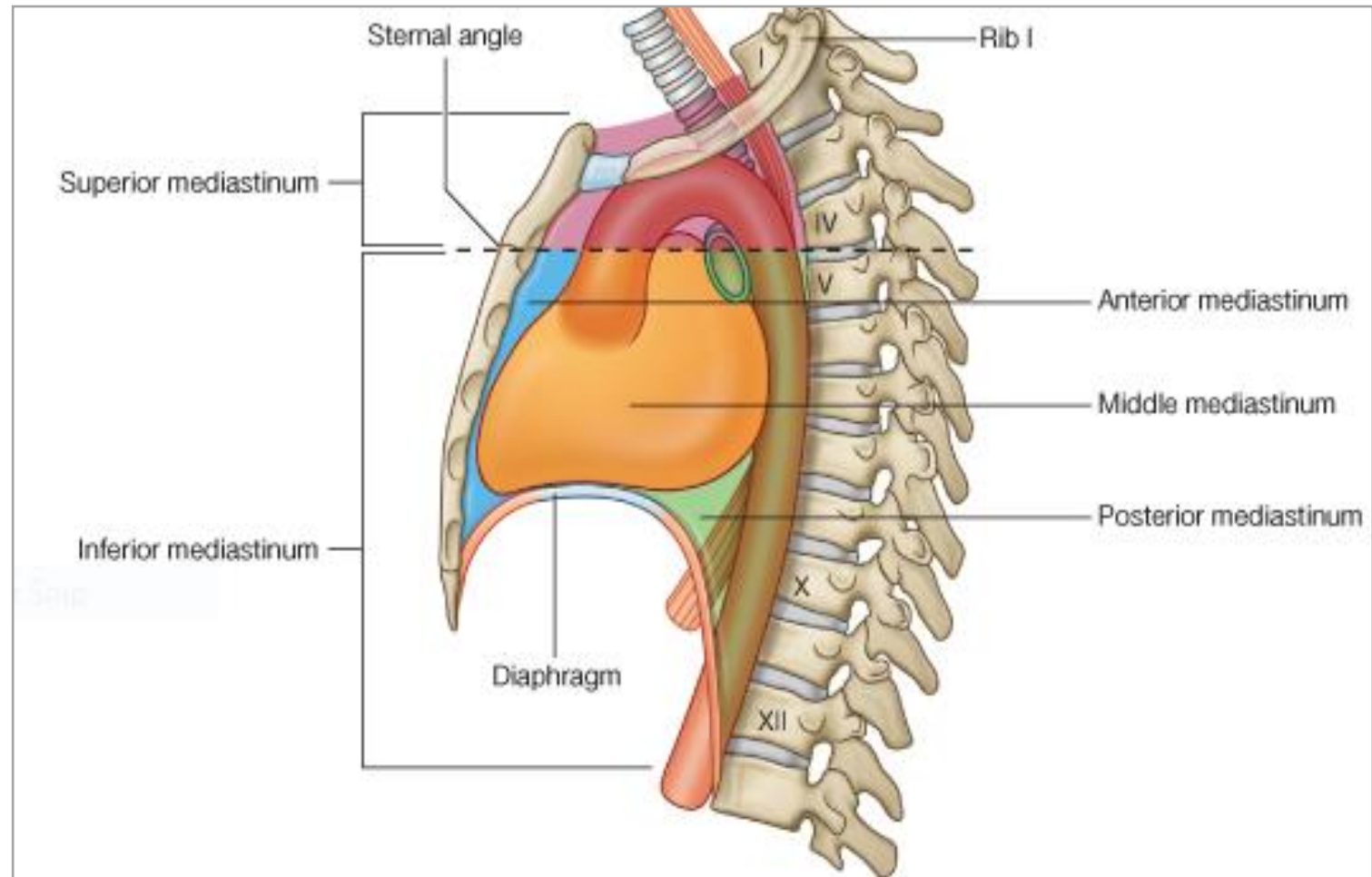
- ✓ anterior -- pericardium
- ✓ posterior -- pericardium
- ✓ lateral -- mediastinal pleura
- ✓ superior -- plane of sternal angle
- ✓ inferior -- diaphragm



Middle mediastinum,,,

❑ The contents of middle mediastinum includes:

- ✓ the pericardium,
- ✓ heart
- ✓ roots of its great vessels(vessels enter or leave the heart(ascending aorta, pulmonary trunk, SVC and IVC)

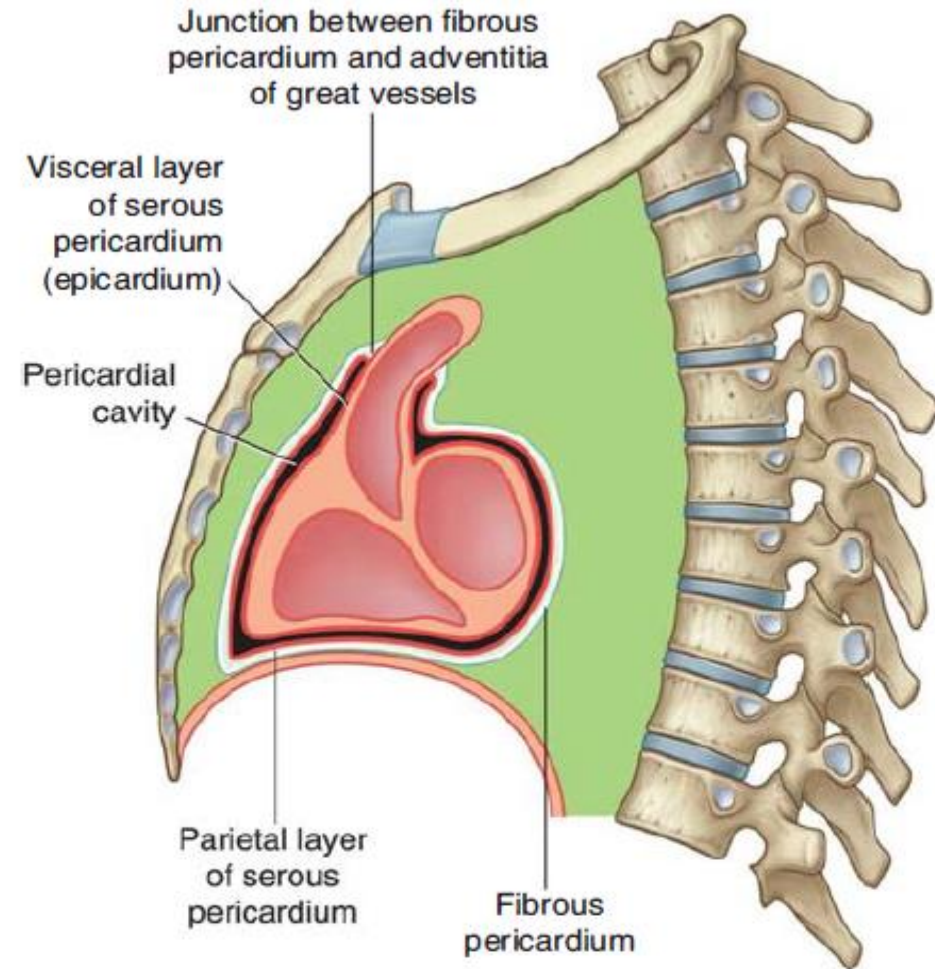
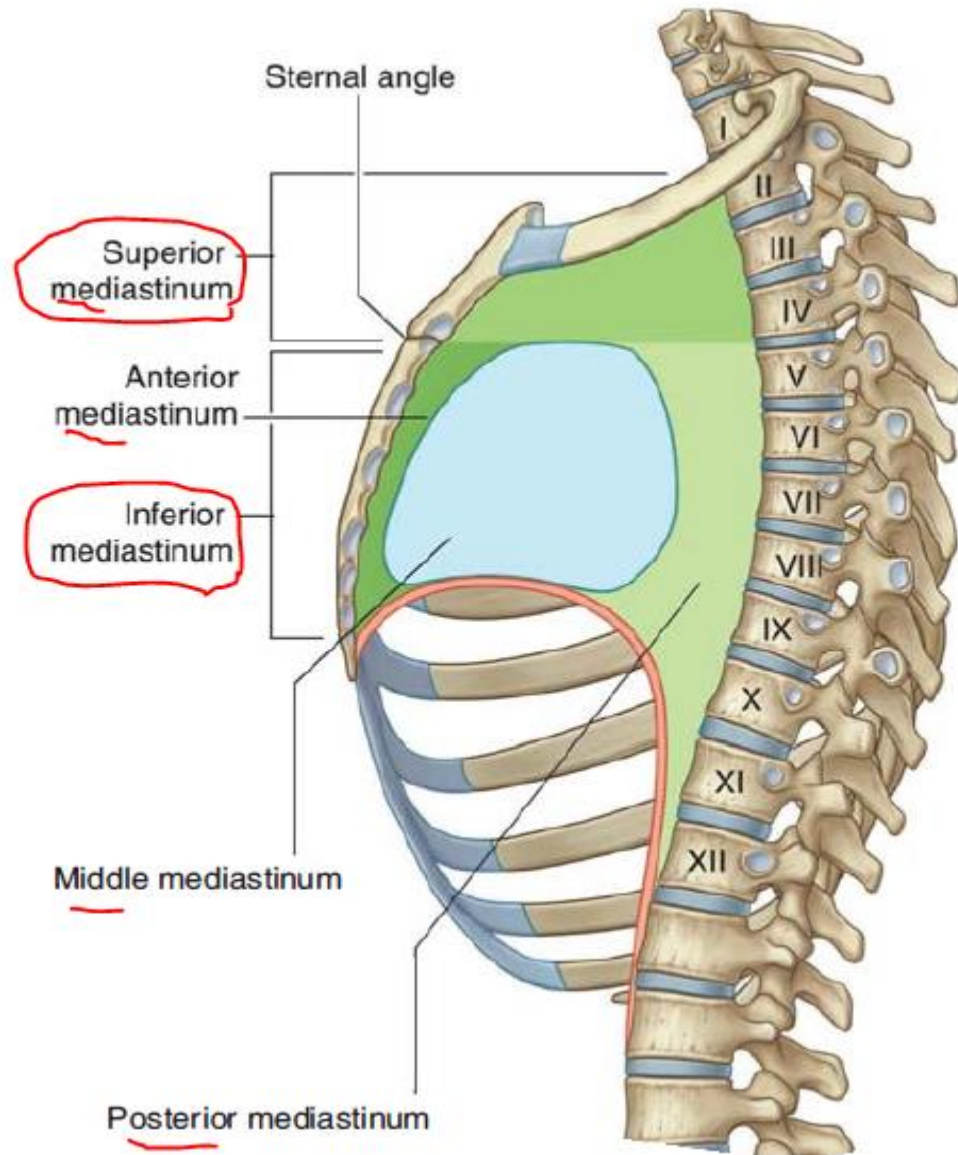


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Figure 3.5 Subdivisions of the mediastinum.

Middle mediastinum,,

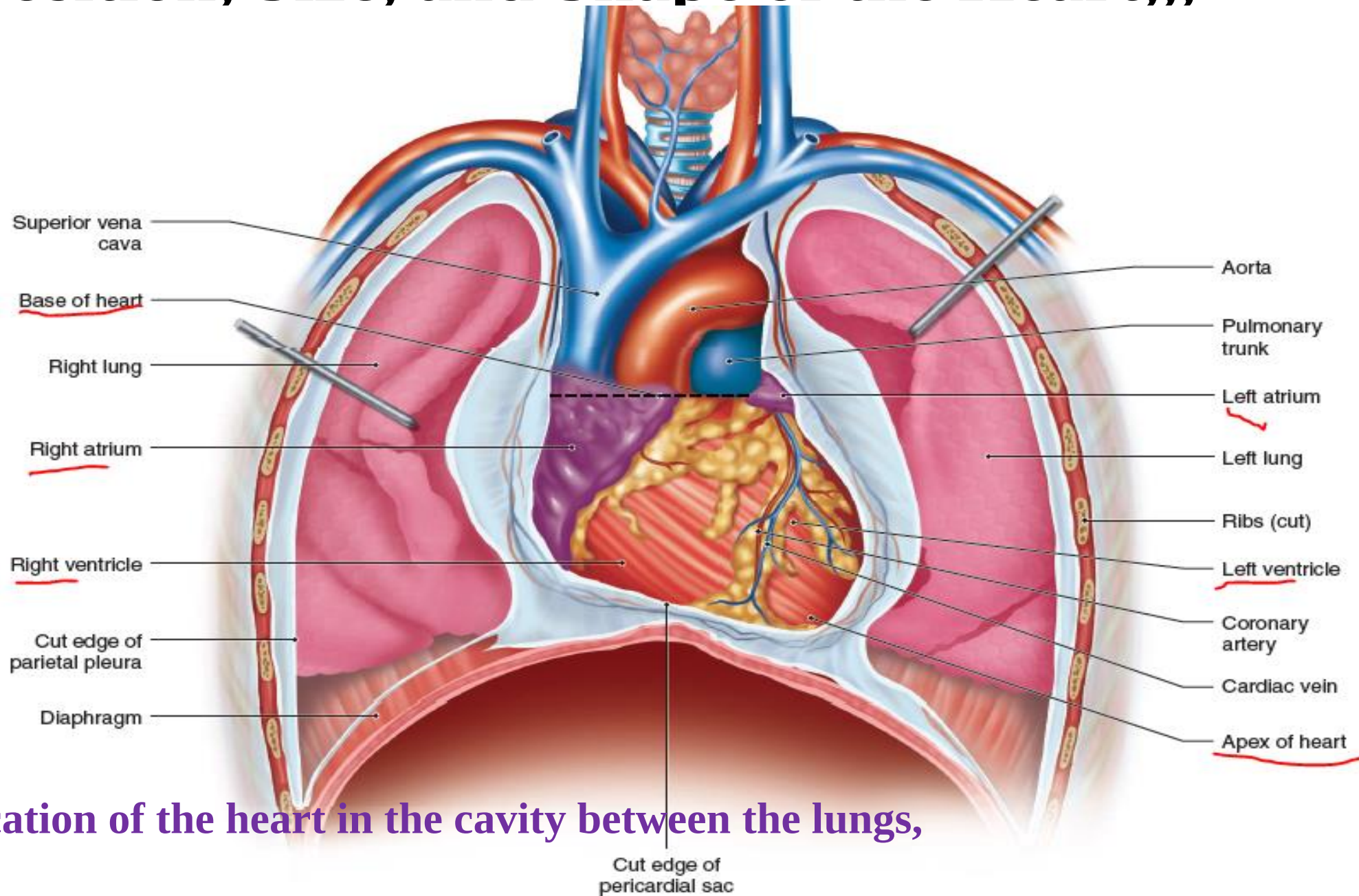


1 Sagittal section of the pericardium.

Position, Size, and Shape of the Heart

- The heart lies obliquely in the mediastinum, with its base **directed posteriorly and slightly superiorly** & its apex **directed anteriorly and slightly inferiorly**.
- The apex is also directed to the left, so that approximately **two-thirds** of the heart's mass lies to the left of the midline of the sternum.
- **The base of the heart** is located deep to the sternum and extends to the level of the 2nd intercostal space and **the apex is deep** to the left 5th intercostal space.
- The adult heart is about 9 cm wide at the base, 13 cm from base to apex, and 6 cm from anterior to posterior at its thickest point—roughly **the size of one's fist**.
- **It weighs about 300 g.**

Position, Size, and Shape of the Heart,,

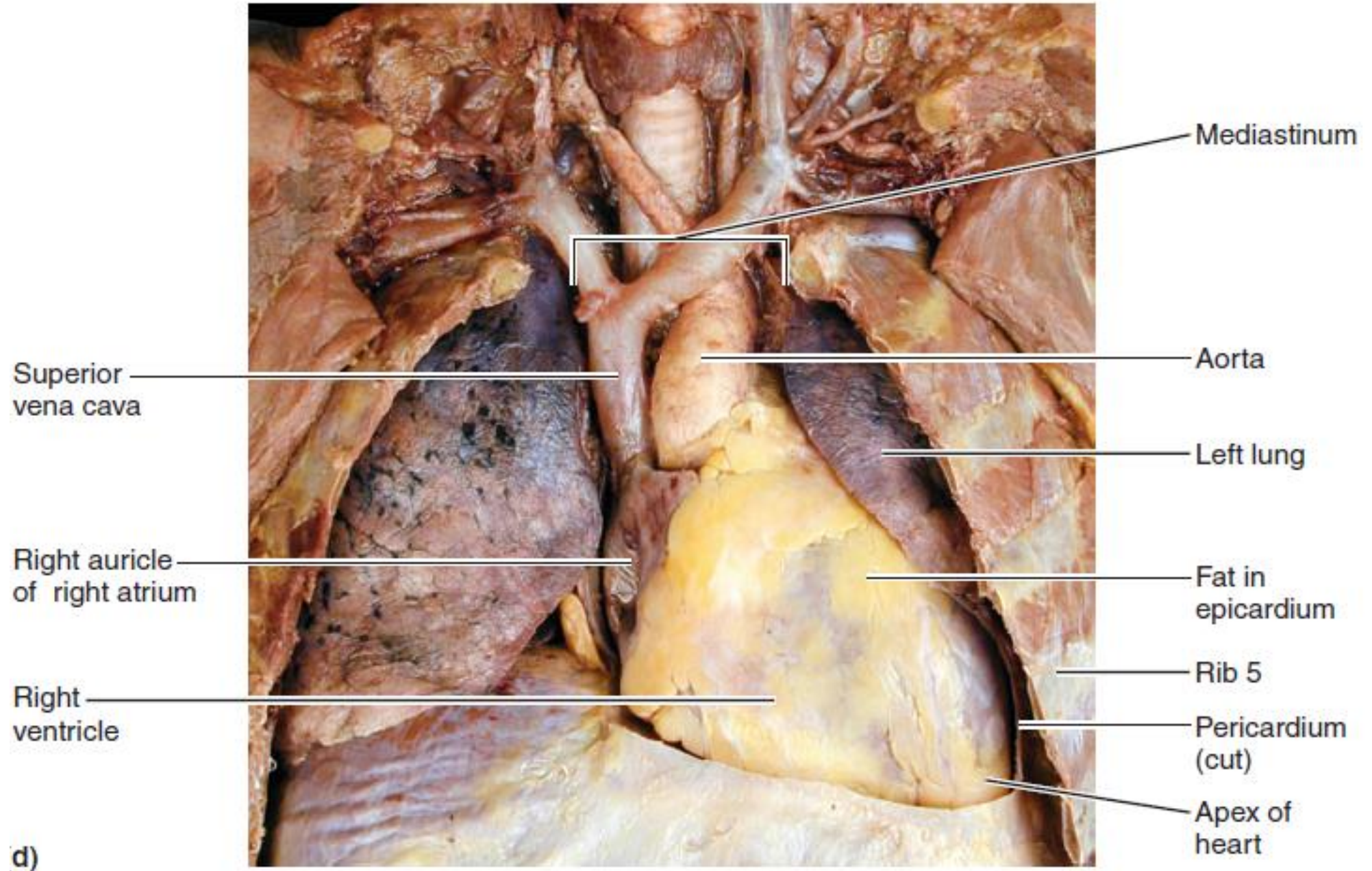


❑ Location of the heart in the cavity between the lungs,

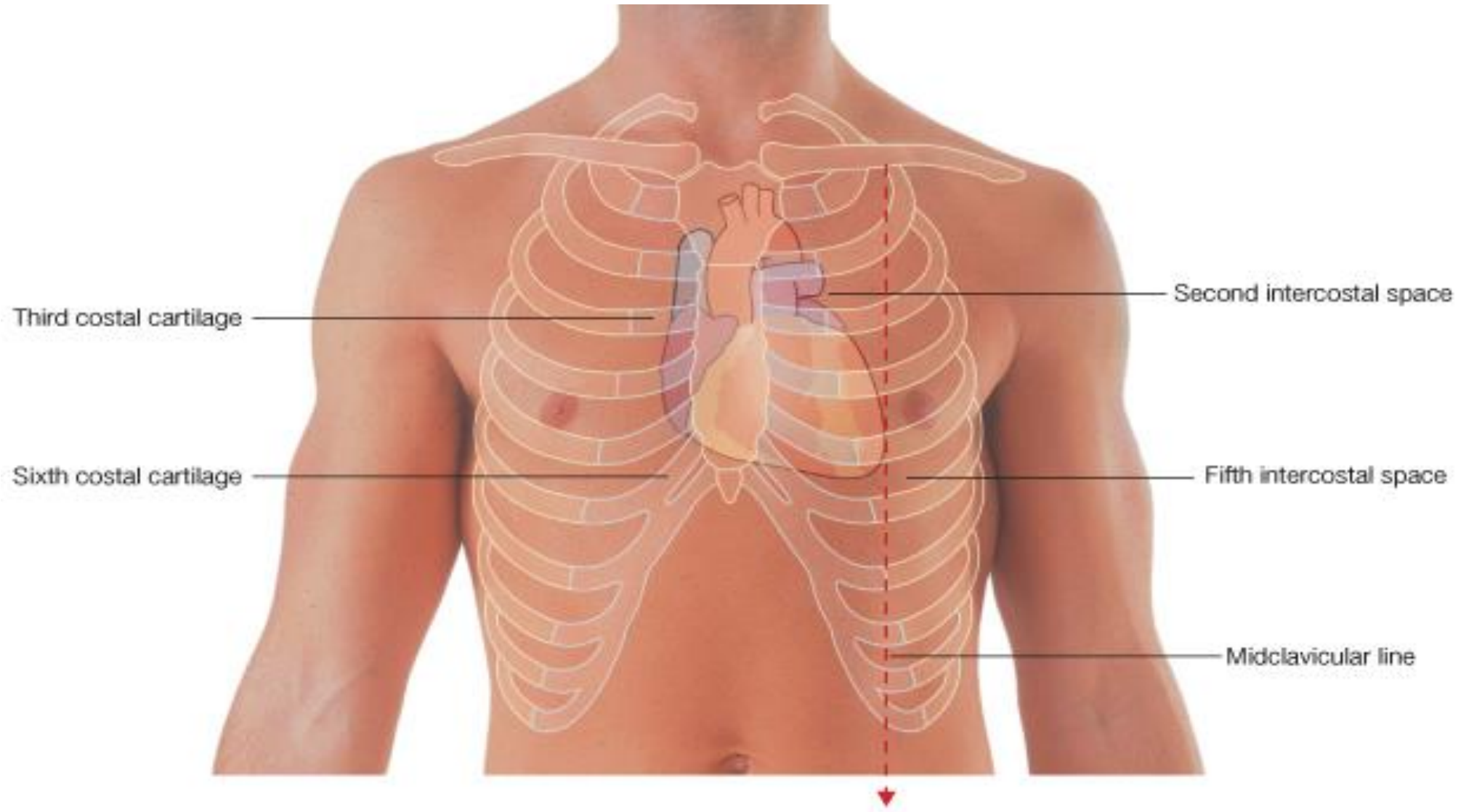
Position, Size, and Shape of the Heart,,

- ❑ The dorsal surface of the heart lies near the bodies of the vertebrae, and its anterior surface sits deep to the sternum and costal cartilages.
- The great veins, **the superior and inferior venae cavae**, and
- The great arteries, the **aorta and pulmonary trunk**, are attached to the superior surface of the heart, **called the base**.
- The right side of the heart is deflected anteriorly, and the left side is deflected posteriorly

Position, Size, and Shape of the Heart,,



Position, Size, and Shape of the Heart,,



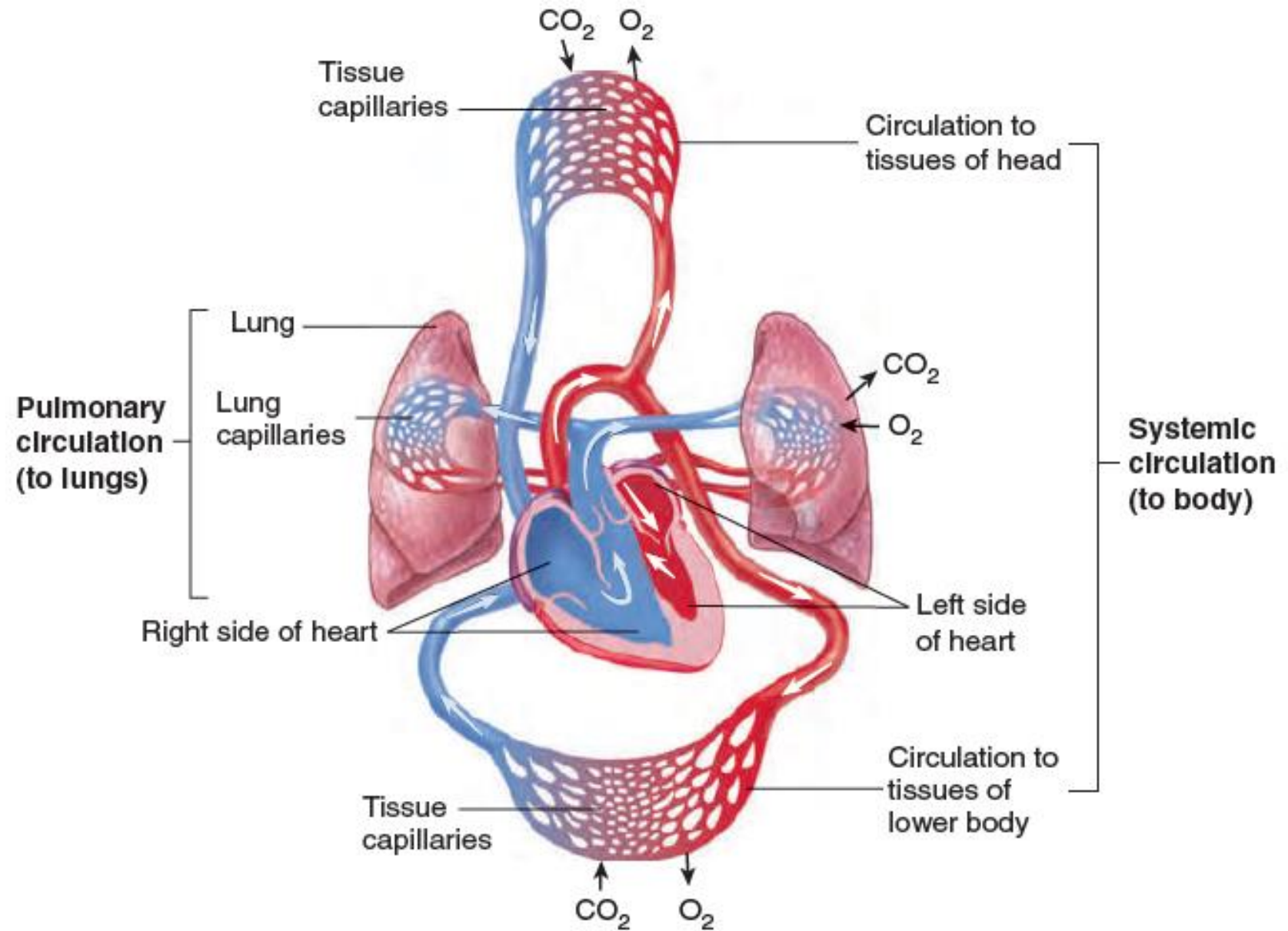
Functions of the heart

- The heart is actually two pumps in one.

❑ **The right side** of the heart pumps blood to the lungs and back to the left side of the heart through vessels of the **pulmonary circulation**.

❑ **The left side** of the heart pumps blood to all other tissues of the body and back to the right side of the heart through vessels of the **systemic circulation**.

Functions of the heart,,



Functions of the heart,,

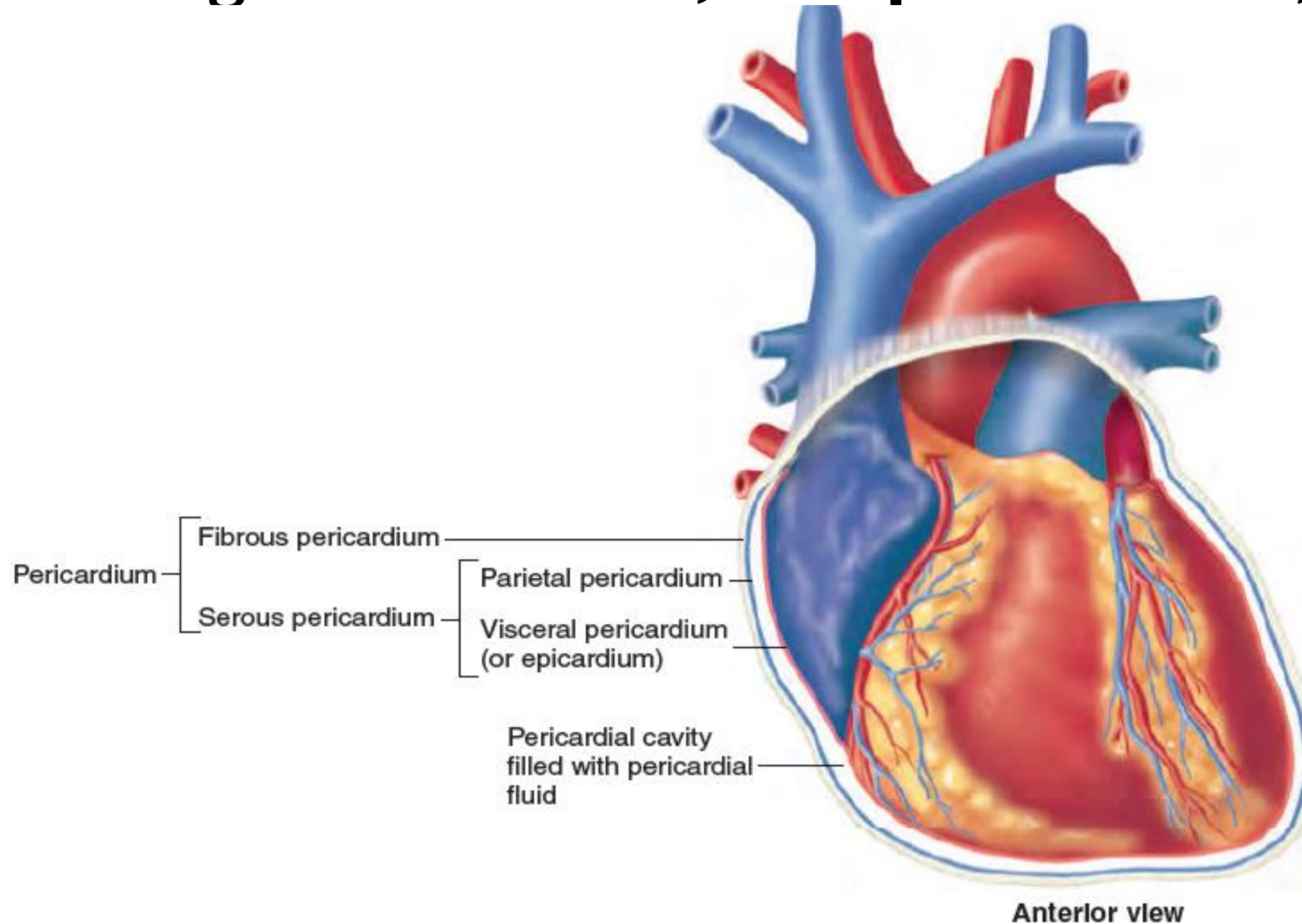
❖ As the heart pumps the blood through the pulmonary and systemic vessels, it performs these functions:

- I. keeps O₂-poor blood separate from O₂-rich blood;
- II. keeps the blood flowing in one direction—blood flows away from and then back to the heart in each circuit;
- III. creates blood pressure, which moves the blood through the circuits;
- IV. regulates the blood supply based on the current needs of the body.

Coverings of the Heart; The pericardium

- The heart and the great vessels are enclosed by the pericardium.
- ❑ **The pericardium consists of two layers.**
- The tough, fibrous connective tissue outer layer is called the **fibrous pericardium**, and
- The inner layer of flat epithelial cells, with a thin layer of connective tissue, is called the **serous pericardium**.
- The **serous pericardium** is composed mainly of mesothelium, a single layer of flattened cells forming an epithelium.

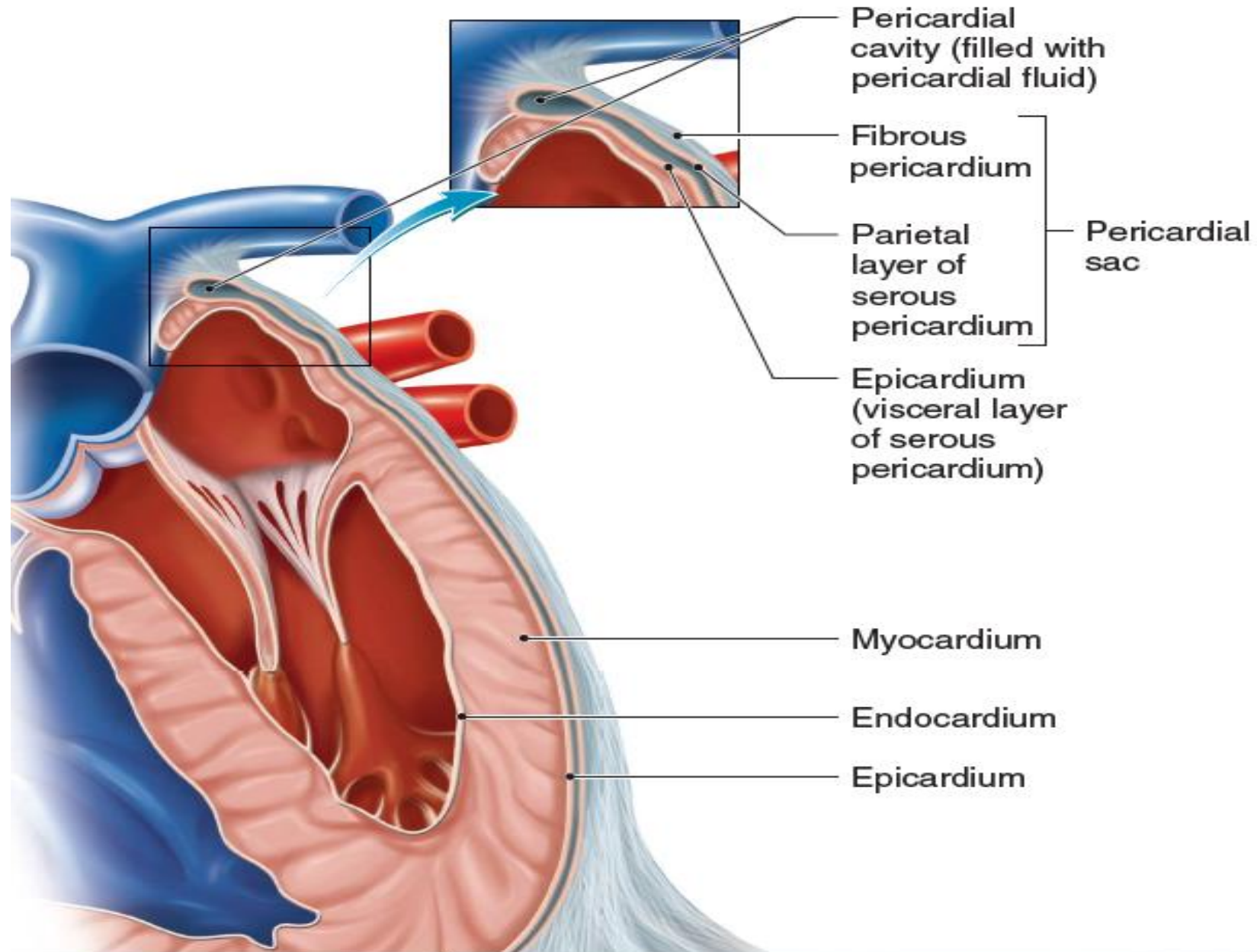
Coverings of the Heart; The pericardium,,



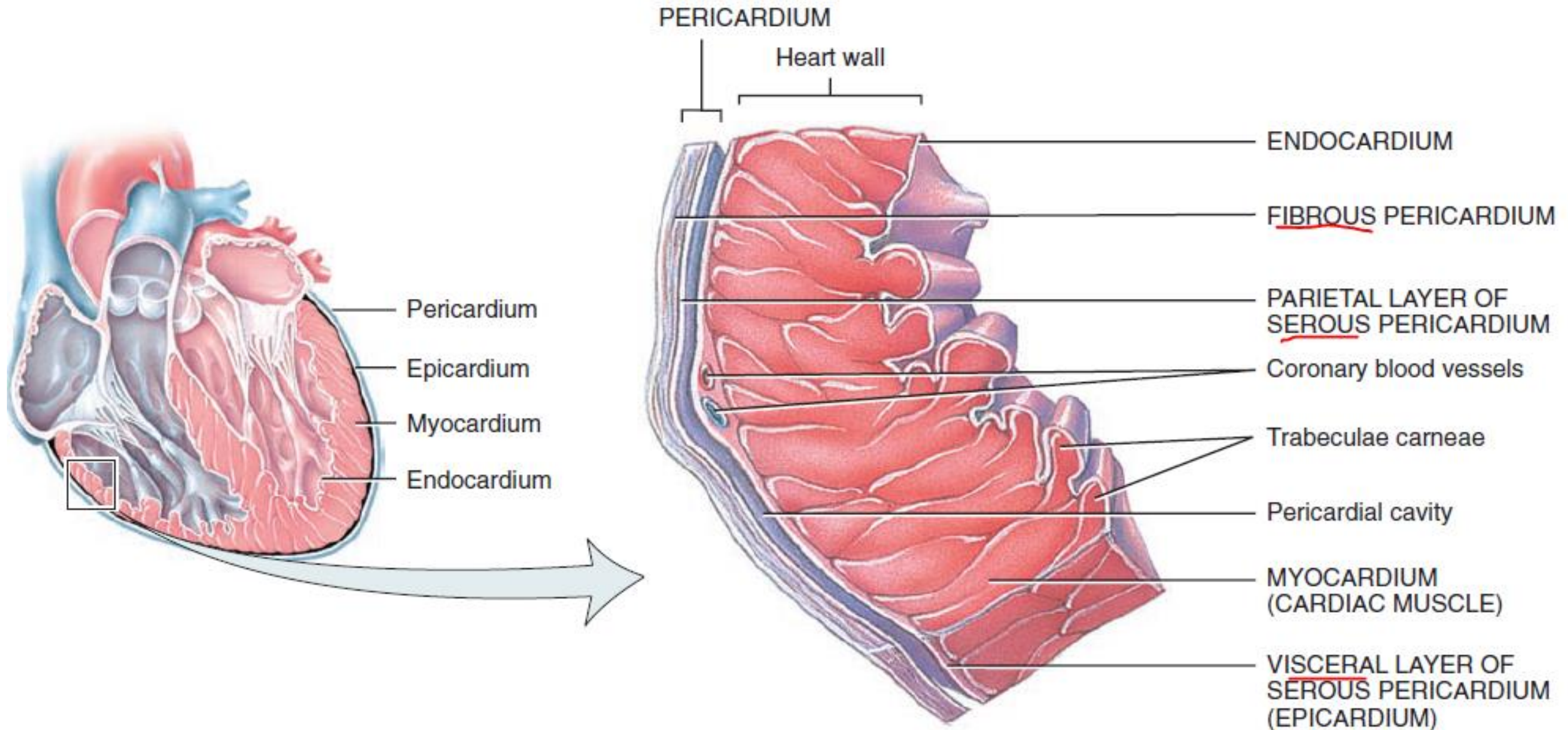
Serous pericardium

- The deeper **serous pericardium** is a thinner, more delicate membrane that forms a double layer around the heart.
- ✓ The outer **parietal layer** of the serous pericardium is fused to the fibrous pericardium.
- ✓ The inner **visceral layer** of the serous pericardium, also called the **epicardium** reflects onto **the great vessels & surface of the heart.**
- Between the parietal and visceral layers of the serous pericardium is a thin film of **lubricating serous fluid.**
- This slippery secretion of the pericardial cells, known as **pericardial fluid**, reduces friction between the layers of the serous pericardium as the heart moves.

Serous pericardium,,

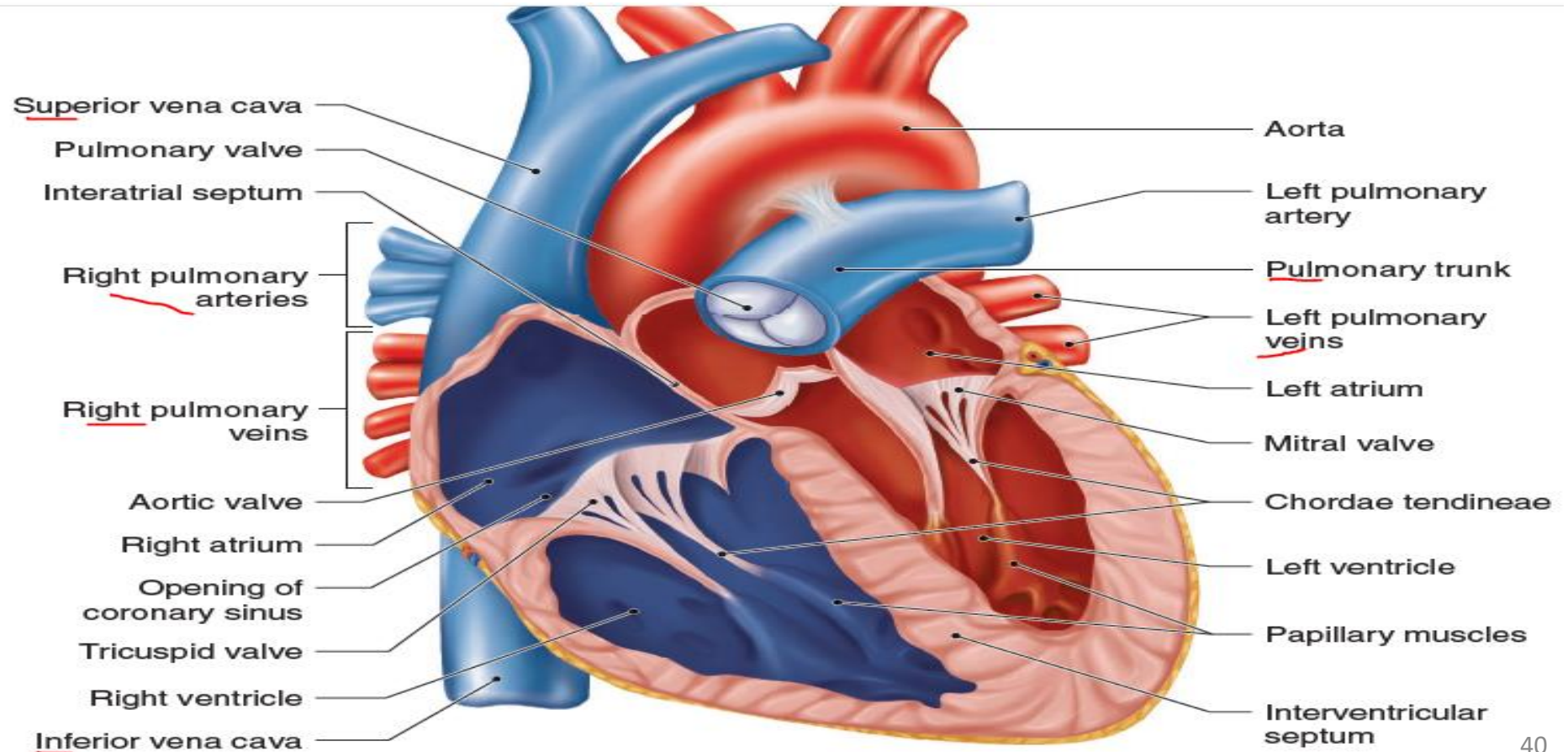


Serous pericardium,,

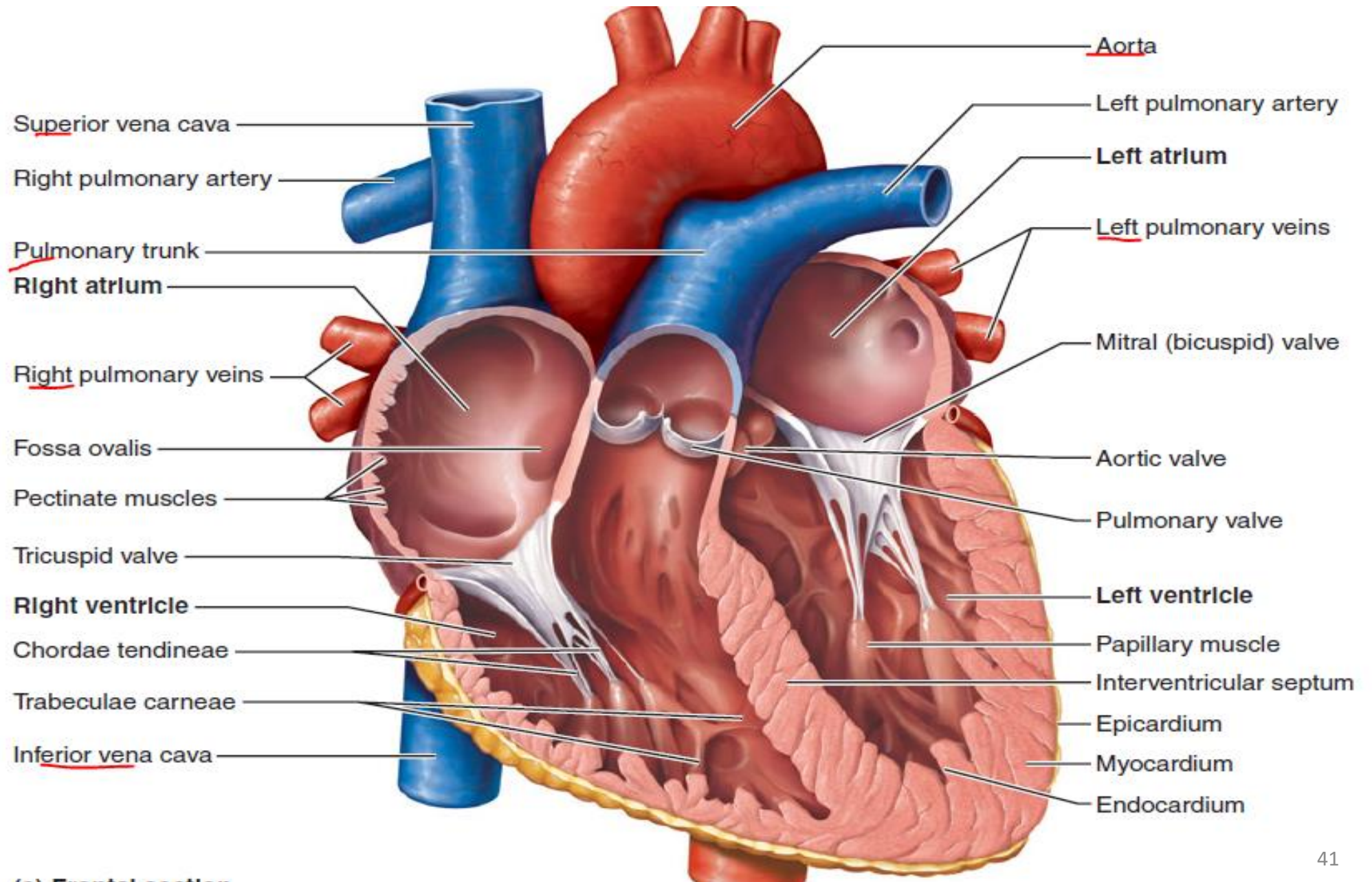


Serous pericardium,,,

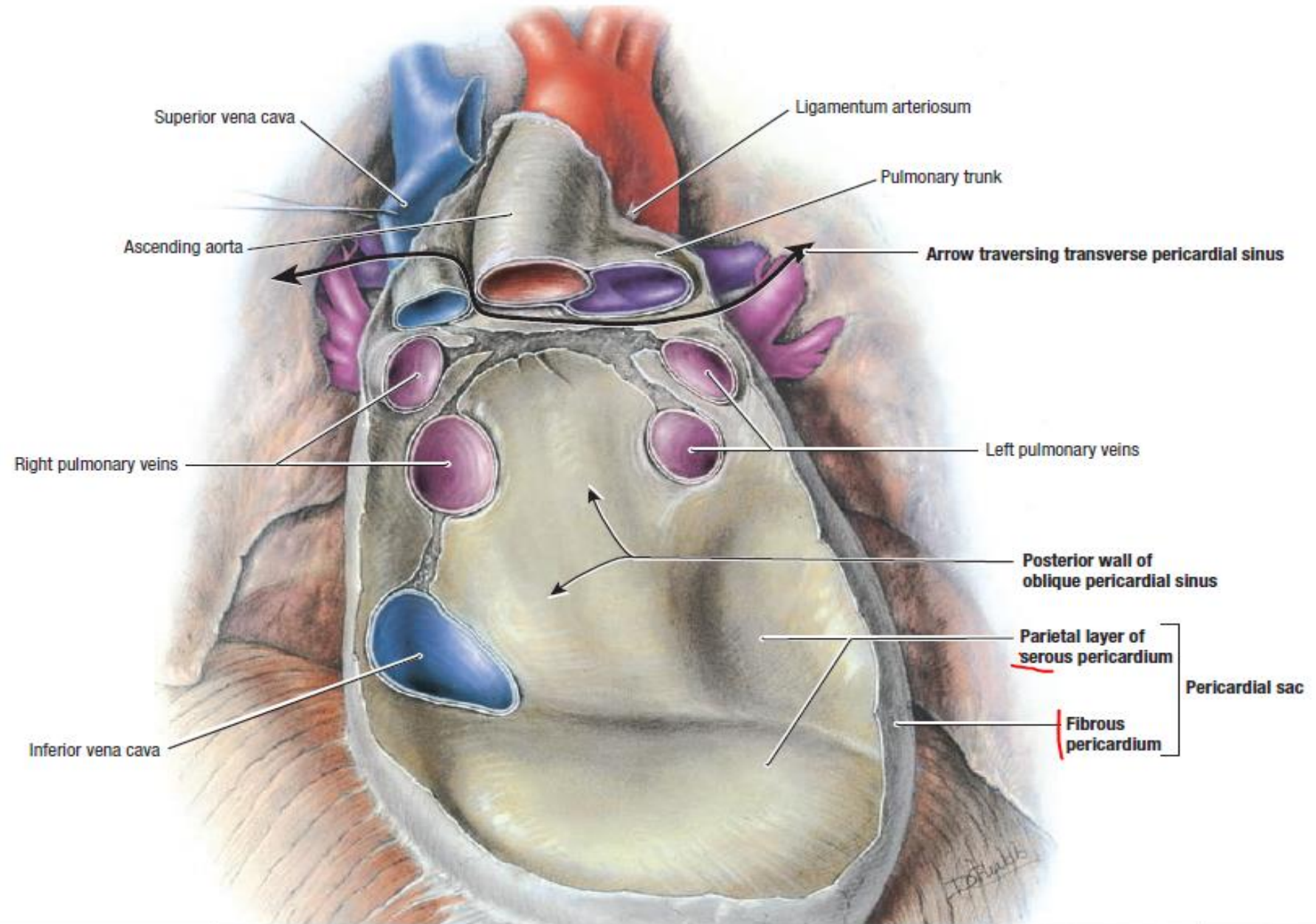
- ❑ The *visceral layer of serous pericardium* extends onto the beginning of the great vessels, becoming continuous with the parietal layer of serous pericardium (1) where the aorta and pulmonary trunk leave the heart and (2) where the SVC, inferior vena cava (IVC), and pulmonary veins enter the heart.



Serous pericardium,,



Serous pericardium,,,



Fibrous pericardium

- Fibrous pericardium is a tough, membrane composed of dense irregular CT.
- The fibrous pericardium prevents overstretching of the heart, provides protection, and anchors the heart in the mediastinum.
- The inferior wall (floor) of the fibrous pericardial sac is firmly attached and confluent (partially blended) centrally with the central tendon of the diaphragm.
- The site of continuity has been referred to as the **pericardiophrenic ligament**.
- These attachments help to retain the heart in its position in the thoracic cavity.

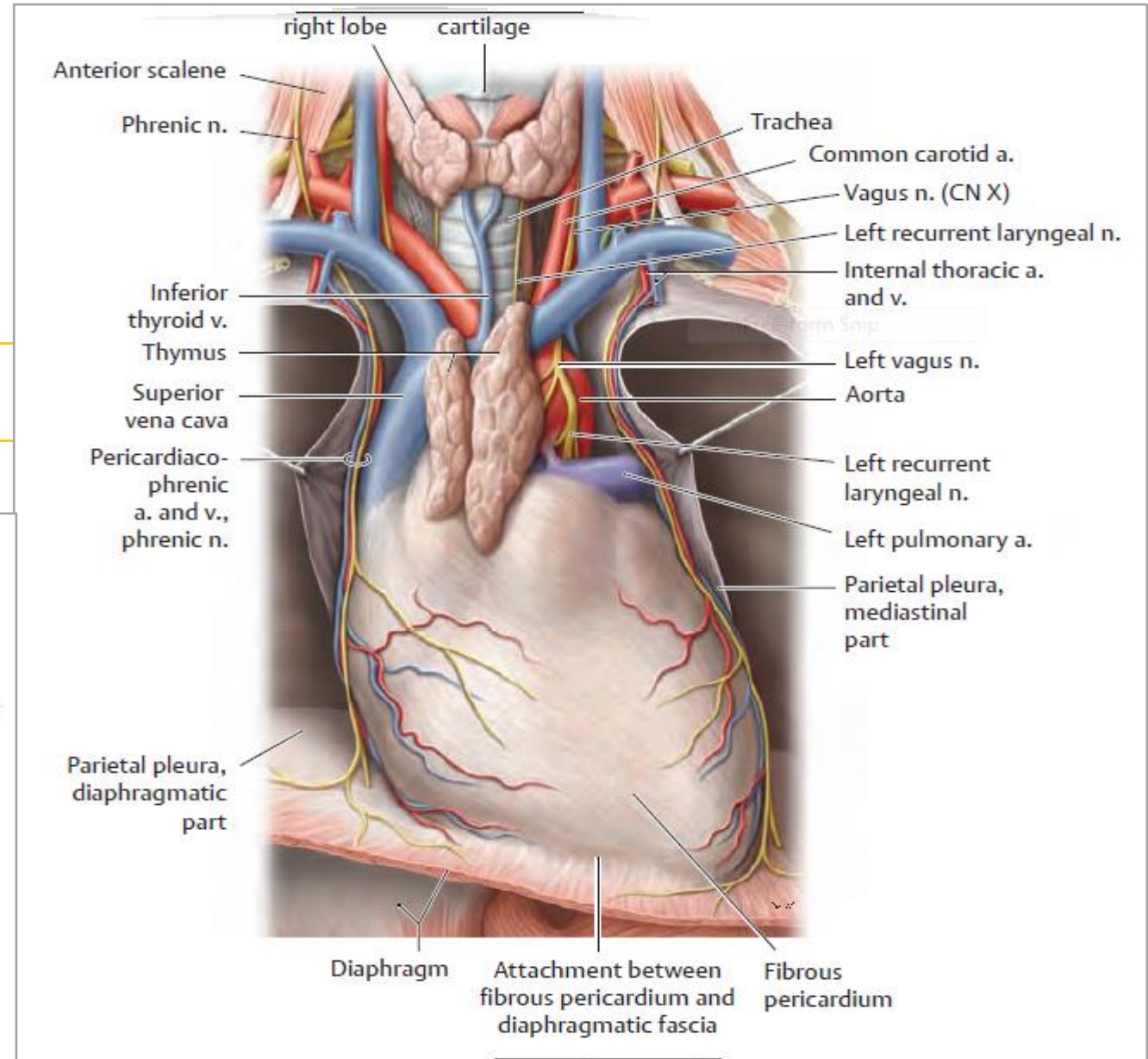
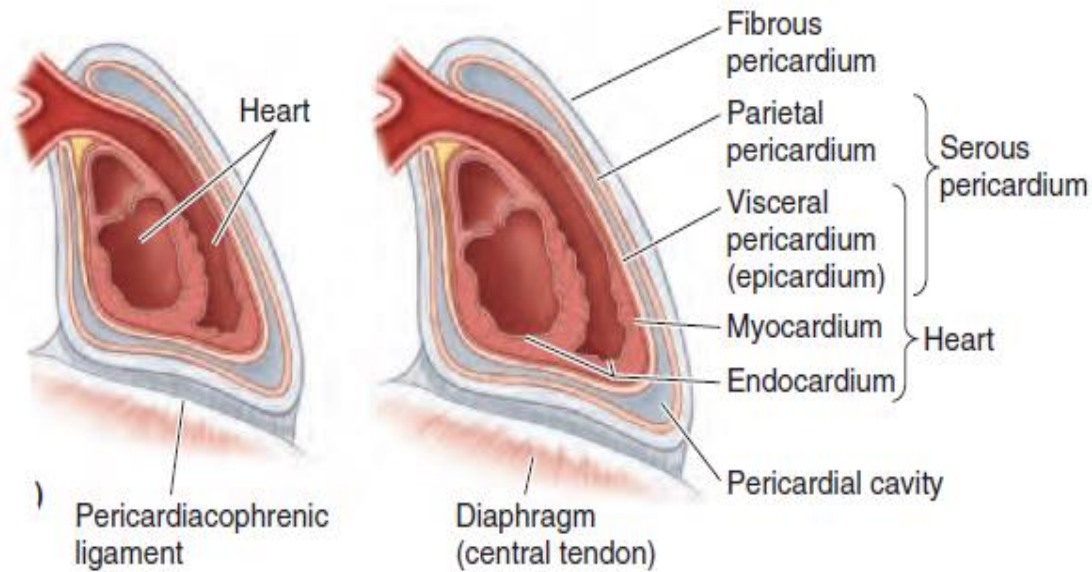
Fibrous pericardium,,

❖ The fibrous pericardium is:

- ✓ Continuous superiorly with the **tunica adventitia** of the great vessels entering & leaving the heart.
- ✓ Anteriorly, it is attached to the posterior surface of the sternum by **sternopericardial ligaments**.
- ✓ Bound posteriorly by loose **connective tissue** to structures in the posterior mediastinum.
- ✓ Continuous inferiorly with the **central tendon of the diaphragm** (**pericardiophrenic ligament**).

Fibrous pericardium,,

Fibrous pericardium,,



Pericardial Sinuses

- The pericardial sinuses form during development of the heart as a consequence of the folding of the primordial heart tube
- The *visceral layer of serous pericardium* forms the *epicardium* and **extends onto** the beginning of the great vessels.
- **The transverse sinus** is **posterior** to the **intrapericardial** parts of the **pulmonary trunk** and **ascending aorta**, anterior to the **SVC, and superior to the atria of the heart**.
- The **transverse pericardial sinus** is a transversely running passage within the pericardial cavity b/n these 2 groups of vessels and the reflections of serous pericardium around them.

Pericardial Sinuses,,

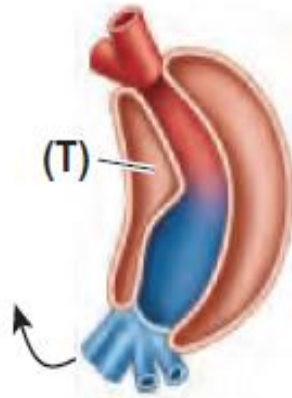
Primordial heart tube in pericardial sac



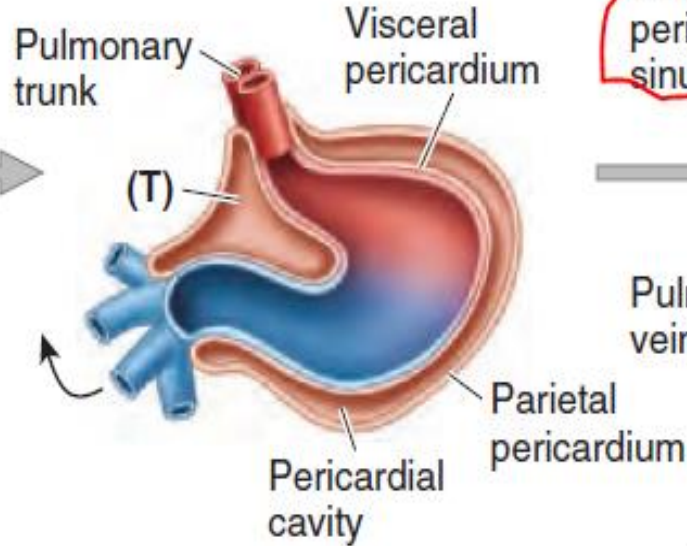
Arterial end

Venous end

Primordial transverse pericardial sinus (T)



Heart loops ventrally



Primordial arterial and venous ends of the developing heart are brought together, forming transverse pericardial sinus.

Transverse pericardial sinus

Pulmonary veins

Aorta

SVC

Visceral pericardium

Area of oblique pericardial sinus

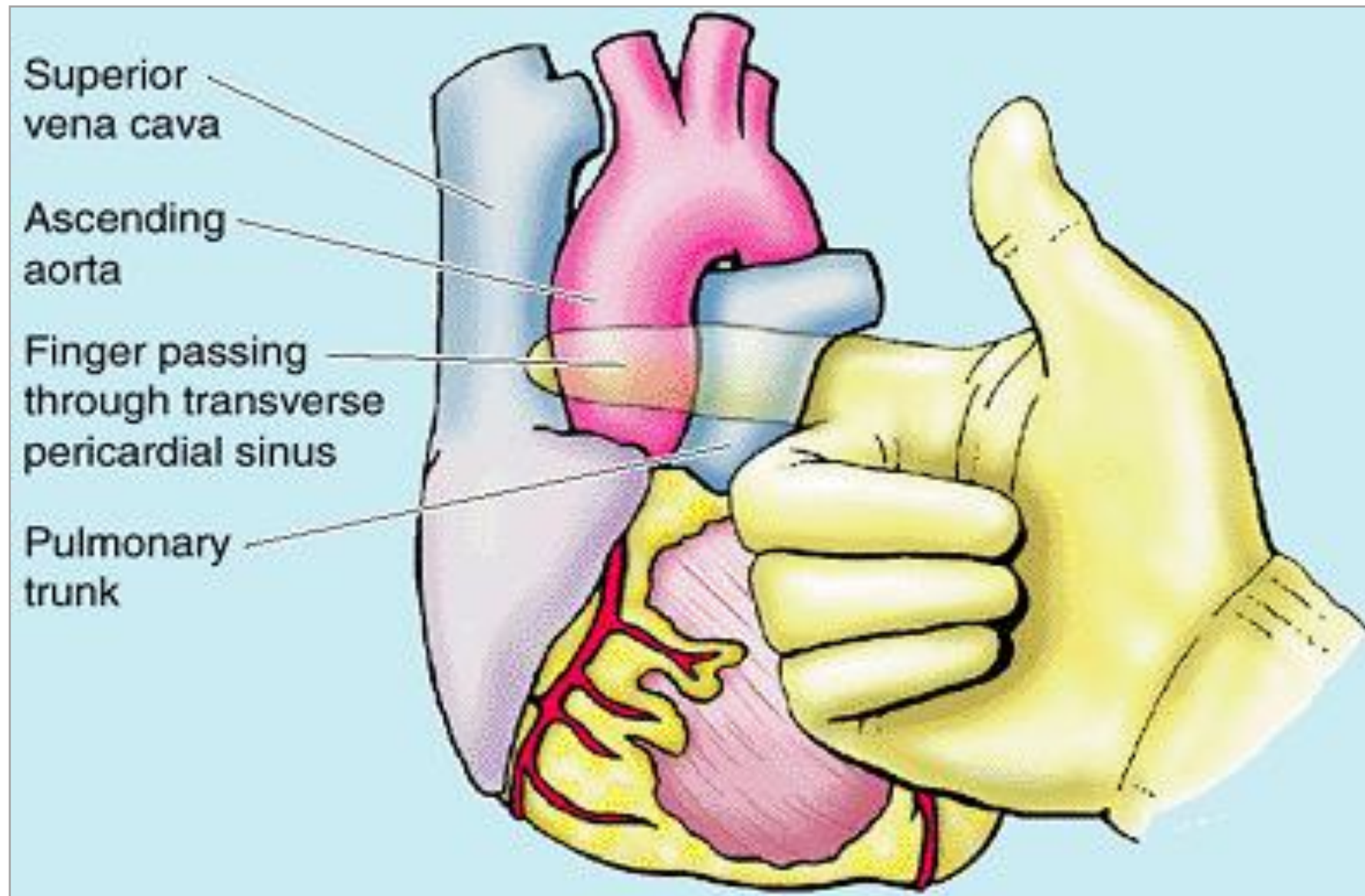
IVC

Posterior view of adult heart

Veins expand and pericardial reflection is carried out around them to form oblique pericardial sinus.

Lateral views of embryonic heart

Pericardial Sinuses,,

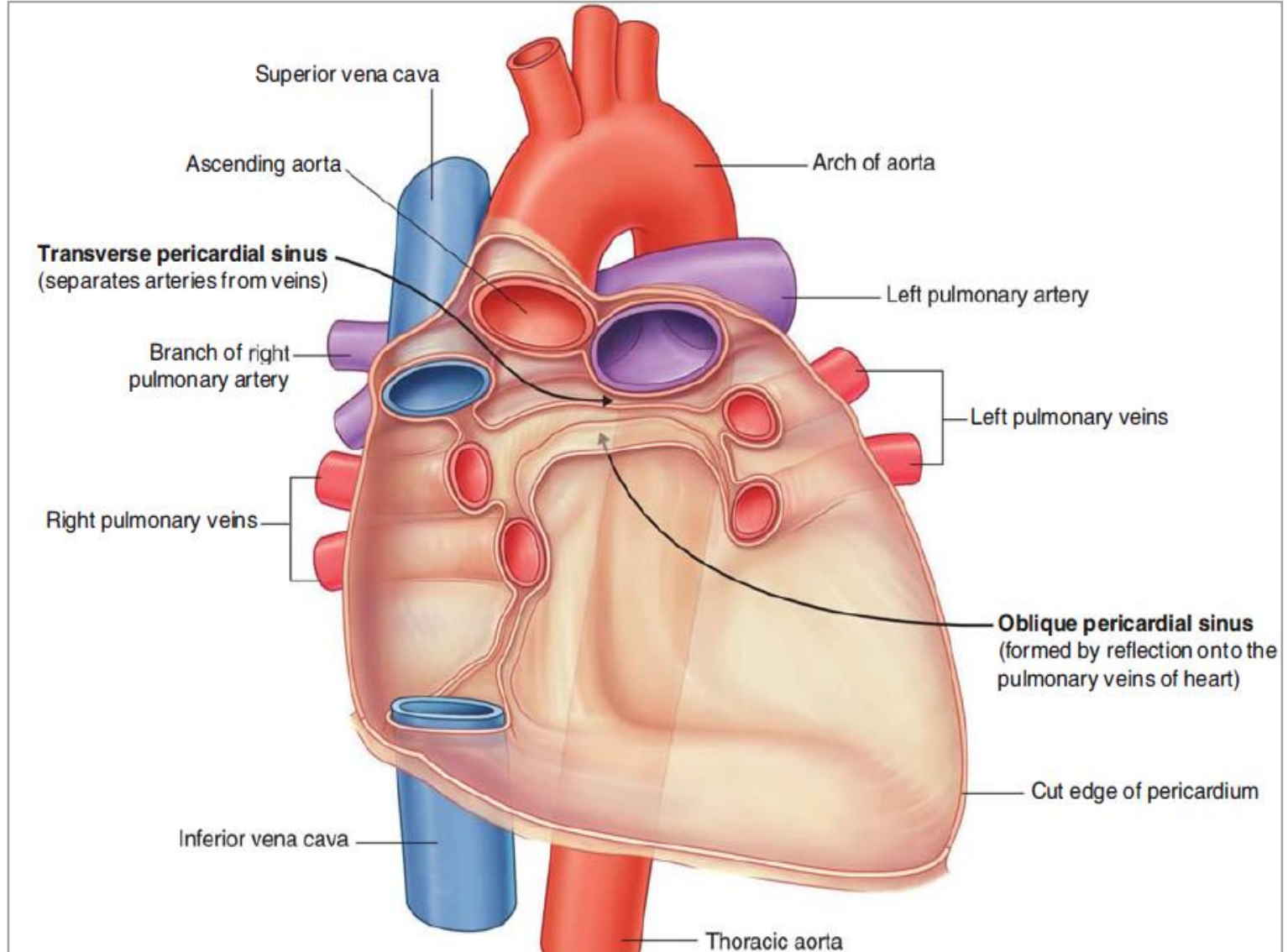


Pericardial Sinuses,,,

- The reflection of the serous pericardium around the second group of vessels defines the *oblique pericardial sinus*.
- As the heart tube folds, its venous end moves posterosuperiorly so that the venous end of the tube lies adjacent to the arterial end, **separated only by the transverse pericardial sinus**.
- As the veins of the heart develop and expand, a pericardial reflection surrounding them forms the **oblique pericardial sinus**, a wide pocket-like recess in the pericardial cavity posterior to the base of the heart.

Pericardial Sinuses,,,

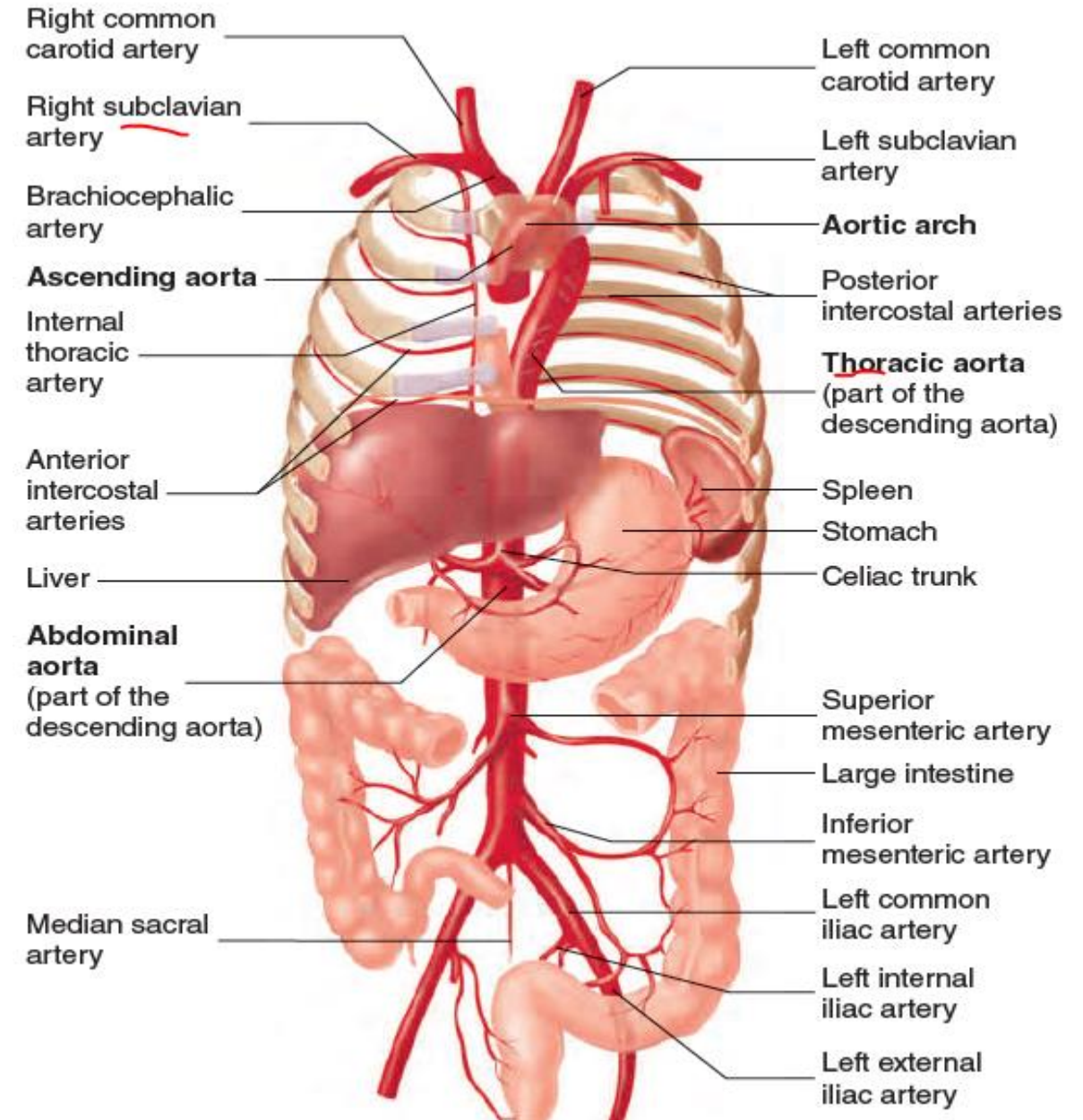
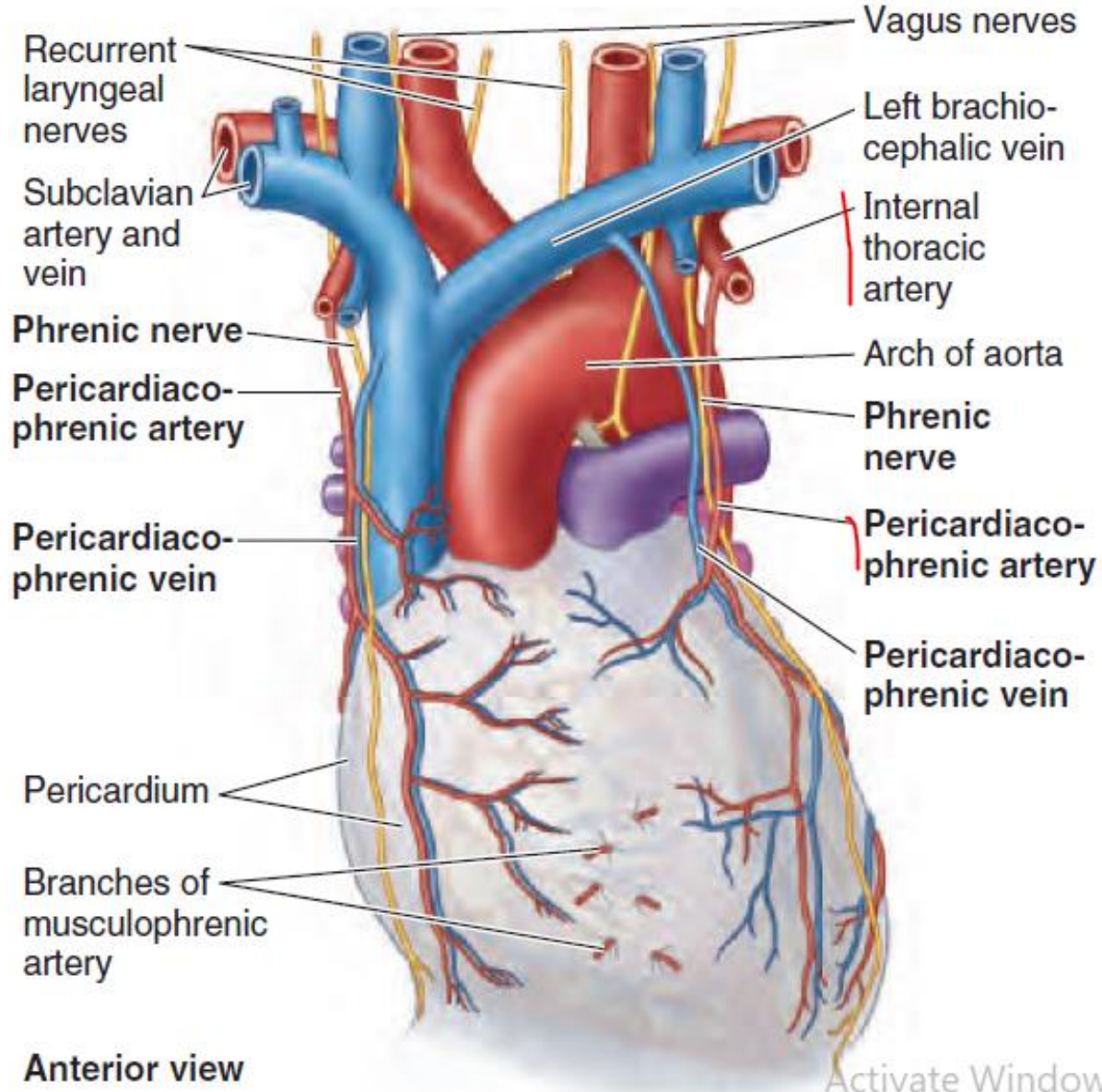
- **The oblique sinus is** bounded laterally by the pericardial reflections surrounding the **pulmonary veins and IVC** and posteriorly by the pericardium overlying the anterior aspect of the **esophagus**.



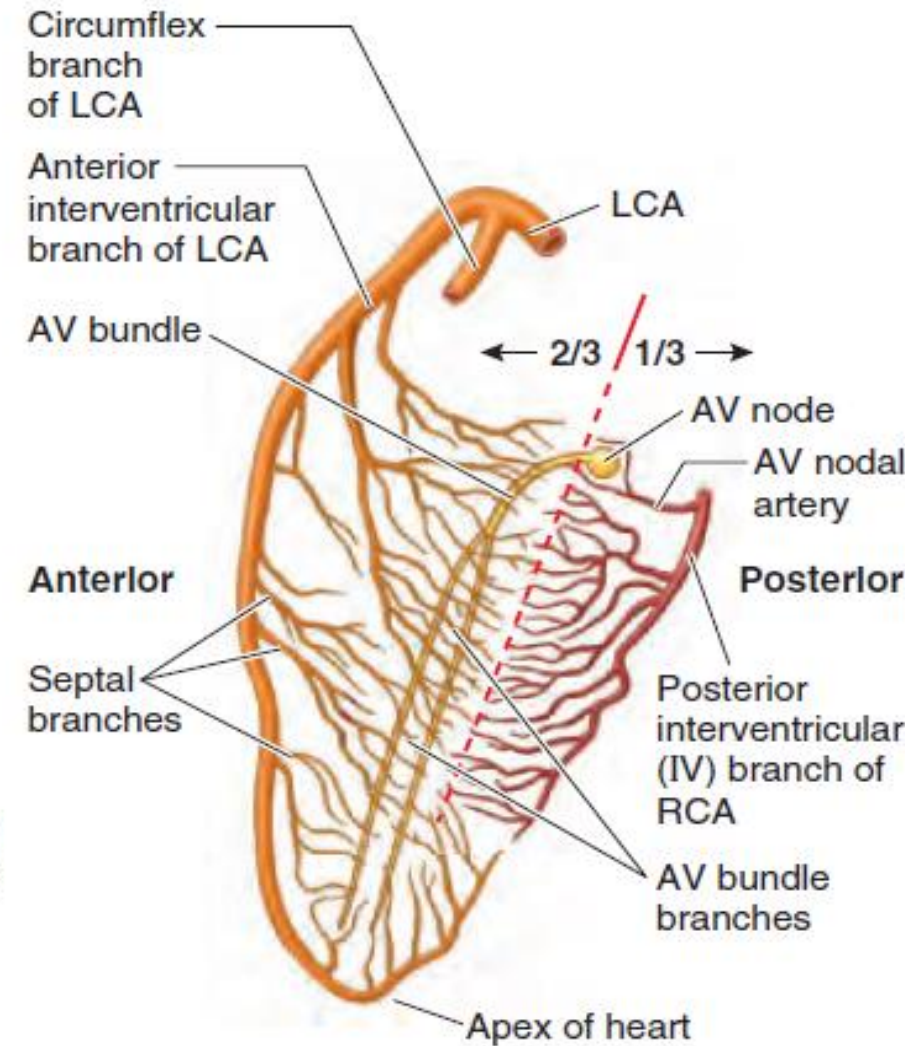
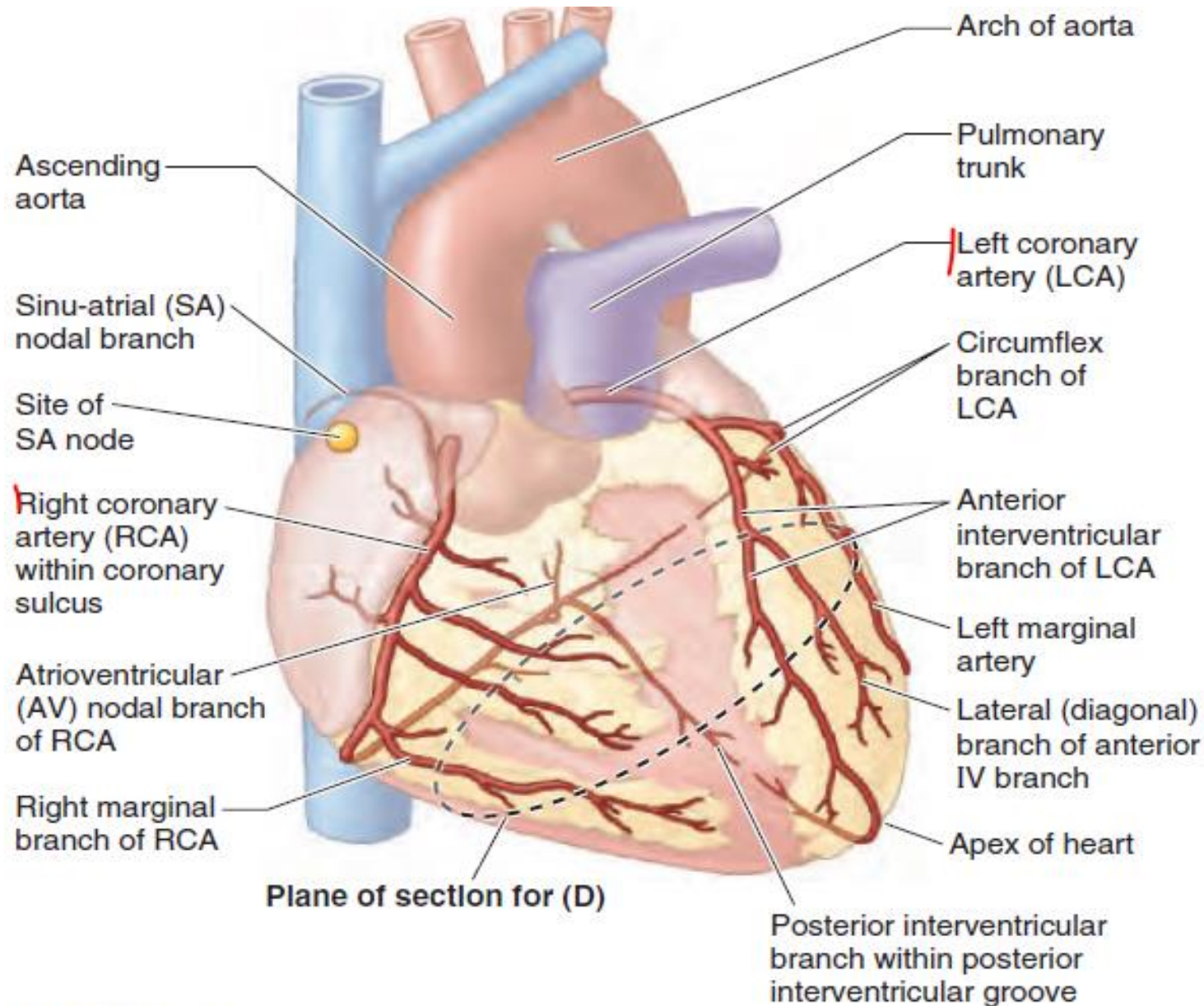
The arterial supply of the pericardium

- Is mainly from branch's of the **internal thoracic artery, the pericardiophrenic artery.**
- Smaller contributions of blood come from the:
 - ✓ **Musculophrenic artery**- a terminal branch of the internal thoracic artery
 - ✓ **Bronchial, esophageal, and superior phrenic arteries**, branches of the thoracic aorta
 - ✓ **Coronary arteries** (visceral layer of serous pericardium only), the first branches of the aorta.
- ❖ **The internal thoracic arteries are branches of the subclavian arteries.**

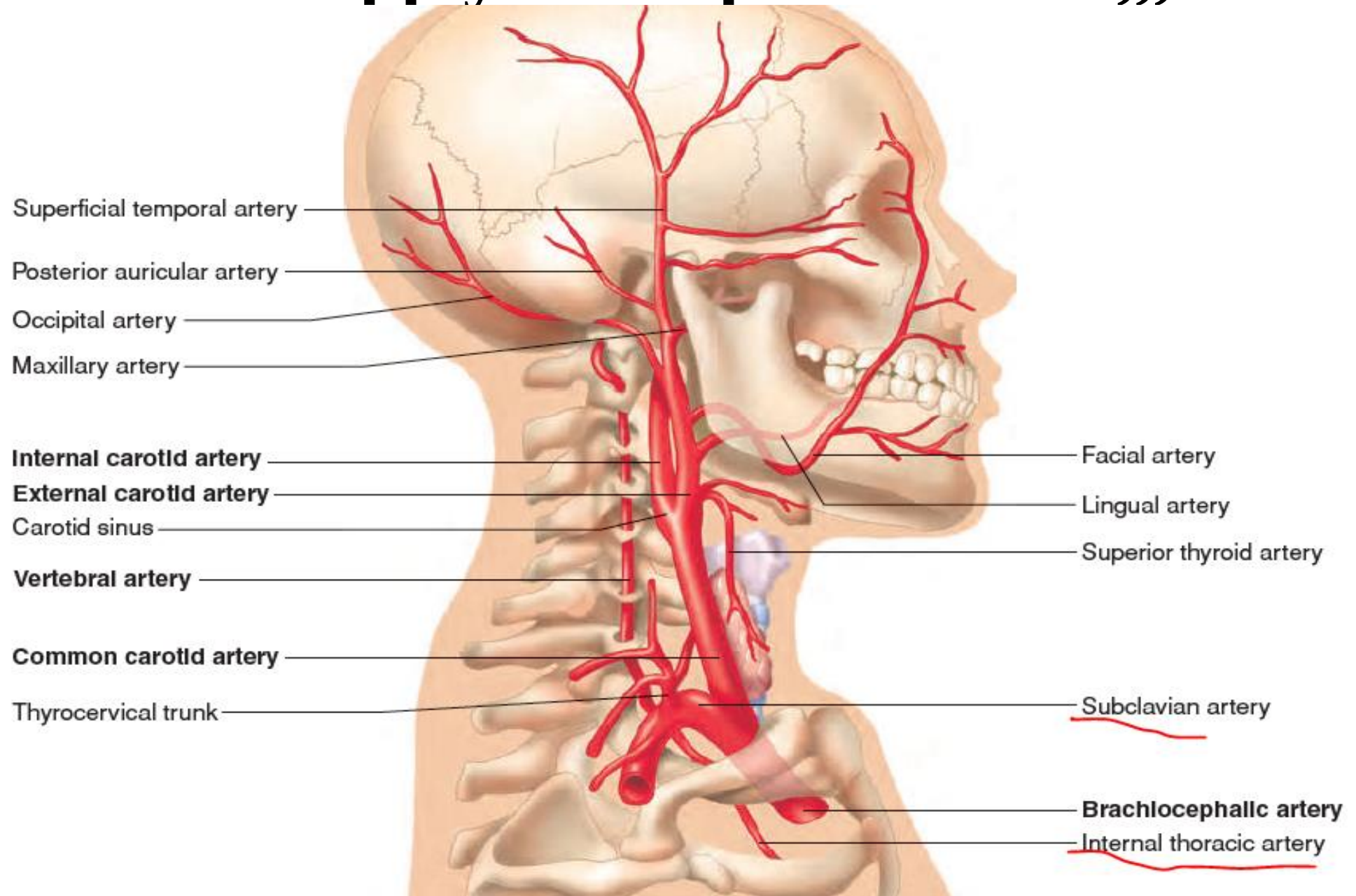
The arterial supply of the pericardium,,,



The arterial supply of the pericardium,,,

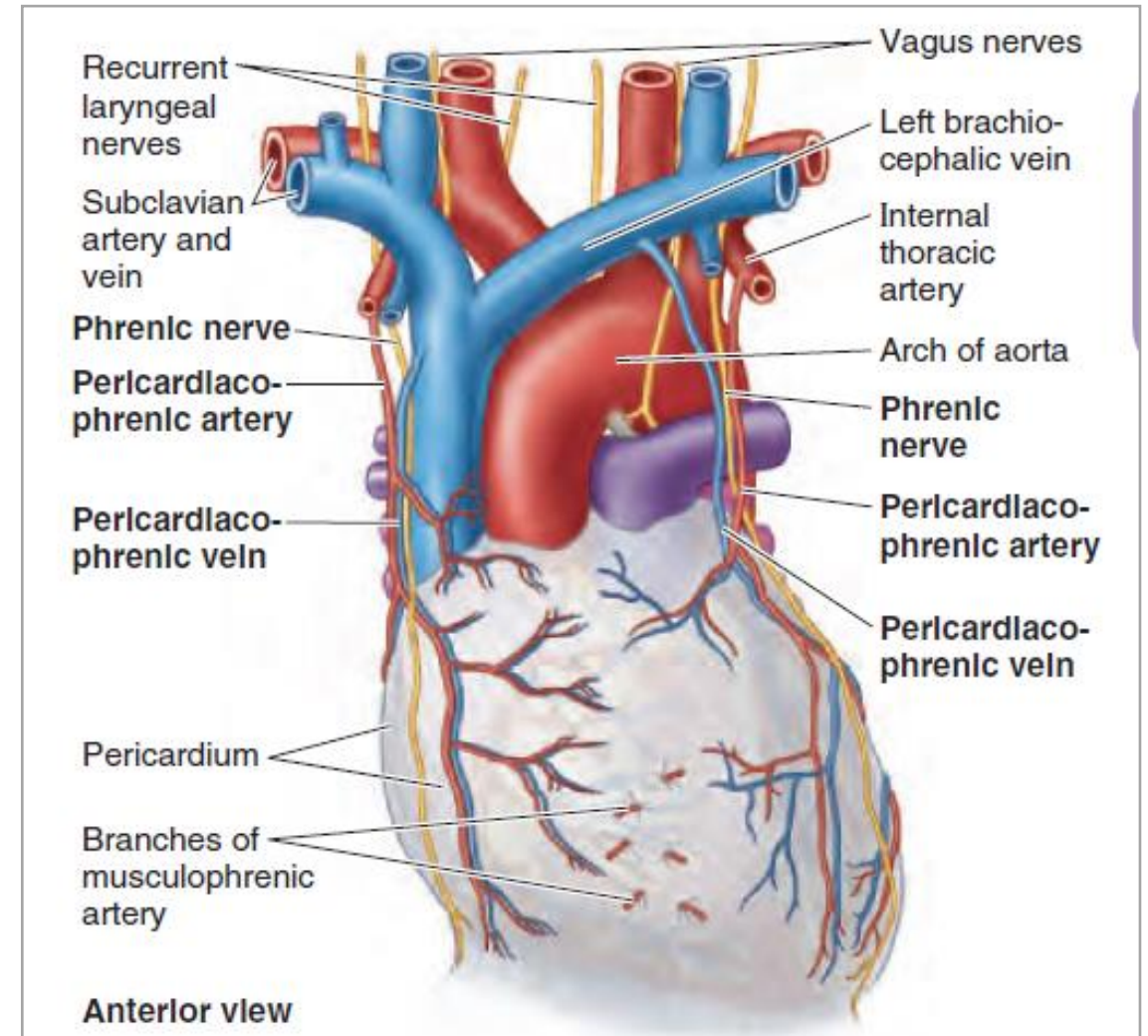


The arterial supply of the pericardium,,,



The venous drainage of the pericardium

- The venous drainage of the pericardium is into the:
- *Pericardiophrenic veins*, tributaries of the **brachiocephalic** (or internal thoracic) veins.
- Variable tributaries of the *azygos venous system*.



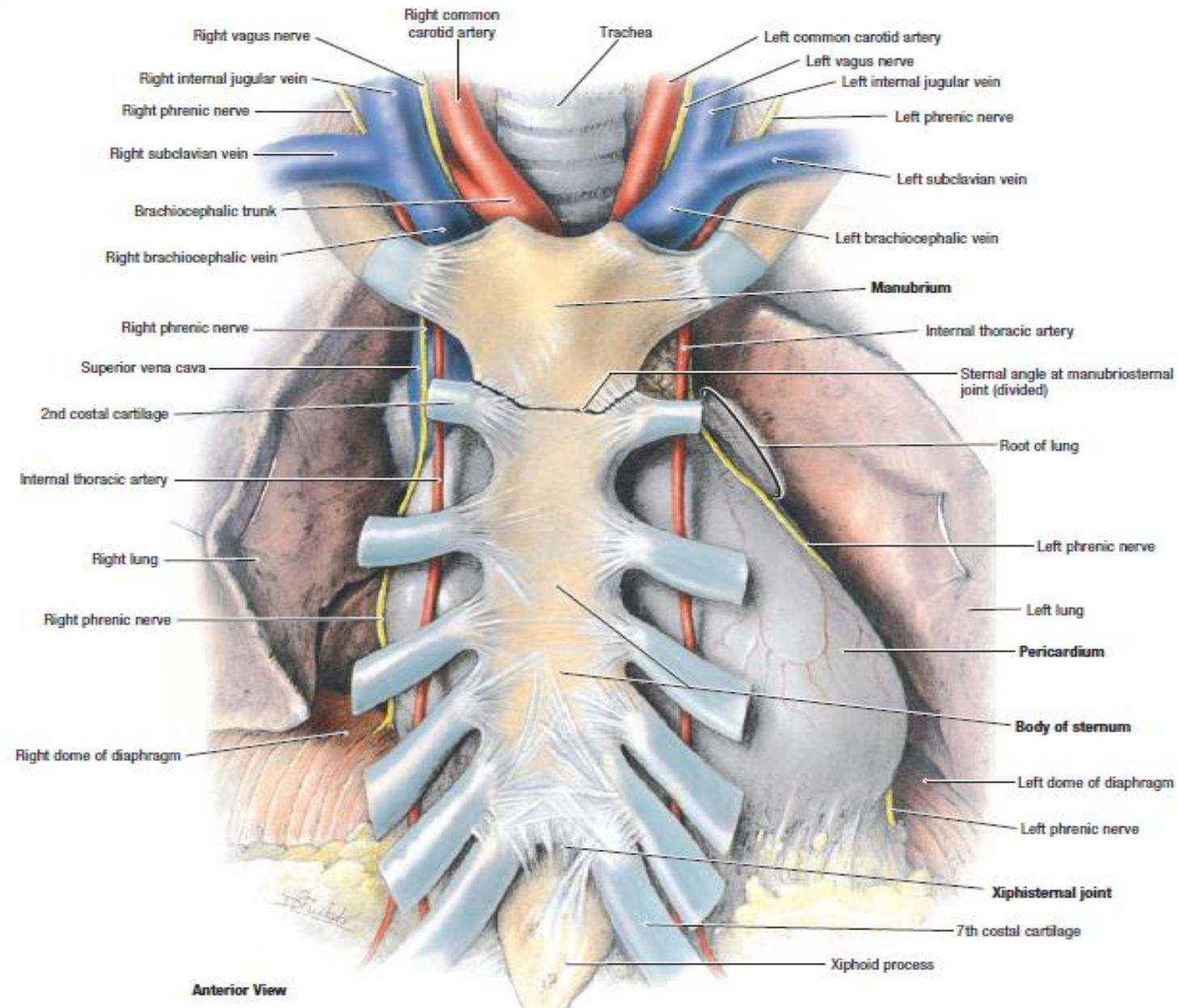
Arterial supply and venous drainage of pericardium

The nerve supply of the pericardium

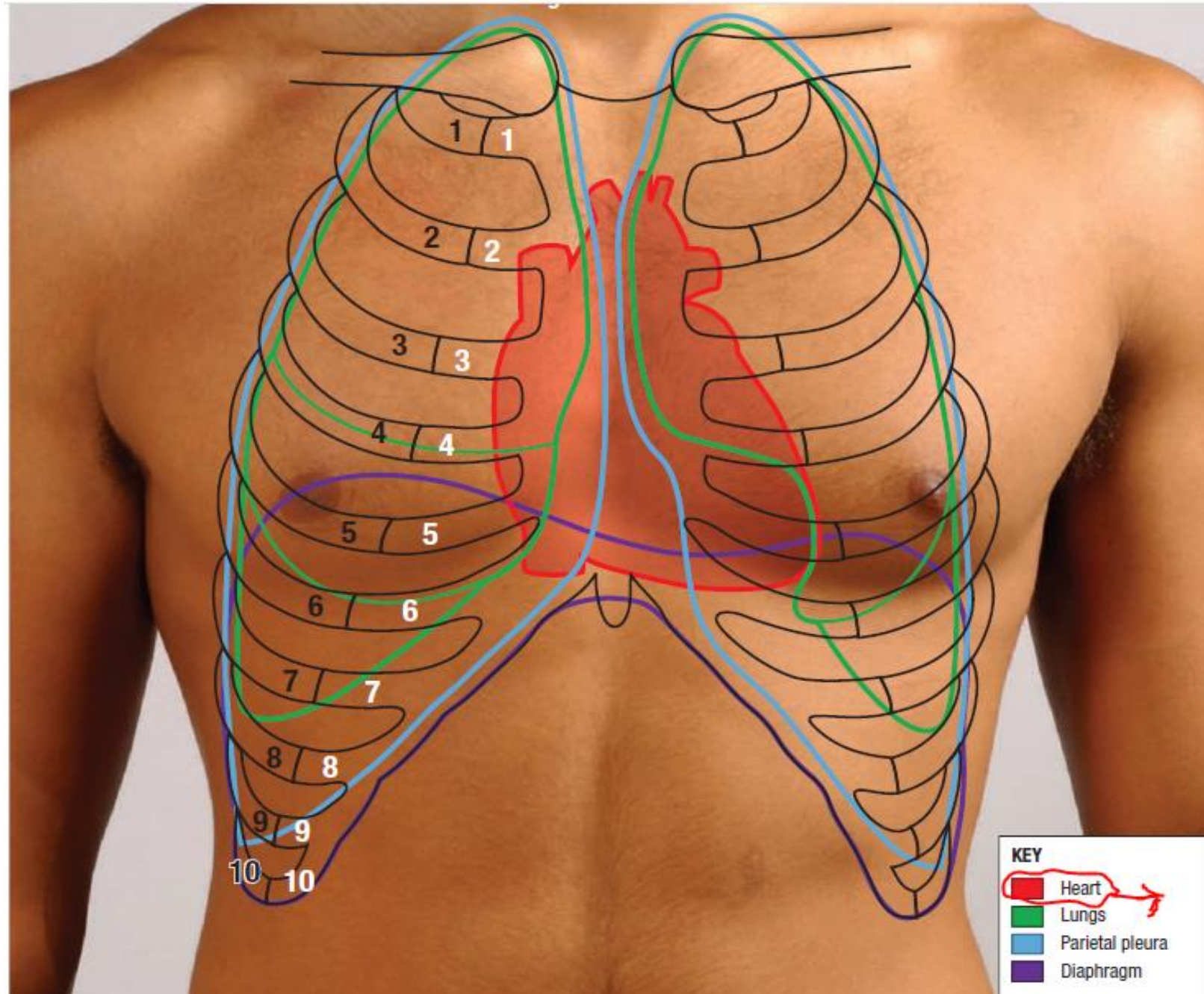
The nerve supply of the pericardium is from the:

- ***Phrenic nerves (C3–C5)***, primary source of sensory fibers; pain sensations conveyed by these nerves are commonly referred to the skin (C3–C5 dermatomes) of the ipsilateral supraclavicular region (top of the shoulder of the same side).
- ***Vagus nerves***, function uncertain.
- ***Sympathetic trunks***, vasomotor

The nerve supply of the pericardium,,



Gross Anatomy of the Heart

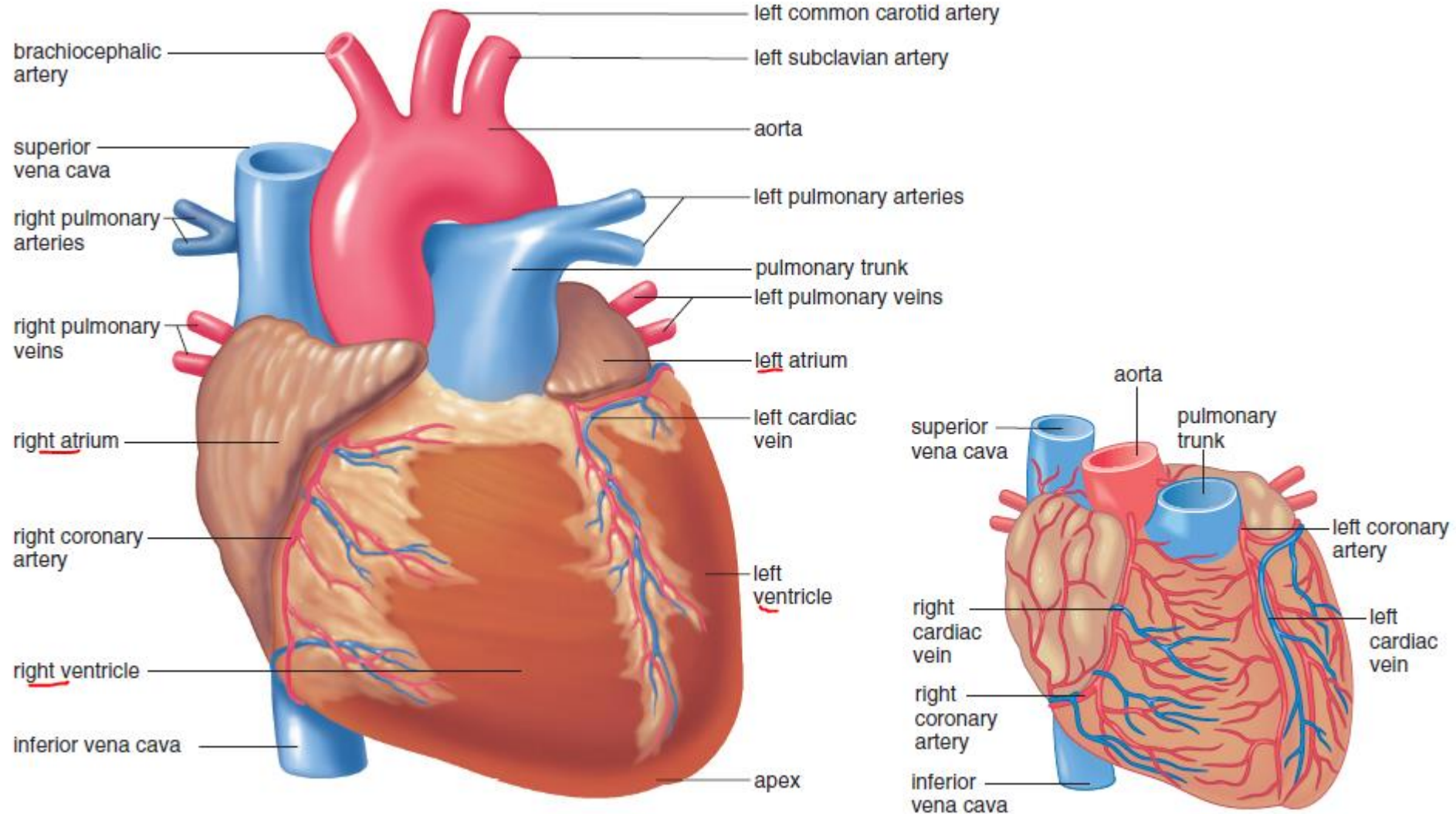


Gross Anatomy of the Heart

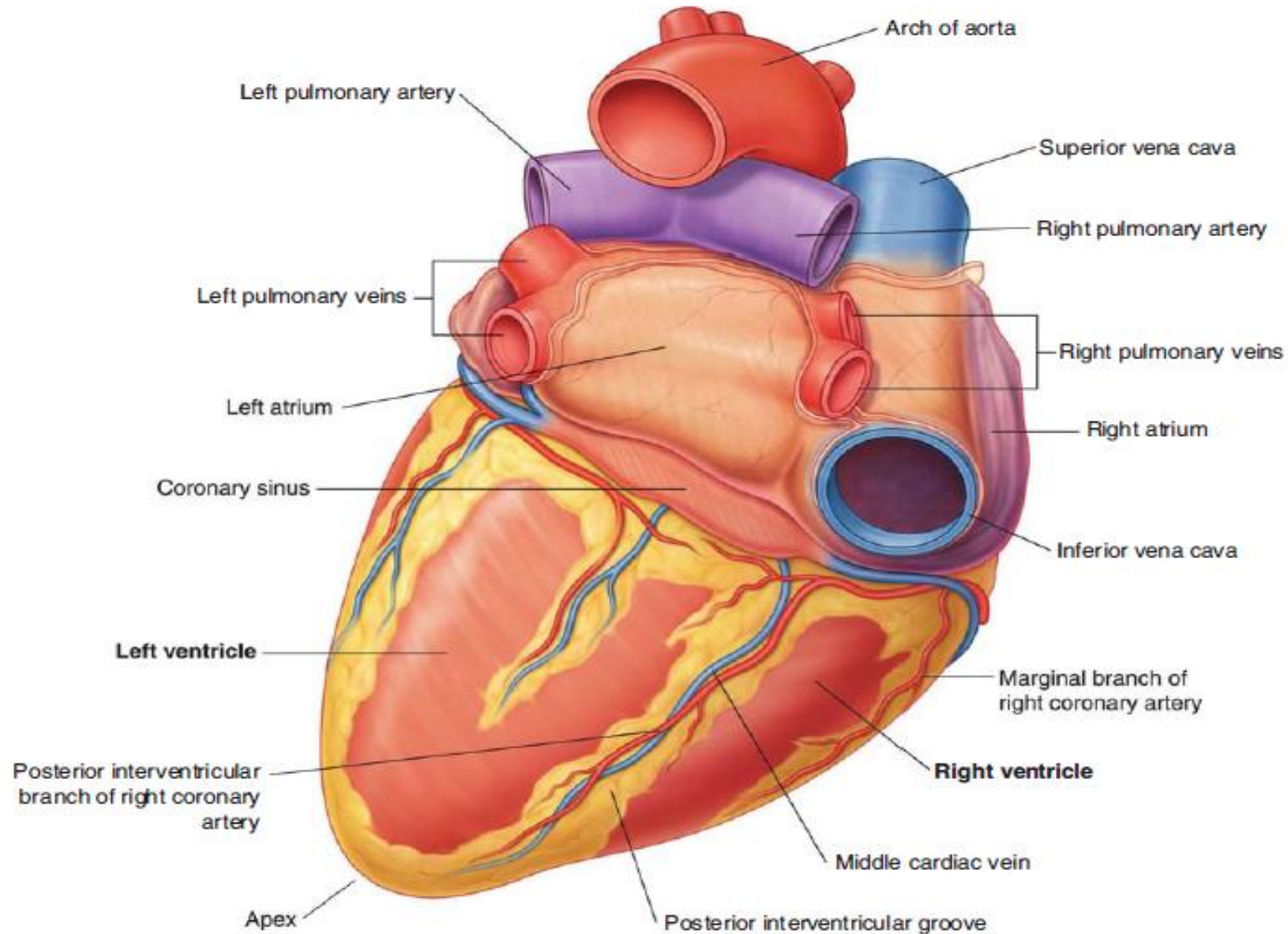
- The heart is a muscular pump that keeps blood flowing through the vessels.
- It is a four-chambered muscular pumping organ of the **cardiovascular system**.
- The general shape of the **heart pyramidal**.
- The right side of the heart receives **poorly oxygenated blood** through the SVC and IVC and pumps it through the pulmonary trunk to the lungs for oxygenation.
- The left side of the heart receives **well-oxygenated (arterial)** blood from the lungs through the **pulmonary veins** & pumps it into the aorta for distribution.

Chambers of the heart

- The heart has 4 chambers: **right & left atria & right and left ventricles.**

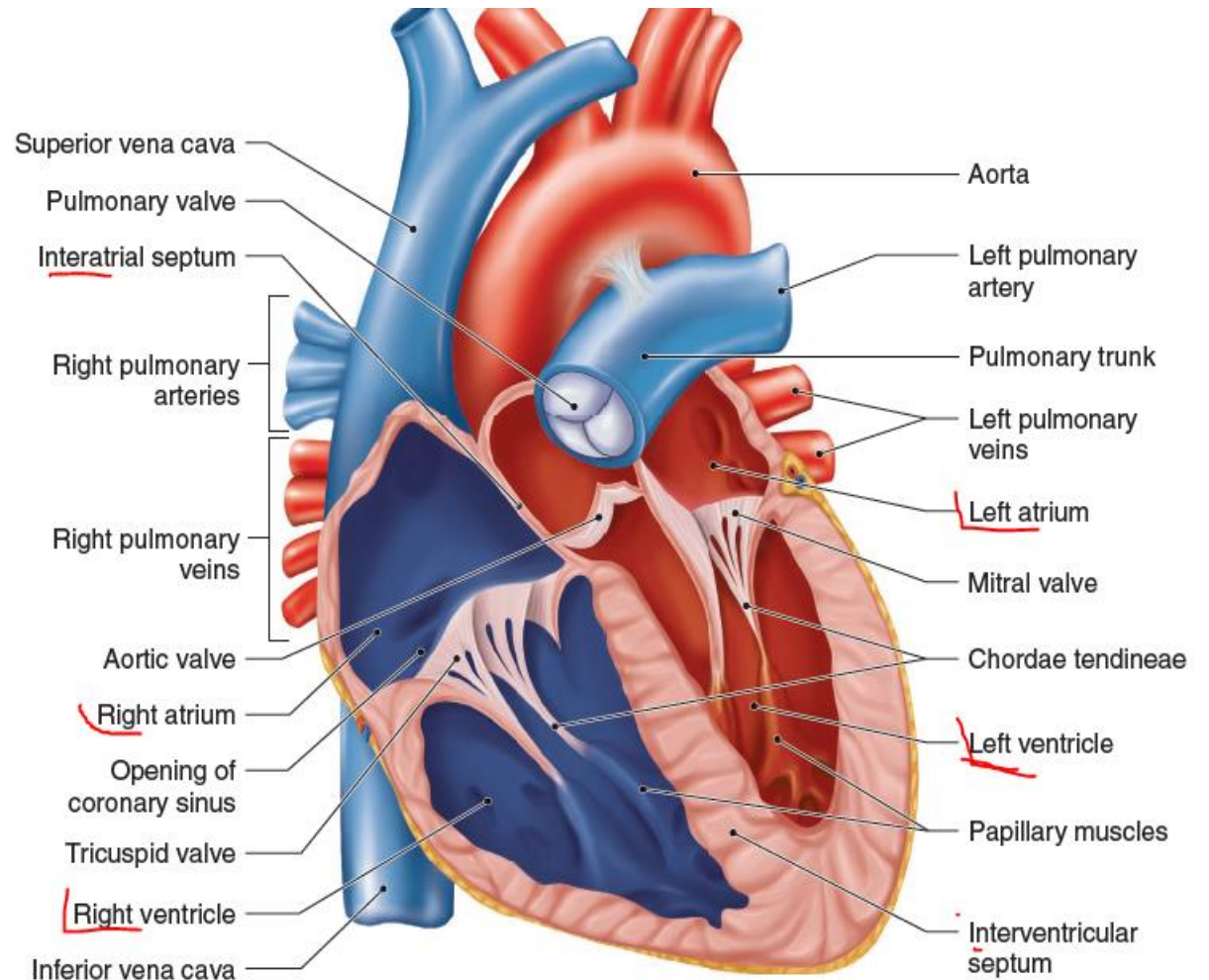


Chambers of the heart,,

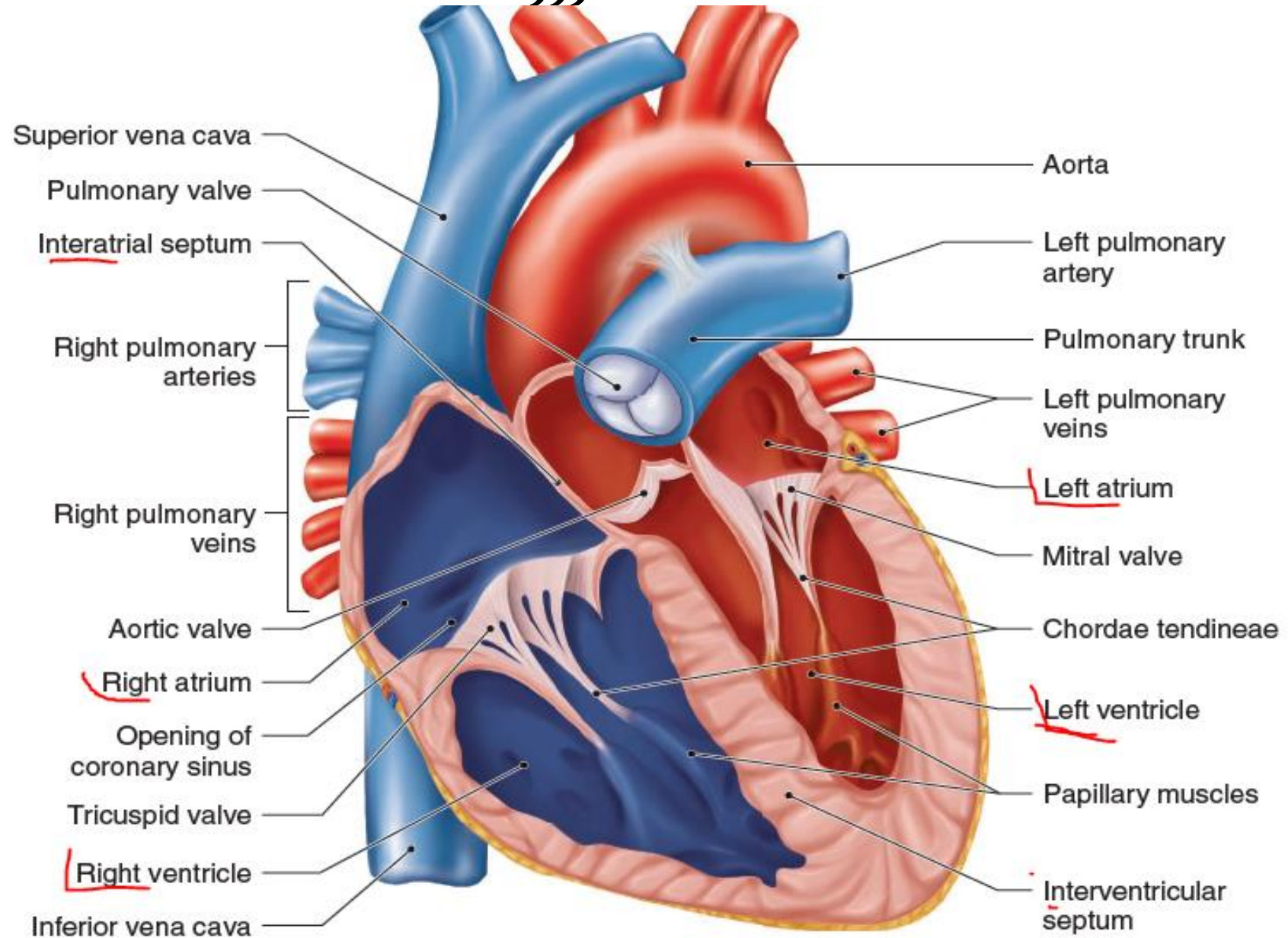


Chambers of the heart,,

- The right and left **atria** are located at the base of the heart, the right & left **ventricles** extend from the base of the heart toward the apex.
- The right and left **atria**, which have relatively thin walls and are separated by a common wall of myocardium called the **interatrial septum**.
- The right and left **ventricles**, which have thicker walls and are separated by the **interventricular septum**.



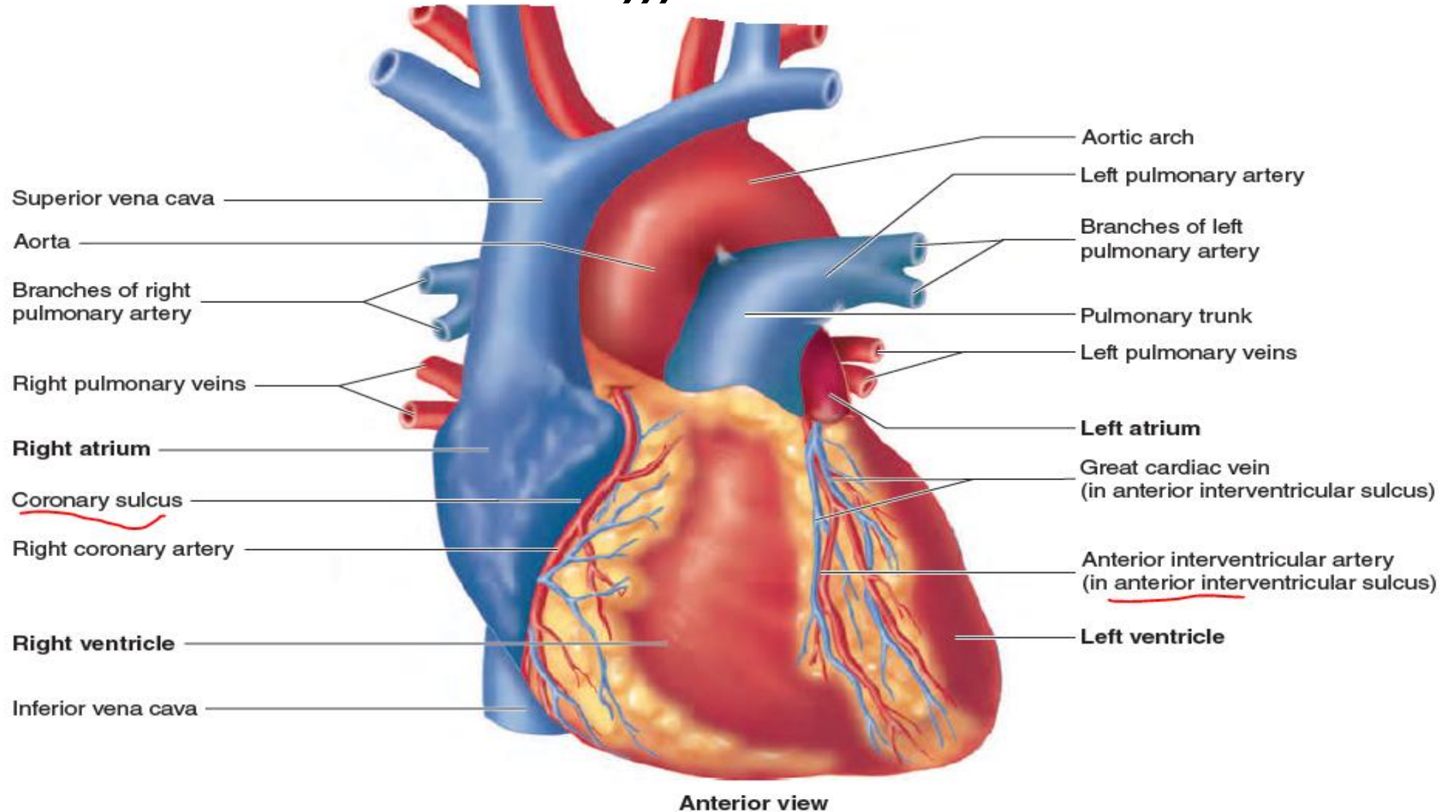
Chambers of the heart,,,



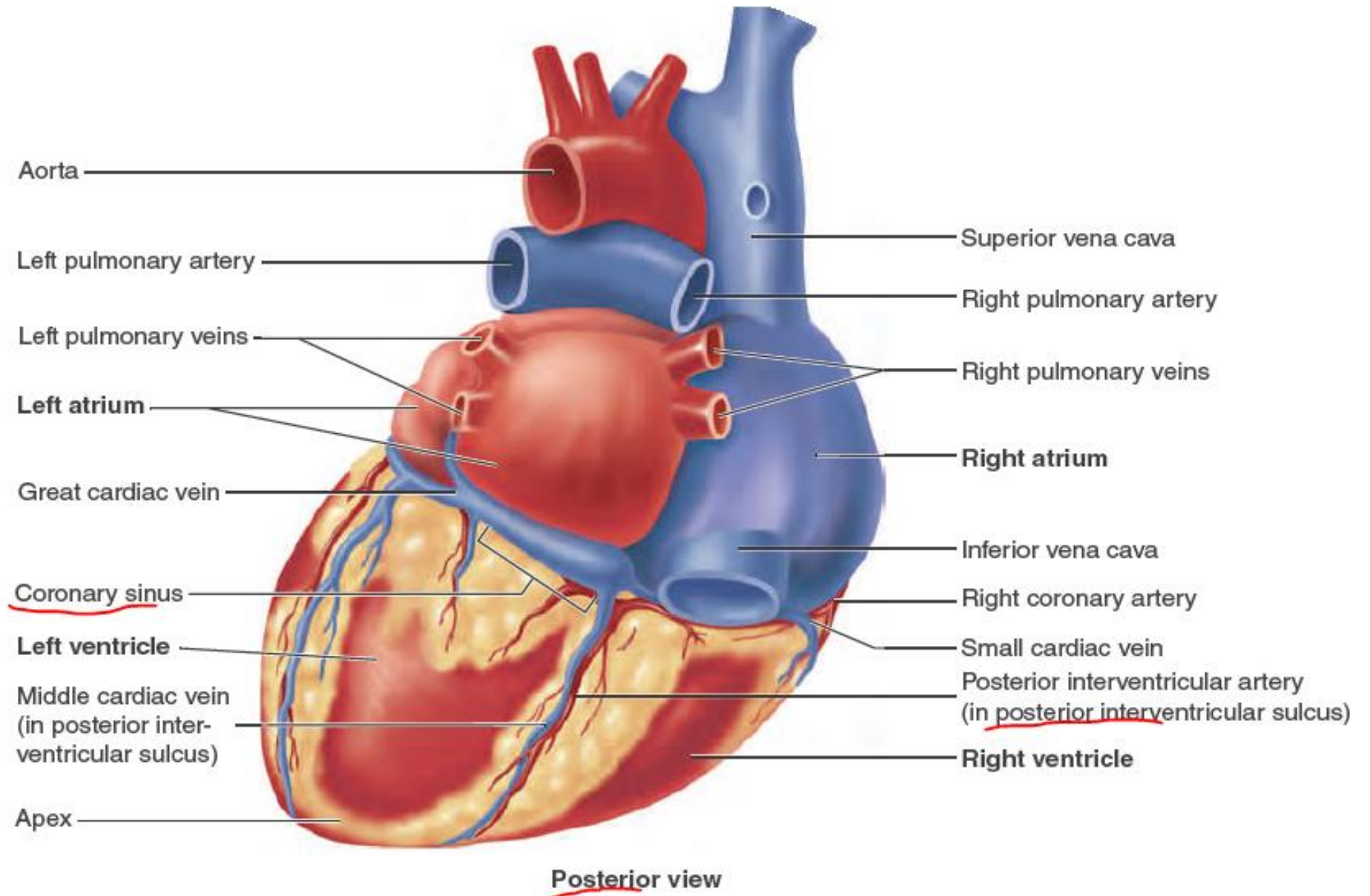
Chambers of the heart,,,

- **On the surface**, the boundaries of the four chambers are marked by **three sulci (grooves)**, which are largely filled by fat and the coronary blood vessels.
- The **coronary(atrioventricular) sulcus** encircles the heart near the base and separates the atria above from the ventricles below.
- **The other two sulci** extend obliquely down the heart from the coronary sulcus toward the apex—one on the front of the heart called the **anterior interventricular sulcus** and one on the back called the **posterior interventricular sulcus**.
- The coronary sulcus and two interventricular sulci harbor the largest of the coronary blood vessels.

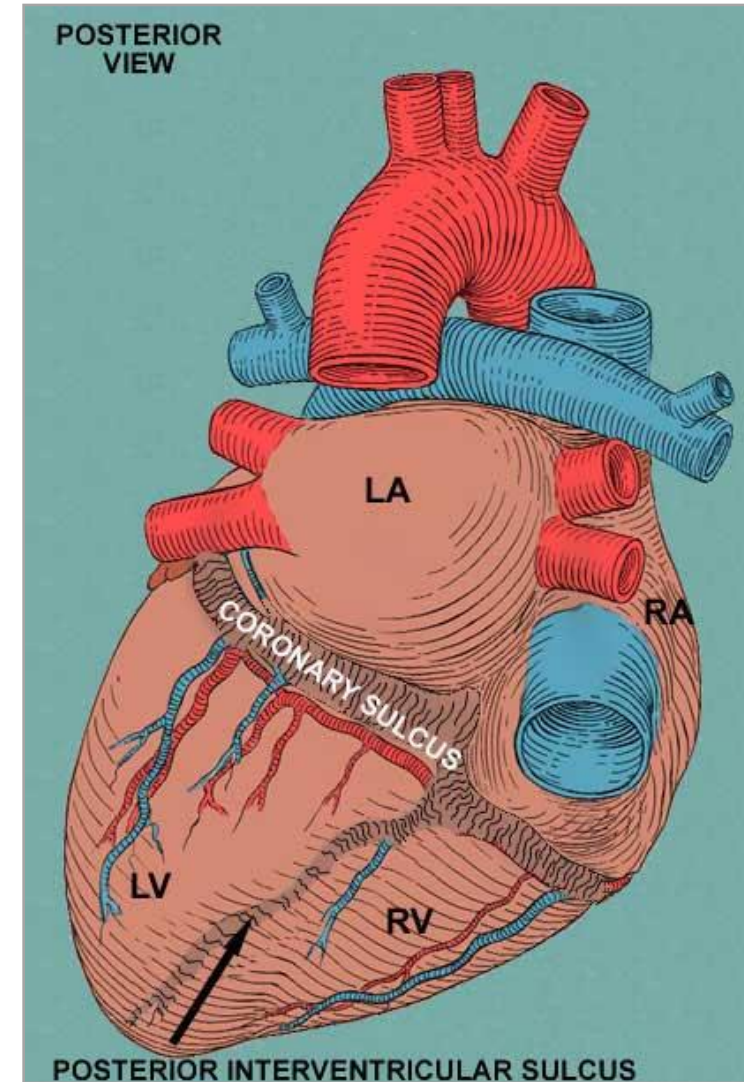
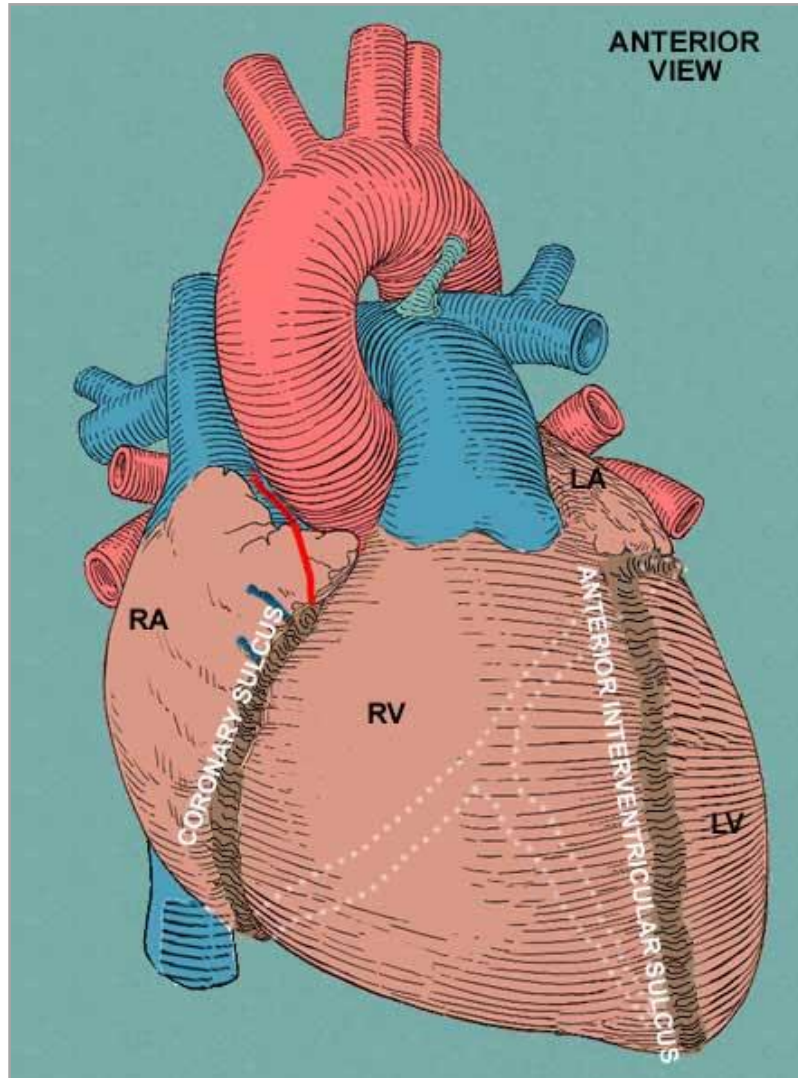
Chambers of the heart,,



Chambers of the heart,,



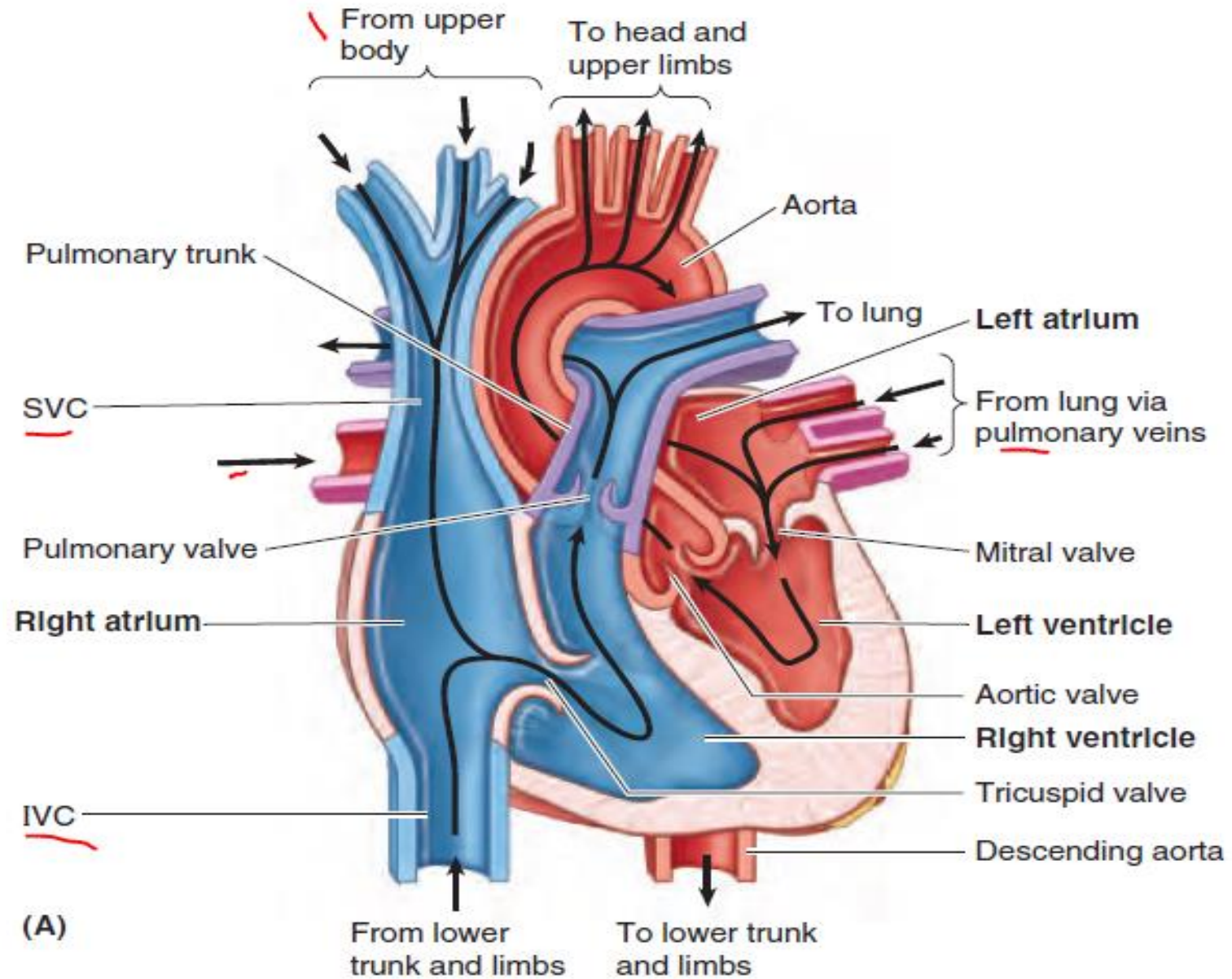
Chambers of the heart,,,



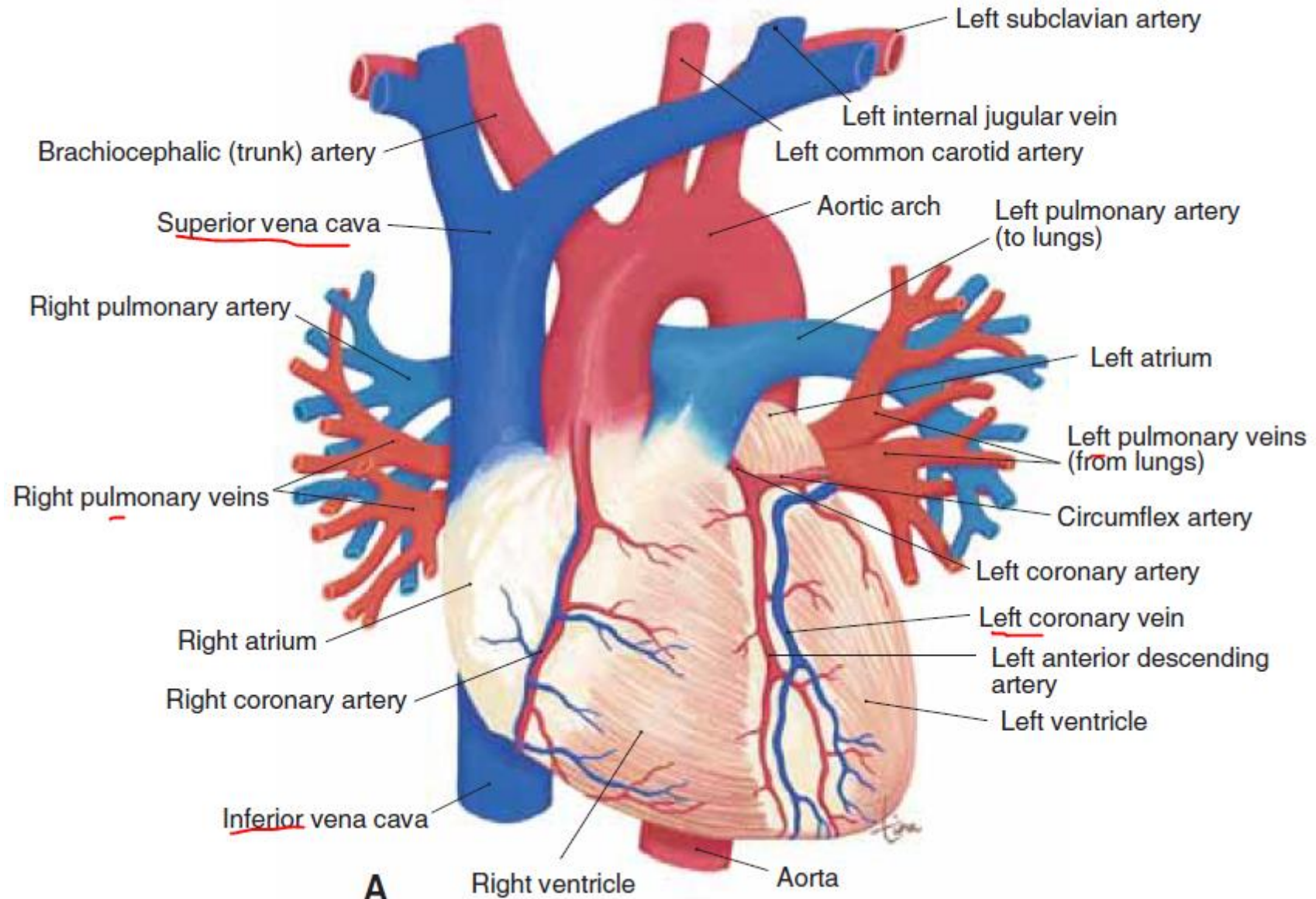
Chambers of the heart,,

- The **atria**, are thin-walled receiving chambers function **primarily as reservoirs**, where blood returning from veins collects before it enters the ventricles.
 - **The right atrium receives blood through** SVC and the IVC from most of the body, & the **smaller coronary sinus** from most of the heart muscle.
 - **The 2 inferior** chambers are the **ventricles**, which are thicker, and they pump blood into blood vessels that travel to all parts of the body.
- **The left ventricle has a thicker wall than the right ventricle;**
- **The right ventricle pumps blood to the lungs, which are nearby.**
 - **The left ventricle pumps blood to all the other parts of the body.**

Chambers of the heart,,



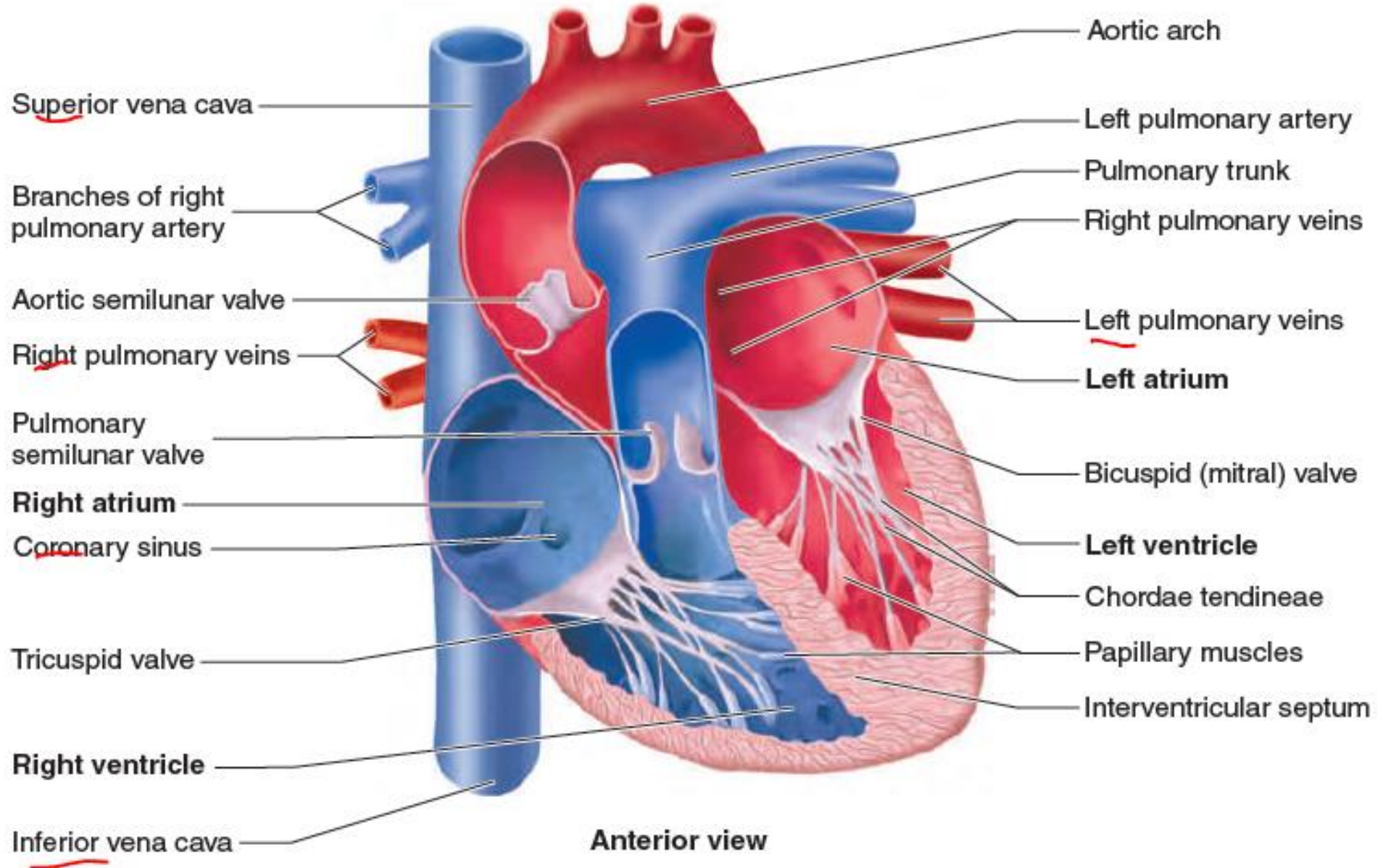
Chambers of the heart,,



Right atrium

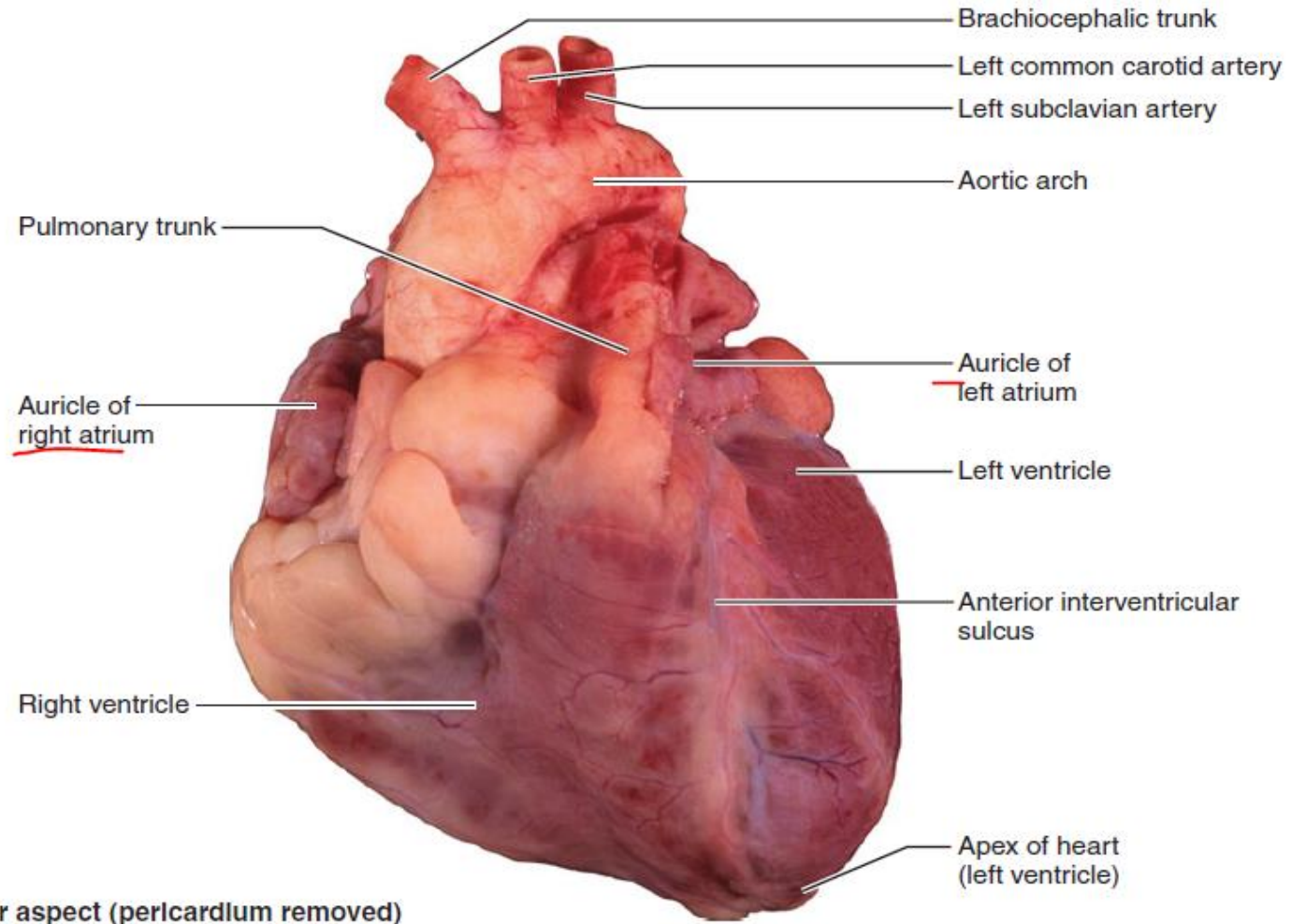
- ❑ **The right atrium receives blood via three veins:** the *superior vena cava* from the upper body part, *inferior vena cava* from the lower body part and the *coronary sinus* from the heart itself
- ✓ **SVC** enters *upper posterior* portion of ***Rt atrium***.
- ✓ **IVC & coronary sinus** enter the *lower posterior* portion of ***Rt atrium***.
- ✓ From Rt atrium, blood passes into Rt ventricle thru **Rt atrioventricular orifice**.
- ✓ **Rt atrioventricular orifice** is closed during ventricular contraction by *tricuspid valve*.
- ❖ **The primordial atrium is represented in the adult by the right auricle.**

Right atrium,,



Right atrium,,

- Externally, the **right auricle**, a small flap shaped like a dog's ear (*auricle* little ear) projects anteriorly from the superior corner of the atrium.

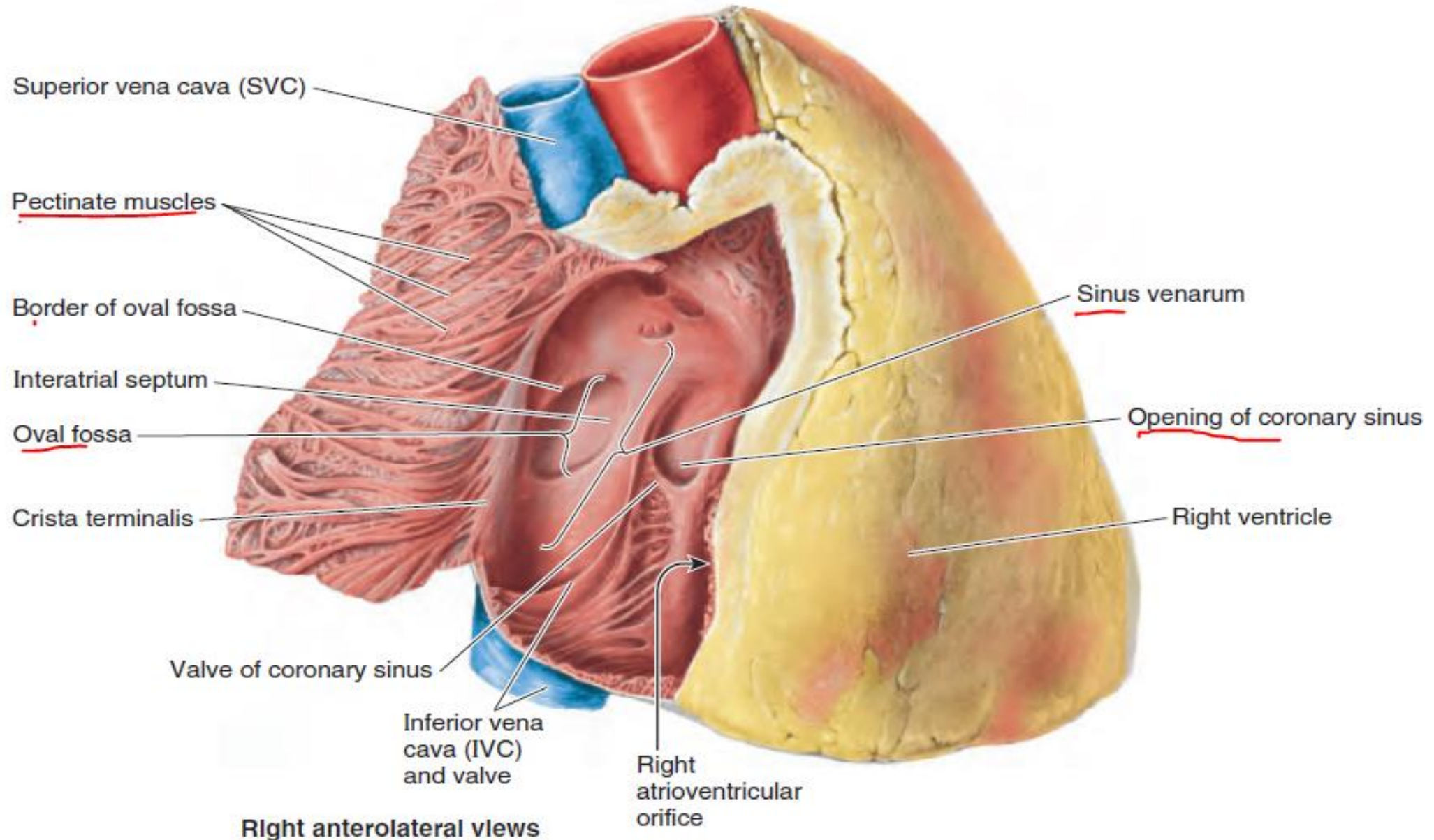


The *interior of the right atrium*

❑ Interior of Rt atrium is divided into two continuous parts :

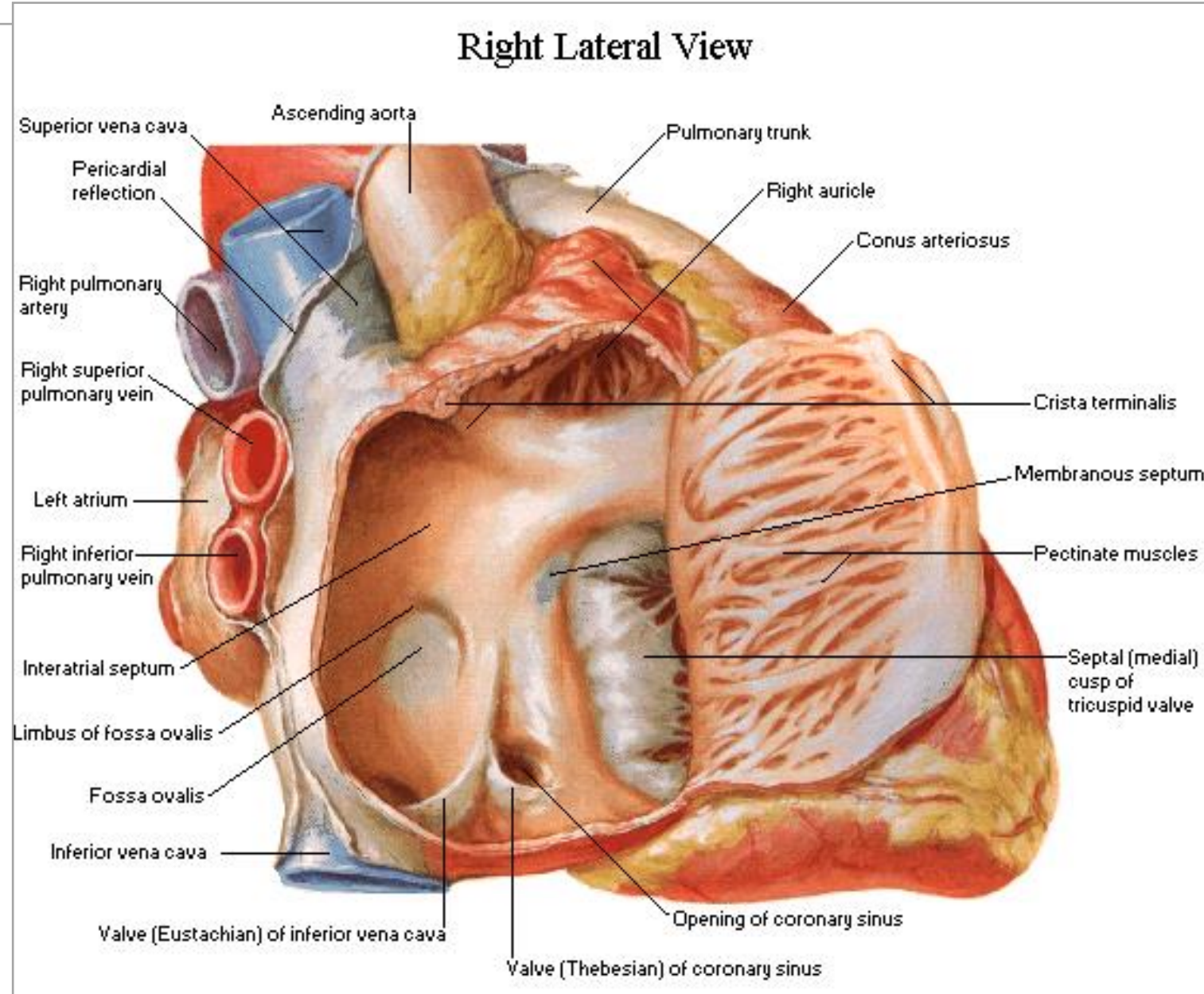
- 1) **Smooth, thin-walled, posterior part** (the **sinus venarum**) on w/c the SVC, IVC & coronary sinus open.
- 2) **Rough, muscular anterior wall** composed of pectinate muscles.
 - The smooth & rough parts are separated externally by the **sulcus terminalis** & internally by the **crista terminalis**.
 - The SVC opens into the superior part of the Rt atrium at the level of Rt 3rd **costal cartilage**.
 - The IVC opens into the inferior part of the right atrium almost in line with the SVC at approximately the level of the **5th costal cartilage**.

The *interior of the right atrium*,,



The *interior of the right atrium*,,,

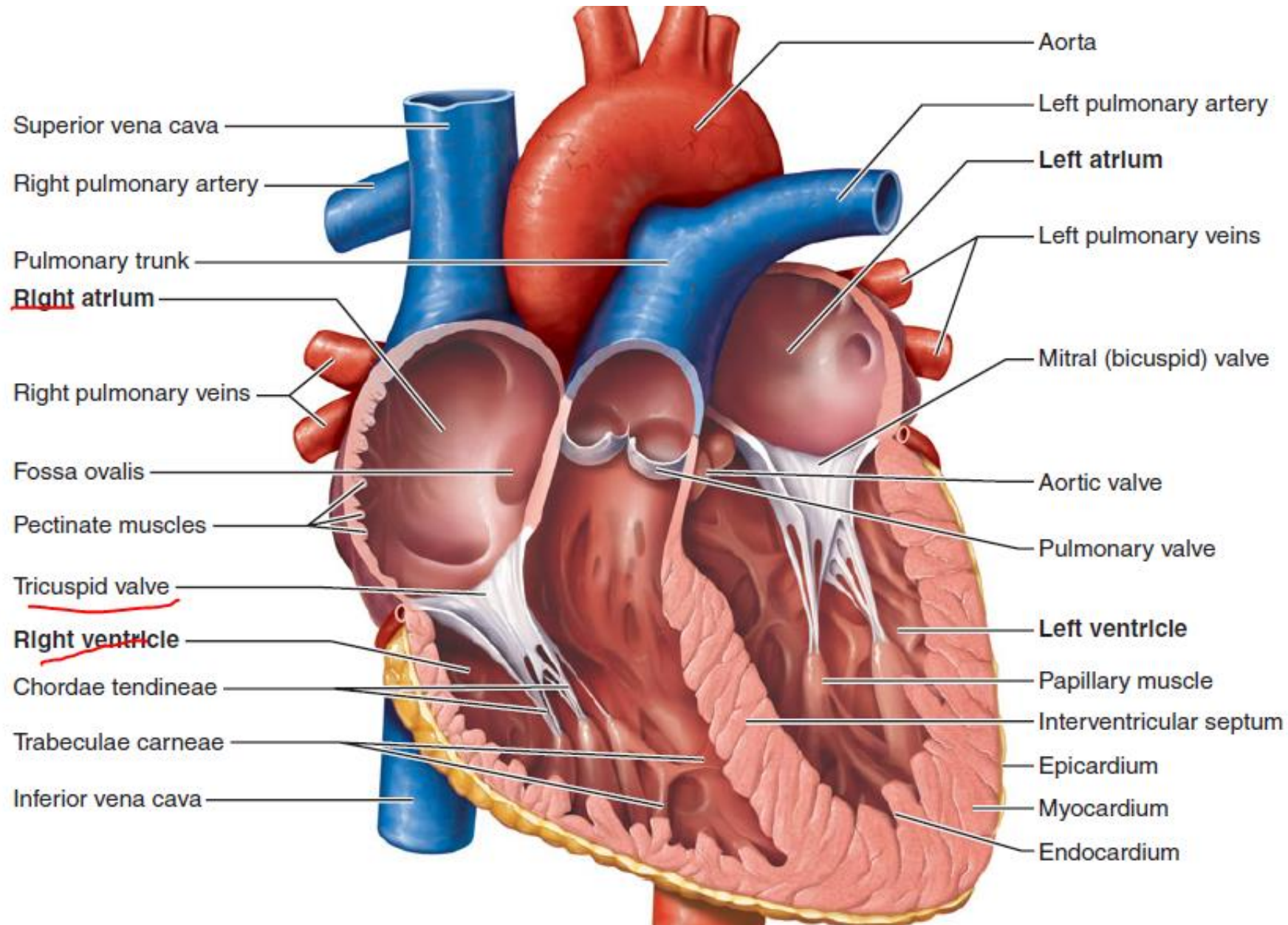
- The **coronary sinus**, opens b/n the orifices of Rt AV& IVC.
- The **interatrial septum** separating the **atria** has an oval, depression, the **oval fossa** (L. *fossa ovalis*), w/c is a remnant of the **oval foramen** (L. *foramen ovale*) & its valve in the fetus.



The interior of the right atrium,,,

- Between the right atrium and left atrium is a thin partition called the **interatrial septum**.
- A prominent feature of this septum is an oval depression called the **fossa ovalis**, the remnant of the *foramen ovale*, an opening in the interatrial septum of the fetal heart that normally closes soon after birth.
- Blood passes from the right atrium into the right ventricle through a valve that is called the **tricuspid valve**

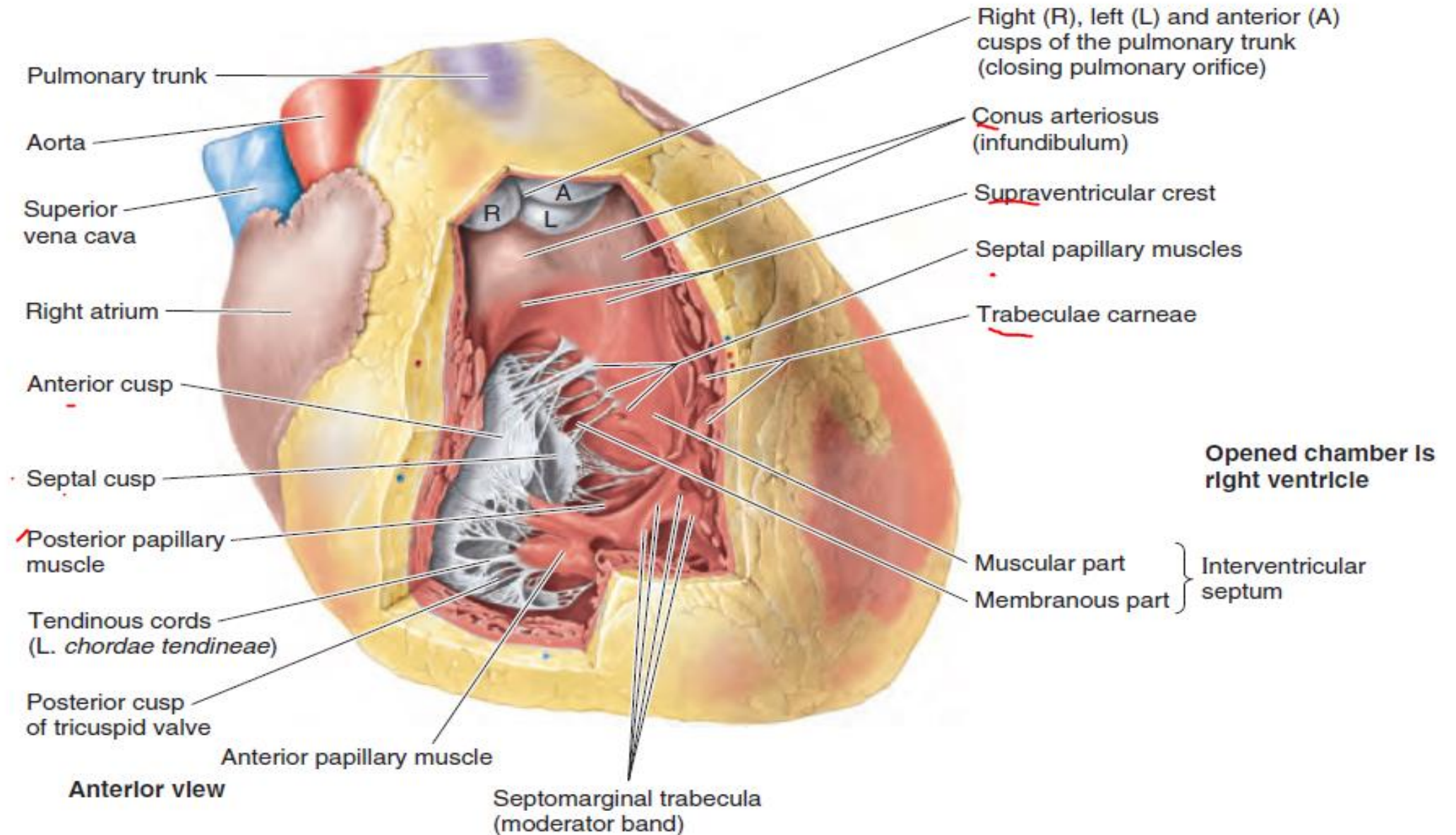
The *interior* of the right atrium,,,



Right ventricle

- The Rt ventricle forms the largest part of the anterior surface of the heart, a small part of the diaphragmatic surface & most inferior border of the heart.
- The outflow tract of the right ventricle, which leads to the pulmonary trunk, is the **conus arteriosus (infundibulum)**.
- **The interior of the Rt ventricle** has a series of ridges formed by raised bundles of cardiac muscle fibers called **trabeculae carneae**.
- A thick muscular ridge, the **supraventricular crest**, separates the **muscular wall** of the **inflow** part, from the **smooth wall** of the conus arteriosus, or **outflow** part.

Right ventricle,,

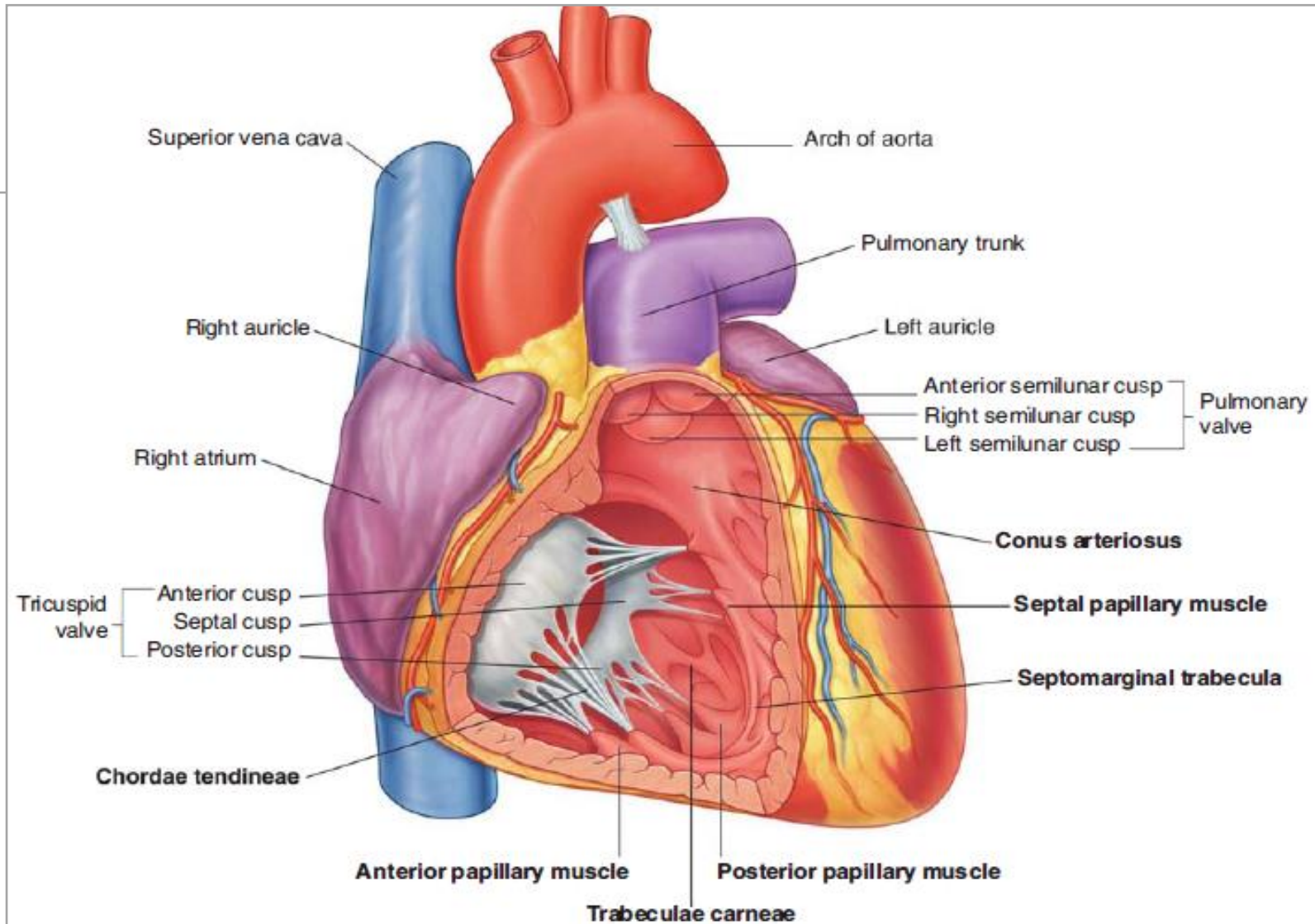


Right ventricle,,,

- **The inflow part** of the ventricle receives blood from the right atrium through the **Rt AV (tricuspid) orifice** located at the level of the 4th& 5th intercostal spaces.
- The walls of the inflow part of the Rt ventricle have numerous muscular, irregular structures called **trabeculae carneae** .
- A few **trabeculae carneae (papillary muscles)** have only one end attached to the ventricular surface, while the other end serves as the point of attachment for tendon-like fibrous cords (the **chordae tendineae**), which connect to the free edges of the cusps of **the tricuspid valve**.

Right ventricle,,,

- **Tendinous cords** attach to the anterior, posterior & septal cusps.
- **The tendinous** cords arise from the apices of **papillary muscles**.

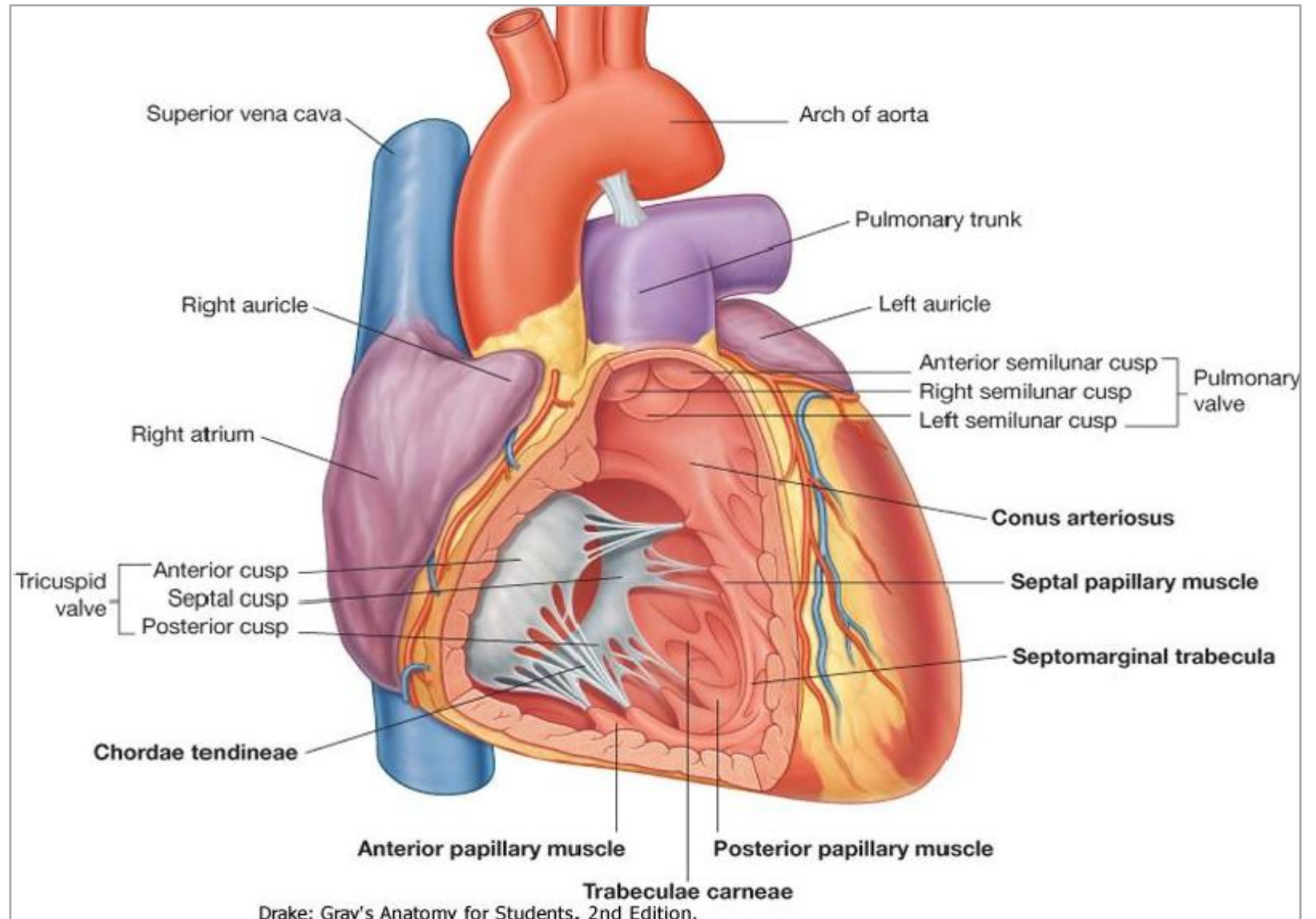


Right ventricle,,

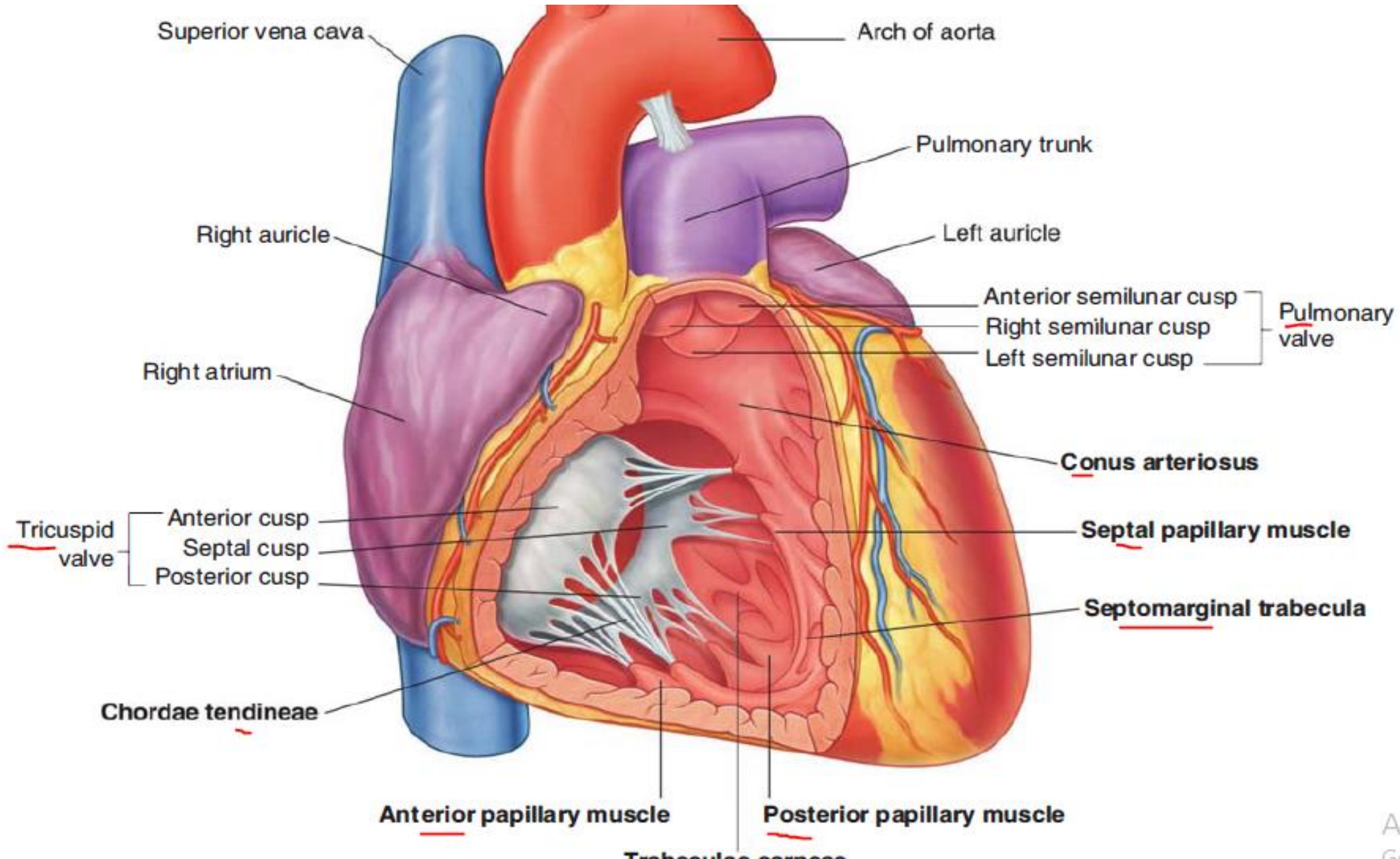
- There are 3 papillary muscles in the Rt ventricle.
- They are named relative to their point of origin on the ventricular surface. **These are:**
 1. The **anterior papillary muscle** is the largest & most prominent, arises from the anterior wall of the right ventricle; **its tendinous cords attach to the anterior and posterior cusps of the tricuspid valve.**
 2. The **posterior papillary muscle**, smaller than the anterior muscle,; it arises from the **inferior wall of the right ventricle**, and its tendinous cords attach to the **posterior and septal cusps** of the tricuspid valve.

Right ventricle,,

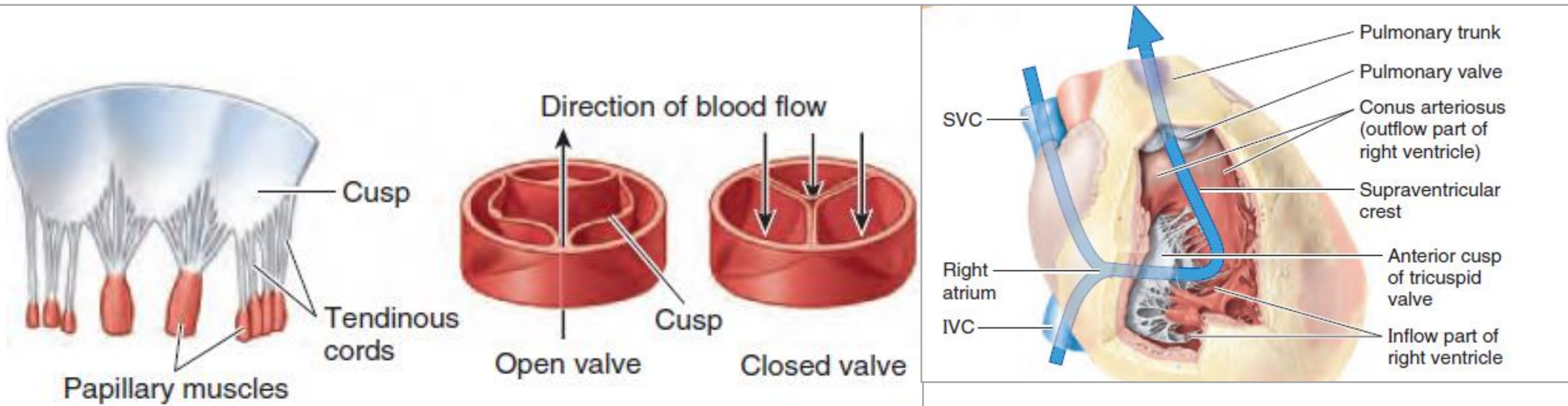
3. The **septal papillary muscle** arises from the interventricular septum, and its tendinous cords attach to the anterior and septal cusps of the tricuspid valve.



Right ventricle,,



Right ventricle,,

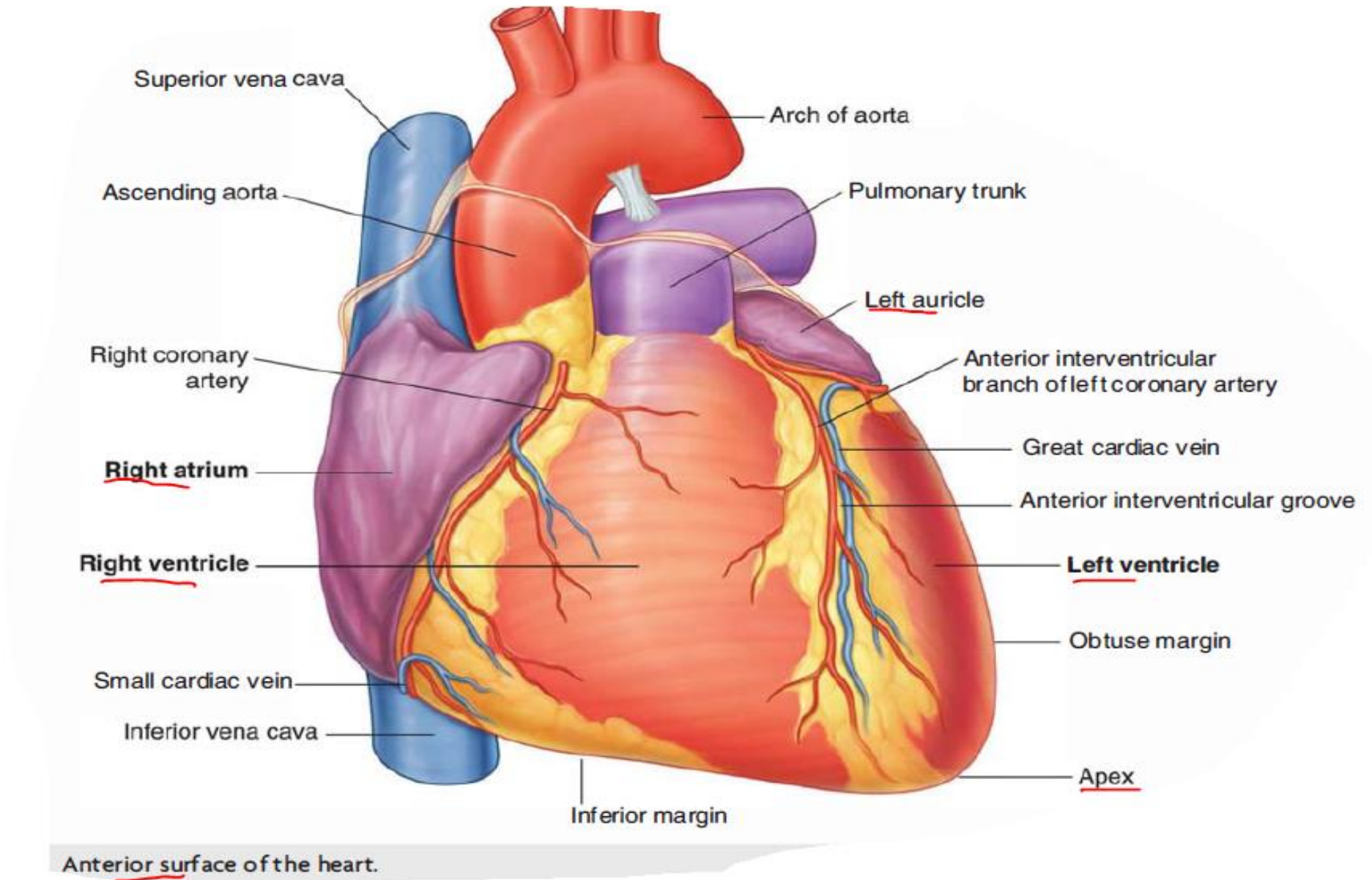


❖ *The **right ventricle** receives blood from the right atrium and pumps it into the pulmonary circuit via an artery called the **pulmonary trunk**.*

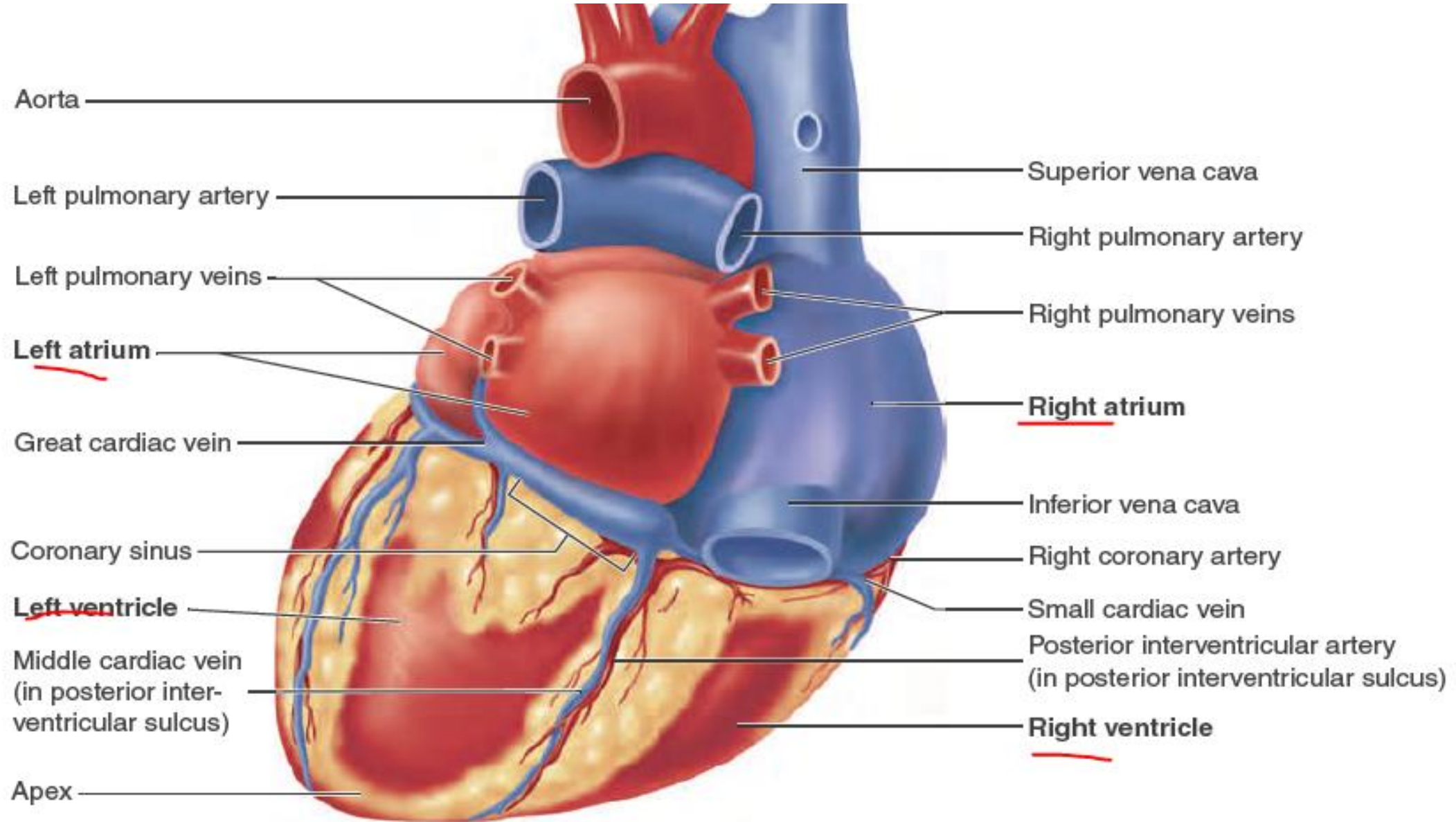
Left atrium

- The left atrium forms most of the base of the heart and has same thickness as the right atrium.
- The valveless pairs of Rt and Lt pulmonary veins enter the **smooth-walled atrium**.
- In the embryo, there is only one common pulmonary vein, just as there is a single pulmonary trunk.
- The part of the wall derived from the **embryonic pulmonary vein** is smooth walled.

Left atrium,,,



Left atrium,,,



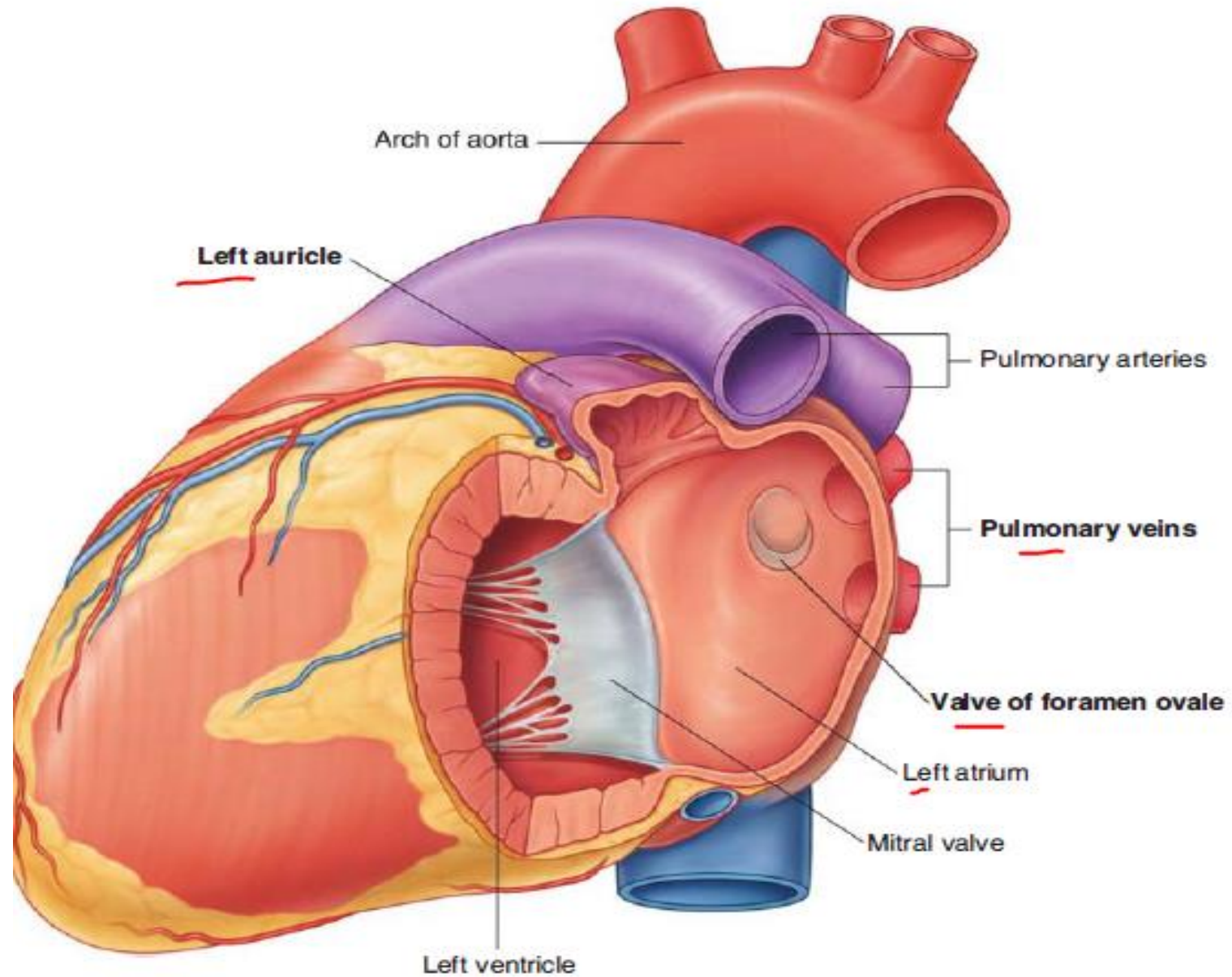
(c)

Posterior view

Left atrium,,

- **The tubular**, muscular **left auricle**, its wall trabeculated with *pectinate muscles*, forms the superior part of the left border of the heart and overlaps the root of the *pulmonary trunk*.
- It represents the remains of the left part of the primordial atrium.
- A *semilunar depression* in the interatrial septum indicates the floor of the oval fossa; the surrounding ridge is the valve of the oval fossa (*foramen ovale*).
- A larger **smooth**-walled part and a smaller **muscular auricle** containing pectinate muscles.
- ❖ **Four pulmonary** veins (two superior and two inferior) entering its smooth posterior wall.
- A slightly thicker wall than that of the right atrium.

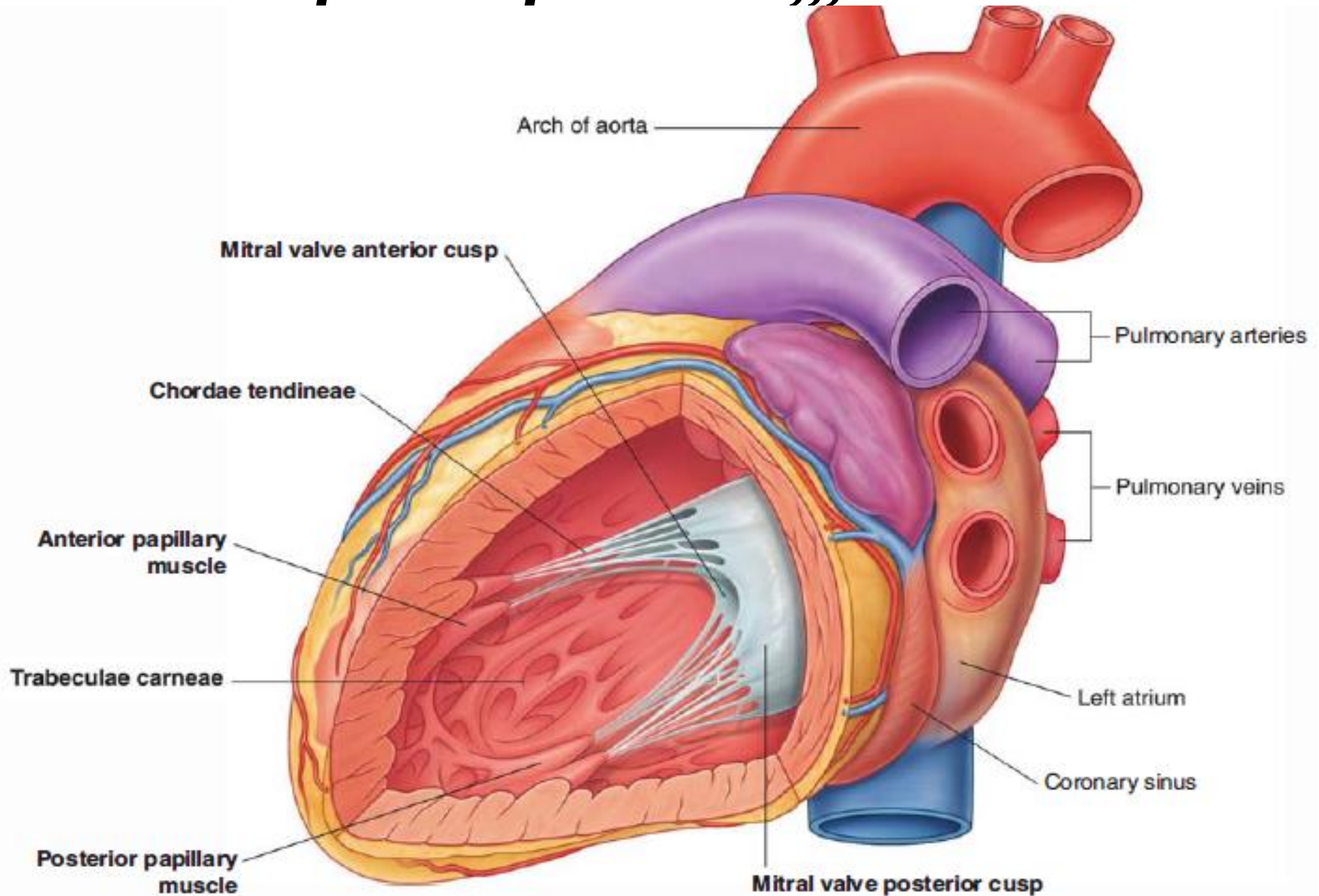
Left atrium,,,



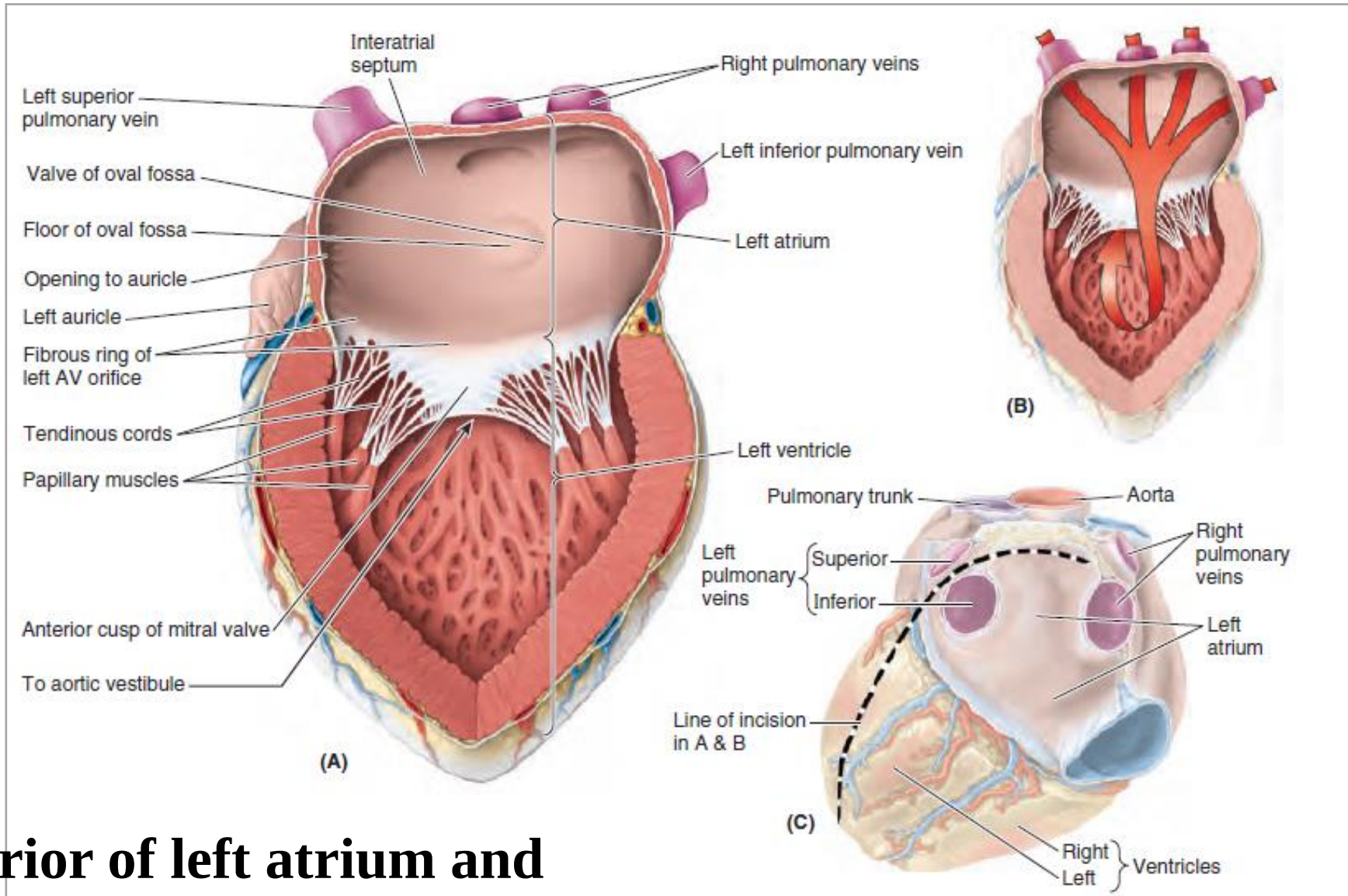
The *interior of the left atrium*

- An interatrial septum that slopes posteriorly and to the right.
- A left AV orifice through which the left atrium discharges the oxygenated blood it receives from the pulmonary veins into the left ventricle.
- Blood passes from the left atrium into the left ventricle through the **bicuspid (mitral) valve**, which, as its name implies, has two cusps.
- The term *mitral* refers to the resemblance of the bicuspid valve to a bishop's miter (hat), which is two-sided. It is also called the *left atrioventricular valve*

The *interior* of the left atrium,,,

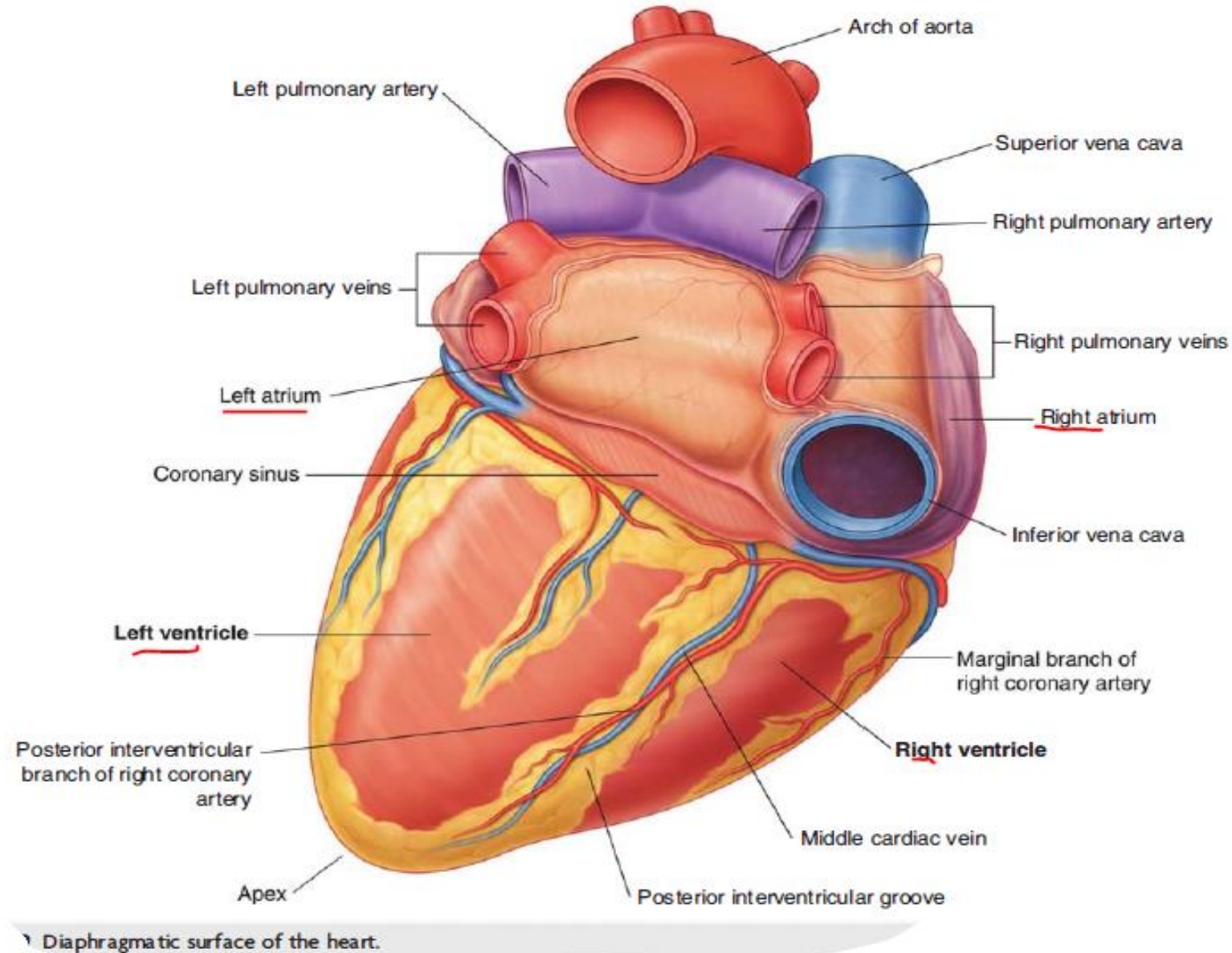


The *interior of the left atrium,,,*



**Interior of left atrium and
ventricle of heart.**

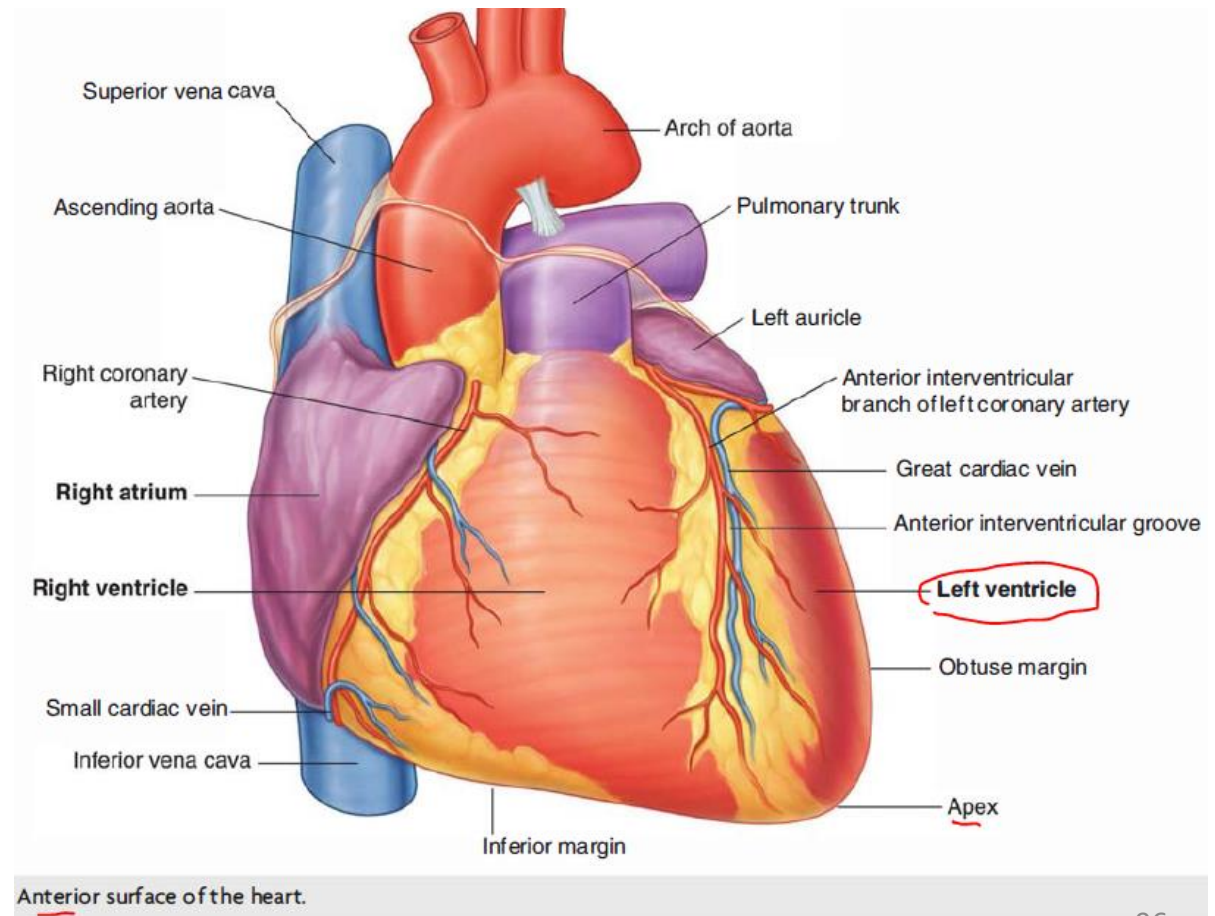
The *interior* of the left atrium,,,



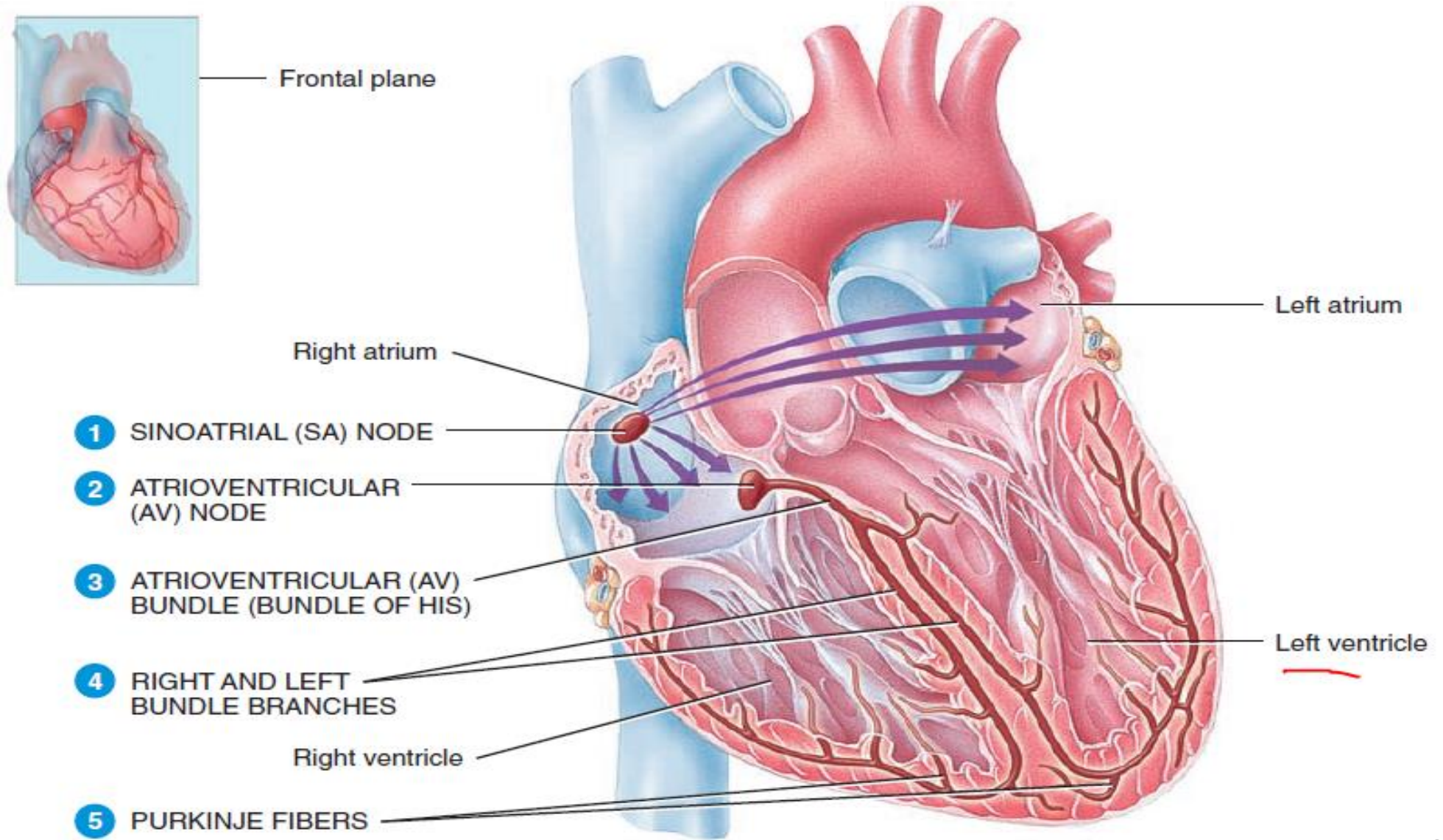
Left ventricle

❑ The left ventricle forms the apex of the heart, nearly all its left (pulmonary) surface and border, and most of the diaphragmatic surface.

❑ The **left ventricle** is the **thickest** chamber of the heart, like the right ventricle, the left ventricle contains trabeculae **carneae** and has chordae tendineae that anchor the cusps of the bicuspid valve to papillary muscles.



Left ventricle,,

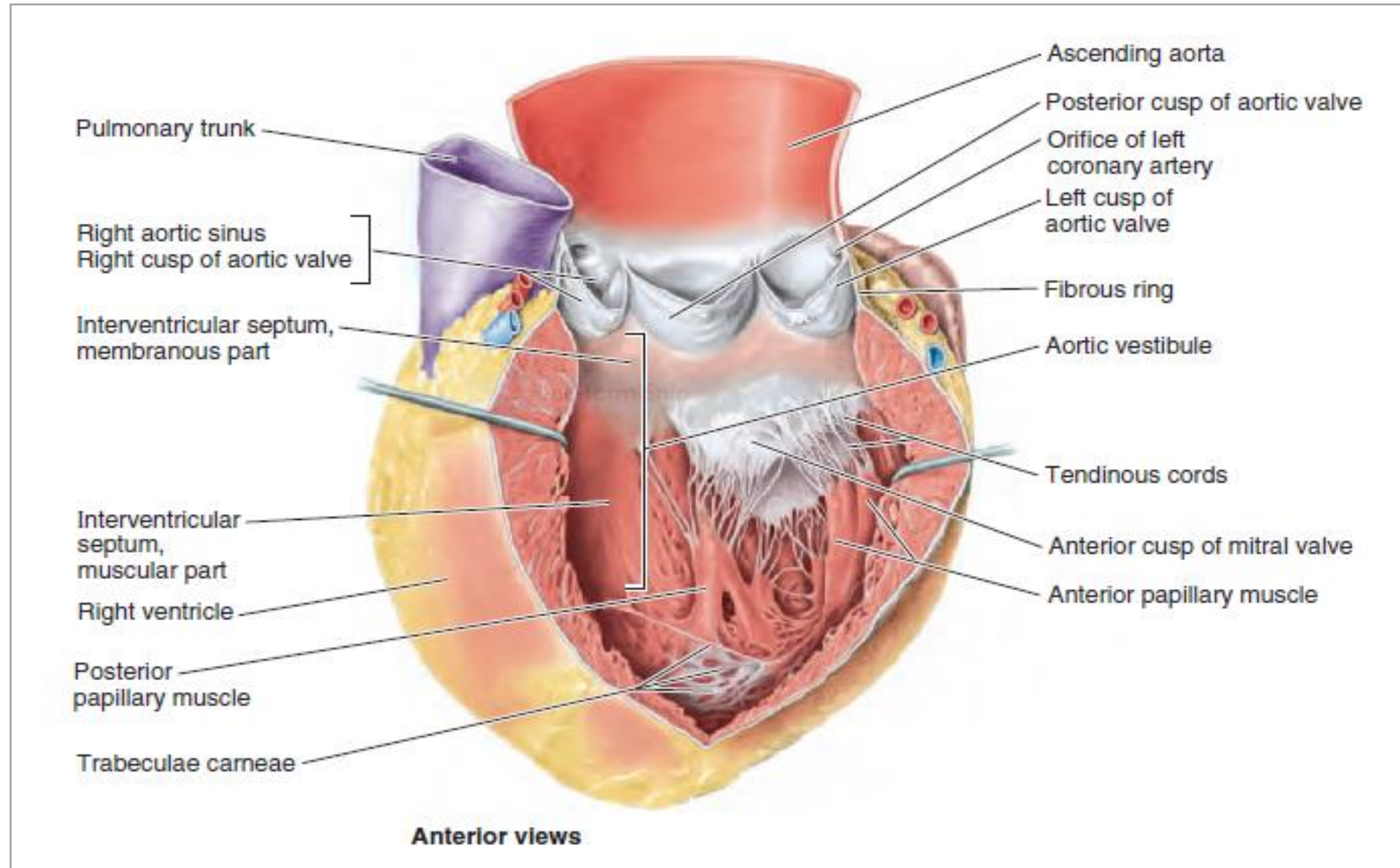


(a) Anterior view of frontal section

Left ventricle,,,

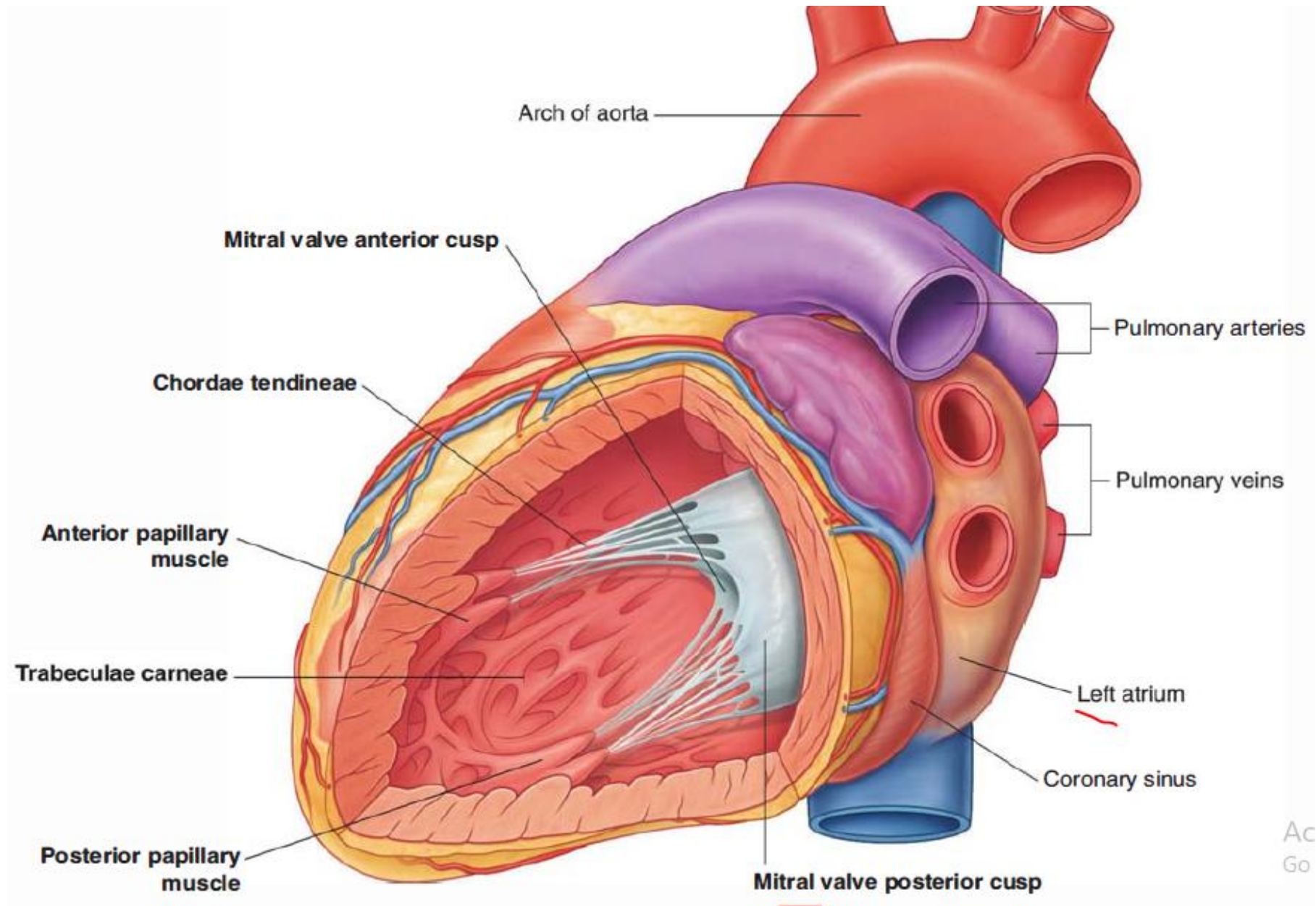
- Walls that are **two to three times** as thick as those of the *right ventricle*.
- A conical cavity that is longer than that of the right ventricle.
- Anterior & posterior *papillary muscles* that are larger than those in the right ventricle.
- A smooth-walled, non-muscular, supero-anterior outflow part, the **aortic vestibule**, leading to the aortic orifice and *aortic valve*.
- A double-leaflet **mitral valve** that guards the left AV orifice.
- An **aortic orifice** that lies in its right posterosuperior part and the ascending aorta begins at the aortic orifice.

Left ventricle,,

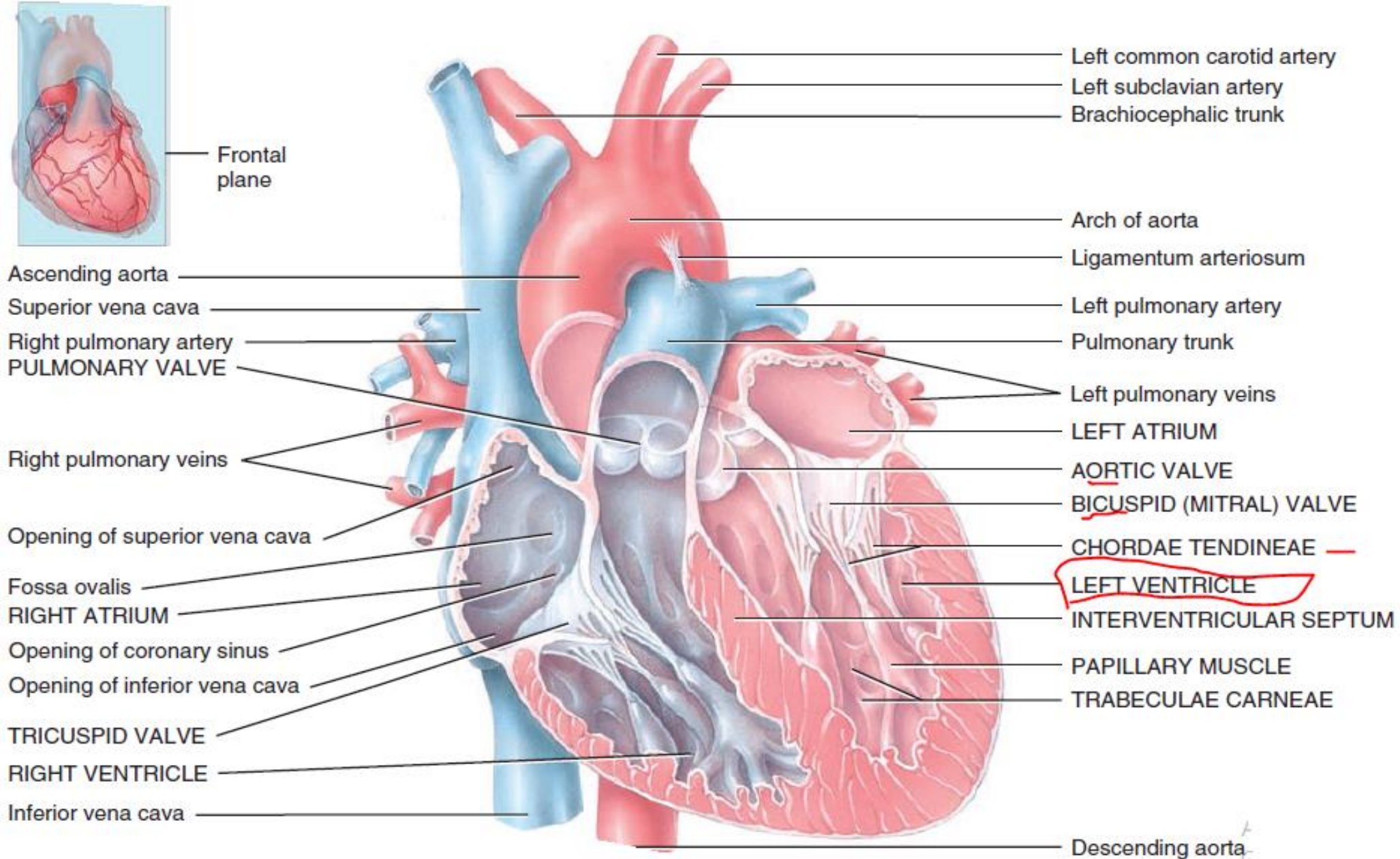


Interior and outflow tract of left ventricle of heart.

Left ventricle,,

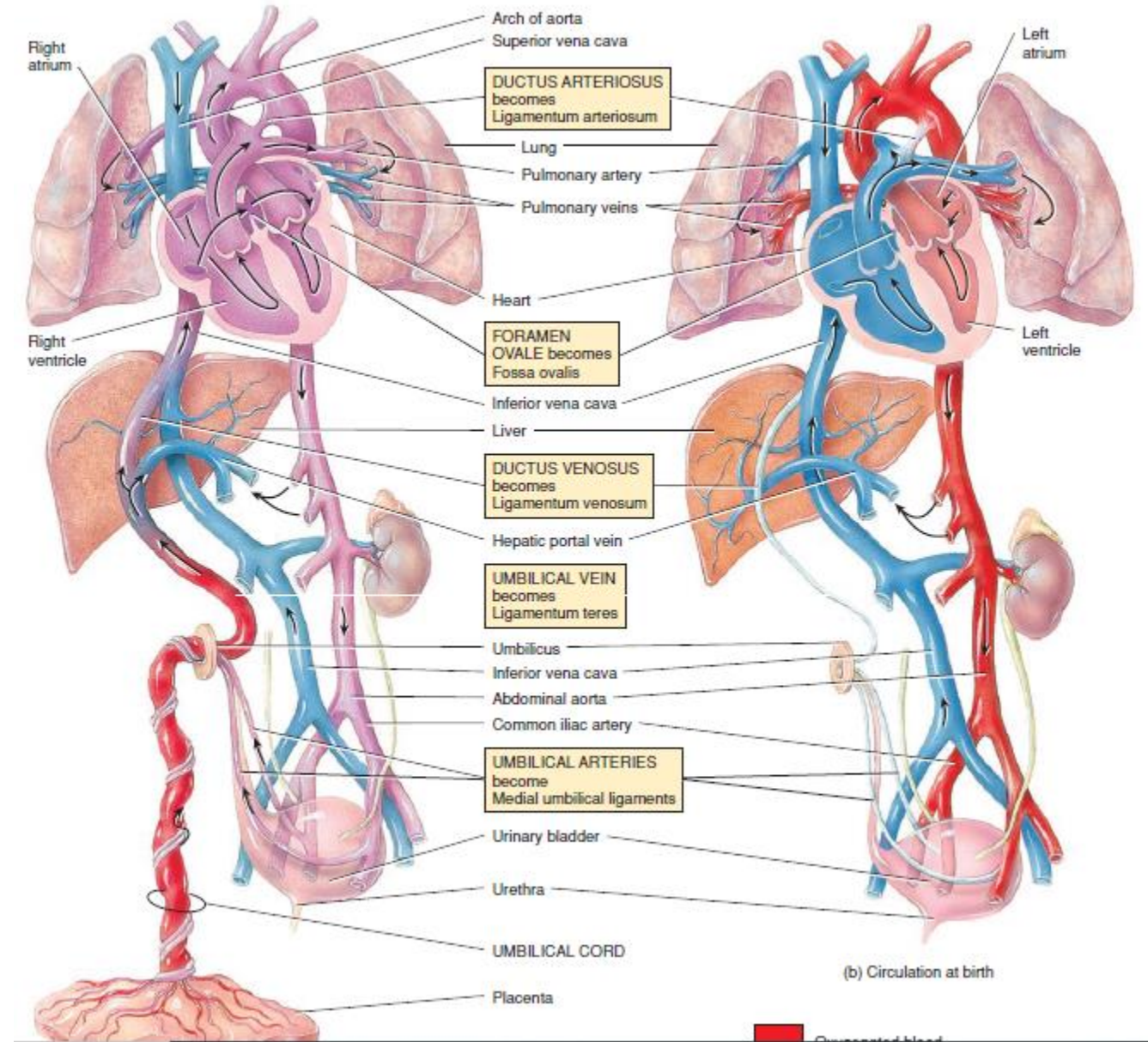


Left ventricle,,,

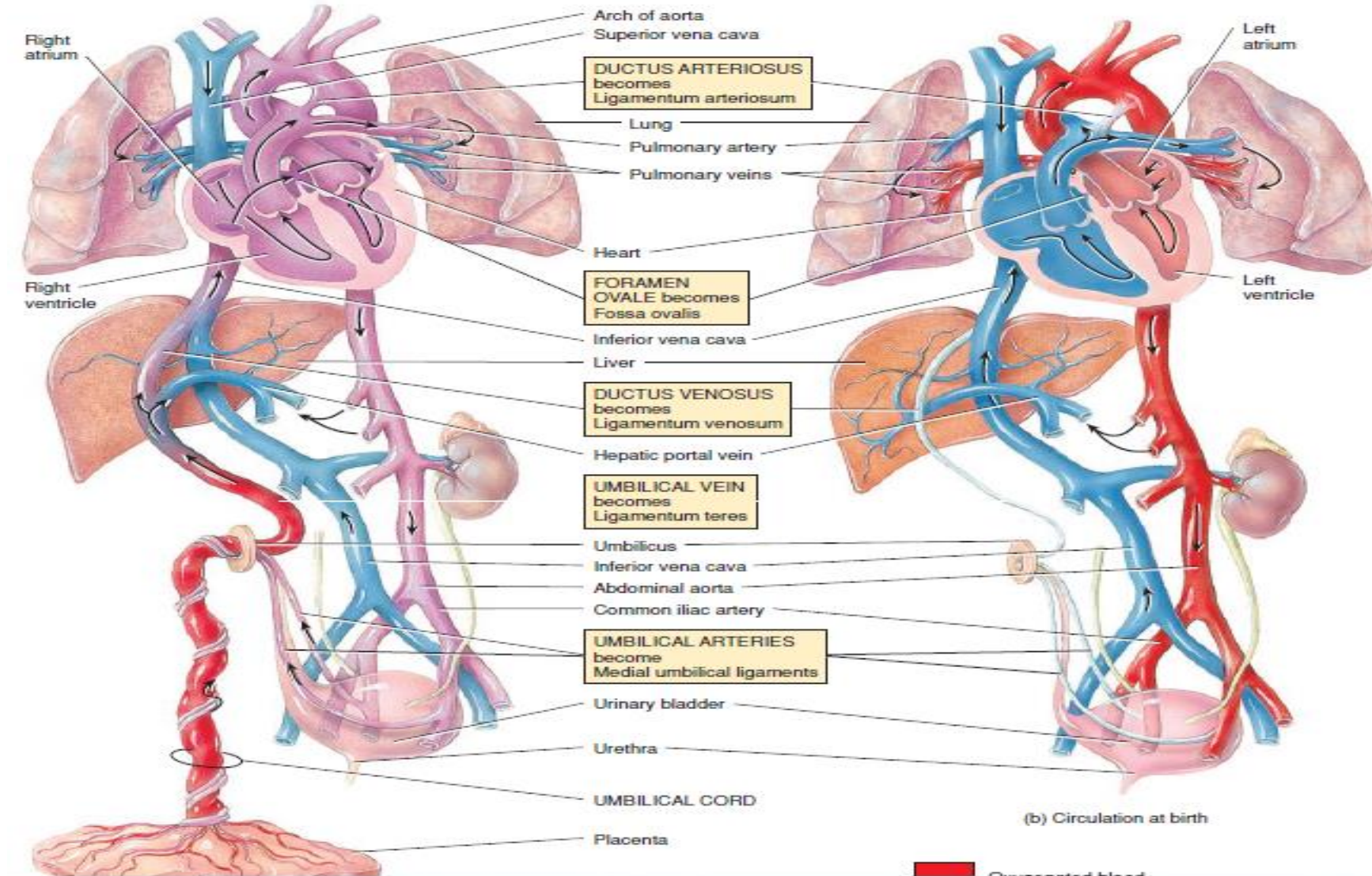


Left ventricle,,,

- During fetal life, a temporary blood vessel, **called the *ductus arteriosus***, shunts blood from the pulmonary trunk into the aorta.
- Hence, only a small amount of blood enters the nonfunctioning fetal lungs.
- The ductus arteriosus normally closes shortly after birth, leaving a remnant known as the **ligamentum arteriosum**, which connects the arch of the aorta and pulmonary trunk



Left ventricle,,



Left ventricle,,,

